

Remote Hardened Controller Planning, Installation, and Service

PM02-530

**Implementation
PM/APM/HPM Optional Devices**

Remote Hardened Controller Planning, Installation, and Service

**PM02-530
Release 530
09/08**

Copyright, Notices, and Trademarks

© Copyright 1997 – 2008 Honeywell Inc.

Revision 03 – September, 2008

While this information is presented in good faith and believed to be accurate, Honeywell disclaims the implied warranties of merchantability and fitness for a particular purpose and makes no express warranties except as may be stated in its written agreement with and for its customer.

In no event is Honeywell liable to anyone for any indirect, special or consequential damages. The information and specifications in this document are subject to change without notice.

TDC 3000, **TotalPlant**, and Process Manager are U.S. registered trademarks of Honeywell Inc.

Other brand or product names are trademarks of their respective owners.

Honeywell
Process Solutions
2500 West Union Hills Drive
Phoenix, AZ 85027
1-800-852-3211

About This Publication

This publication provides Remote Hardened Controller (RHC) enclosure planning, installation, and service information. Generally, the Remote Hardened Controller enclosure consists of a stainless steel dual-access enclosure (cabinet) that contains a High-Performance Process Manager (HPM) with supporting IOPs, FTAs, Universal Control Network Extender (UCNE), and a Power System. The Power System can be optionally supported by backup battery assembly.

Each enclosure is “customized” for the user’s requirements and will not always contain the hardware previously mentioned. For example, the enclosure may contain only IOPs, FTAs, and a Power System.

The publication does not attempt to provide planning, installation, or servicing information for the IOPs, FTAs, I/O Link Extender, UCNE, or the Power System. That information can be found in the reference publications that are listed in this publication.

It is assumed that the user is familiar with the use of a High-Performance Process Manager, IOPs, FTAs, I/O Link Extender, and UCNE and has access to the documentation that is referenced.

This publication supports **TotalPlant** Solution (TPS) system network Release 530 and later software releases. TPS is the evolution of TDC 3000^X.

Although some of the assemblies that are installed in the enclosure may be CE Compliant, the enclosure and its contents together are not CE Compliant.

Support and Other Contacts

Location	Organization	Address	Phone/Facsimile
United States and Canada	Honeywell Solution Support Center	Honeywell TAC, MS P13, 2500 West Union Hills Drive, Phoenix, AZ, 85027	Phone: 1-800 822-7673 602- 313-5558 (In Arizona) Facsimile: (602) 313-3293
Note: Calls are answered by dispatcher between 6:00 A.M. and 4:00 P.M. Mountain Standard Time. Emergency calls outside normal working hours are received by an answering service and returned within one hour.			
Europe	Honeywell TAC-EMEA	TAC-BE02 Hermes Plaza Hermeslaan, 1HB-1831 Diegem, Belgium	Phone: +32-2-728-2732 Facsimile: +32-2-728-2696
Pacific	Honeywell Global TAC – Pacific	Honeywell Limited Australia 5 Kitchener Way Burswood 6100, Western Australia Email: GTAC@honeywell.com	Phone: 1300-300-4822 (toll free within Australia) +61-8-9362-9559 (outside Australia) Facsimile: +61-8-9362-9564
India	Honeywell Global TAC – India	Honeywell Automation India Ltd. 56 and 57, Hadapsar Industrial Estate Hadapsar, Pune –411 013, India Email: Global-TAC- India@honeywell.com	Phone: +91-20- 66039400 Facsimile: +91-20- 66039800
Korea	Honeywell Global TAC – Korea	Honeywell Co., Ltd 17F, Kikje Center B/D, 191, Hangangro-2Ga Yongsan-gu, Seoul, 140- 702, Korea Email: Global-TAC- Korea@honeywell.com	Phone: +82-2-799-6317 +82-11-9227-6324 Facsimile: +82-2-792-9015

Support and Other Contacts

Location	Organization	Address	Phone/Facsimile
People's Republic of China	Honeywell Global TAC – China	Honeywell (China) Co., Ltd 33/F, Tower A, City Center, 100 Zunyi Rd., Shanghai 200051, People's Republic of China Email: Global-TAC-China@honeywell.com	Phone: +86- 21-52574568
Singapore	Global TAC – South East Asia	Honeywell Private Limited Honeywell Building 17, Changi Business Park Central 1, Singapore 486073 Email: GTAC-SEA@honeywell.com	Phone: +65-6580-3500 Facsimile: +65-6580-3501 +65-6445-3033
Taiwan	Global TAC – Taiwan	Honeywell Taiwan Ltd. 17F-1, No. 260, Jhongshan 2nd Road. Cianjhen District Kaohsiung, Taiwan, ROC Email: Global-TAC-Taiwan@honeywell.com	Phone: +886- 7- 536 2567 Facsimile: +886-7-536 2039
Japan	Global TAC – Japan	Honeywell Japan Inc. New Pier Takeshiba, South Tower Building, 20th Floor, 1-16-1 Kaigan, Minato-ku, Tokyo 105-0022, Japan Email: Global-TAC-JapanJA25@honeywell.com	Phone: +81-3-6730-7160 Facsimile: +81-3-6730-7228
Elsewhere	Call your nearest Honeywell office.		
World Wide Web	Honeywell Solution Support Online: http://www.honeywell.com/ps		
Training Classes	Honeywell Automation College: http://www.automationcollege.com		

Table of Contents

SECTION 1 – INTRODUCTION.....	1
1.1 Overview.....	1
1.2 Description.....	2
1.2.1 Overview.....	2
1.2.2 Basic Enclosure Description	8
1.2.3 Basic Enclosure Option Descriptions	12
SECTION 2 – PLANNING	15
2.1 Overview.....	15
2.2 Guidelines.....	15
2.3 AC Power Requirements.....	16
2.3.1 Overview.....	16
2.3.2 AC Power System Design	19
2.3.3 AC Power Distribution Guidelines	24
2.4 Ground Requirements	28
2.4.1 .ii.System Grounds	28
2.4.2 Star Plate	33
2.4.3 PM/APM/HPM Grounding	34
2.5 Hardware Selection	37
2.5.1 Power System Selection	38
2.5.2 Card File Selection	39
2.5.3 HPMM Card Set Selection	40
2.5.4 IOP Selection.....	40
2.5.5 FTA Selection	41
2.5.6 FTA Mounting Channel Selection	42
2.5.7 I/O Link Extender Selection.....	42
2.5.8 Universal Control Network Extender Selection	42
2.6 Cabling Selection	43
2.6.1 24 Vdc Power Cable Selection.....	43
2.6.2 IOP to FTA Cable Selection	43
2.6.3 I/O Link Interface Cable Selection.....	43
2.6.4 I/O Link Extender Cable Selection	44
2.6.5 UCN Cable Selection	44
2.6.6 UCN Extender Cable Selection.....	45
2.7 Conduit Selection	46
2.8 Heater Selection	47
2.9 Enclosure Complexing	48
2.9.1 Thermal Considerations	48
SECTION 3 – INSTALLATION.....	53
3.1 Overview.....	53
3.2 Installation Considerations	54
3.3 Enclosure Mounting.....	55
3.4 Conduit Installation.....	57
3.5 AC Power and Ground Connections	59
3.5.1 Standard Power System.....	59
3.5.2 AC Only Power System.....	61

Table of Contents

3.6	Alarm Contacts Connections	62
3.6.1	Standard Power System	62
3.6.2	AC Only Power System	63
3.6.3	Fan Assembly	64
3.7	I/O Link Interface Cables	65
3.8	I/O Link Interface Extender	66
3.9	Universal Control Network Extender	71
3.10	Card File I/O Link Interface Address Selection.....	77
3.11	Card File UCN Node Address Selection.....	78
3.12	Optional Backup Battery Assembly	80
3.13	Optional Heater Assembly	85
3.14	Enclosure Integrity	88
SECTION 4 – SERVICE		89
4.1	Overview	89
4.2	Fan Assembly Replacement.....	90
4.3	Backup Battery Replacement	91
4.4	Electrical Heater Assembly Replacemen	97
4.5	Preventative Maintenanc	98
SECTION 5 – SPARE PARTS		99
5.1	Overview	99
5.2	Periodic Maintenance Part.....	100
5.3	Optimum Replaceable Units	102
5.4	UCN Parts List	112

Figures

Figure 1-1	Front Interior View of RHC Enclosure	10
Figure 1-2	Rear Interior View of RHC Enclosure.....	11
Figure 2-1	HPM AC Power Distribution—Single Power Source.....	25
Figure 2-2	HPM AC Power Distribution—Separate Power Sources	26
Figure 2-3	Subsystem AC Power and Ground Connections—Multi-Ground System ..	27
Figure 2-4	TPS System Ground System—Upper View	29
Figure 2-5	TPS System Ground System—Lower View	30
Figure 2-6	Example of a Star Plate.....	33
Figure 2-7	HPM Cabinet Grounding	34
Figure 2-8	HPM Cabinet Ground Wiring.....	35
Figure 3-1	Enclosure Base Mounting Dimensions	55
Figure 3-2	Acceptable Enclosure Conduit Entry Areas	58
Figure 3-3	Standard Power System Power and Ground Connections.....	60
Figure 3-4	AC Only Power System Power and Ground Connections	61
Figure 3-5	Standard Power System Power Supply Module	62
Figure 3-6	AC Only Power System Power Supply Module.....	64
Figure 3-7	Fan Assembly Alarm Annunciator Connections.....	64
Figure 3-8	Standard I/O Link Interface Extender Card Pinning.....	68
Figure 3-9	Long Distance I/O Link Interface Extender Card Pinning	70
Figure 3-10	DC-Powered UCNE Device.....	71
Figure 3-11	Exploded View of UCNE Enclosure Installation	72
Figure 3-12	Mounting Panel Attachment Holes.....	74
Figure 3-13	Mounting Panel with UCNE Enclosure and Heater Assembly Attached....	75
Figure 3-14	UCNE Fiber Optic Cable Interconnections.....	76
Figure 3-15	Left 7-Slot Card File I/O Link Interface Address Selection.....	77
Figure 3-16	Left 7-Slot Card File UCN Node Address Selection.....	79
Figure 3-17	Standard Power System Backpanel.....	81
Figure 3-18	Exploded View of Model MU-RBMK01 Backup Battery Mounting Kit.....	83
Figure 3-19	Battery Extension Cable Assembly	84
Figure 3-20	Bottom View of RHC Enclosure	84
Figure 3-21	Heater Kit Parts Pictorial	87
Figure 4-1	Expected Battery Life for Average Ambient Temperature	93

Tables

Table 1-1	Remote Hardened Controller Enclosure Assembly Model Numbers	3
Table 2-1	AC Power System Guidelines.....	20
Table 2-2	Copper Wire Size Based on RMS Current Requirement	22
Table 2-3	Maximum Cable Length in Meters for 240 Vac	23
Table 2-4	Maximum Cable Length in Meters for 120 Vac	23
Table 2-5	Conduit Fittings – Bridgeport Fittings, Inc.....	46
Table 2-6	Model MU-RCHT01/MU-RCHT02 Specifications	47
Table 2-7	Enclosure Complexing Thermal Analysis	50
Table 3-1	RHC Enclosure Mounting Procedure	56
Table 3-2	Multiple Enclosure Conduit Installation Procedure	57
Table 3-3	UCNE Enclosure Installation Procedure.....	73
Table 3-4	Backup Battery Assembly Installation Procedure	82
Table 3-5	Model MU-RCHT01/MU-RCHT02 Heater Kits Parts List	85
Table 3-6	Heater Kit Installation Procedure	86
Table 4-1	Fan Assembly Replacement Procedure	90
Table 4-2	Battery Charge Rate Verification Procedure	94
Table 4-3	Battery End of Life Check Procedure	95
Table 4-4	48 Volt Battery Pack Replacement Procedure	96
Table 4-5	Heater Assembly Replacement Procedure	97
Table 5-1	Periodic Maintenance Parts List	100
Table 5-2	Cables and Corrosion Protected ORU Parts List	103
Table 5-2	Cables and Corrosion Protected ORU Parts List	104
Table 5-3	UCN Parts List	112

Acronyms

FTA.....	Field Termination Assembly
HPM.....	High-Performance Process Manager
HPMM	High-Performance Process Manager Module
I/O.....	Input/Output
IEC.....	International Electrotechnical Commission
IOP	Input/Output Processor
LCN	Local Control Network
LED.....	Light Emitting Diode
NEMA	National Electrical Manufacturers Association
ORU	Optimum Replaceable Units
RHC.....	Remote Hardened Controller
TPC	TotalPlant Configurator
UCN.....	Universal Control Network
UCNE	Universal Control Network Extender

References

Publication Title	Publication Number	Binder Title	Binder Number
<i>Process Manager I/O Installation</i>	PM20-620	Implementation/PM/APM	TPS 3043
<i>High-Performance Process Manager Planning</i>	HP02-610	System Site Planning - 2	TPS 3020-2
<i>High-Performance Process Manager Installation</i>	HP20-600	Implementation/HPM - 3	TPS 3066-3
<i>High-Performance Process Manager Checkout</i>	HP20-510	Implementation/HPM - 3	TPS 3066-3
<i>High-Performance Process Manager Service</i>	HP13-610	PM/APM/HPM Service - 1	TPS 3061-1
<i>Universal Control Network Planning</i>	UN02-501	System Site Planning - 1	TPS 3020-1
<i>Universal Control Network Installation</i>	UN20-500	Installation/Universal Control Network	TPS 3041
TotalPlant <i>System Planning</i>	SW02-550	System Site Planning - 1	TPS 3020-1

Section 1 – Introduction

1.1 Overview

Purpose	The Remote Hardened Controller (RHC) enclosure (cabinet) provides a means of protecting High-Performance Process Manager (HPM) hardware in remotely located Gx (Severe) corrosive environments. A sealed dual-access stainless steel NEMA 4X enclosure (cabinet) provides corrosion protection for the HPMM, IOP, FTA, I/O Link Extender (IOLE), Universal Control Network Extender (UCNE), and Power System components that can be mounted in the enclosure.
Introduction	The RHC enclosure permits an HPM to be installed in a corrosive hostile environment at a remote location. The enclosure has four supporting legs that can be bolted to a concrete pad. A fiber optic Universal Control Network Extender (UCNE) communications link can provide the HPM's interface with a central Universal Control Network trunk cable segment that can be up to 2 kilometers away.
Corrosion protection	All the assemblies in the enclosure meet the G3 (Harsh) environmental product rating as defined for “Environmental Conditions for Process Measurement and Control Systems: Airborne Contaminants” in ANSI/ISA-S71.04-1985. When installed inside the RHC enclosure, the subsystem has an environmental rating of Gx (Severe).
Approval body certification	The Remote Hardened Controller is Factory Mutual approved for use in Class I, Division 2, Groups A, B, C, and D locations.

1.1 Overview, Continued

External ambient temperatures

The assemblies in the enclosure are designed to withstand exterior ambient temperatures between -10°C and 60°C. However, when power is first applied to an enclosure at a temperature near -10°C, a 1 hour warm-up period is necessary before operating the equipment with the assumption that a typical complement of electronics is installed in the enclosure. When the external temperature is below -10°C, an internal heater is available for “ordinary” conditions. For Class I, Division II, Groups A, B, C, and D locations, heat will have to be provided on site by steam tracers or another approved means.

1.2 Description

1.2.1 Overview

Introduction

Two model numbers, MU-RCAB01 (120 Vac) and MU-RCAB02 (240 Vac), identify the basic Remote Hardened Controller (RHC) enclosure that consists primarily of mechanical components such as a dual-access stainless steel enclosure, an air plenum, upper and lower fan assemblies that have rotation sensors, and rails to mount card files, a Power System, and FTA Mounting Channels. The card file(s), Power System, and FTA Mounting Channels are not included with the basis RHC enclosure.

Each enclosure is “customized” by the addition of HPM assemblies which are installed in the factory before shipment. The choice of HPM assemblies consists of 7-Slot and 15-Slot card files, HPMM card set(s), IOPs, FTAs, a UCNE, an IOLE, a Power System (standard or AC Only), and a backup battery assembly that is mounted external to the enclosure when a standard Power System is installed in the enclosure.

1.2.1 Overview, Continued

Hardware model numbers

Excluding IOPs and FTAs, Table 1-1 lists the model numbers of the recommended assemblies that can be installed in the RHC enclosure. Lists and descriptions of IOPs and FTAs with their model numbers can be found in the *Process Manager I/O Specification and Technical Data* and *Process Manager I/O Installation* manuals.

Table 1-1 Remote Hardened Controller Enclosure Assembly Model Numbers

Model Number	Component	Description
MC-PSSX04	Standard Power System	20-ampere Standard Power System Single Power Supply Module
MC-PSRX04	Standard Power System	20-ampere Standard Power System Redundant Power Supply Modules
MU-RBMK01	Backup Battery Mounting Kit	Standard Power System backup battery mounting kit with battery
MC-PAS121	AC Only Power System	16-ampere 120 Vac AC Only Power System Single Power Supply Module
MC-PAR121	AC Only Power System	16-ampere 120 Vac AC Only Power System Redundant Power Supply Modules
MC-PAS221	AC Only Power System	16-ampere 240 Vac AC Only Power System Single Power Supply Module
MC-PAR221	AC Only Power System	16-ampere 240 Vac AC Only Power System Redundant Power Supply Modules
MC-PAS111	AC Only Power System	8-ampere 120 Vac AC Only Power System Single Power Supply Module
MC-PAR111	AC Only Power System	8-ampere 120 Vac AC Only Power System Redundant Power Supply Modules
MC-PAS211	AC Only Power System	8-ampere 240 Vac AC Only Power System Single Power Supply Module
MC-PAR211	AC Only Power System	8-ampere 240 Vac AC Only Power System Redundant Power Supply Modules
MU-FOMK21	UCNE Mounting Kit	UCNE mounting kit
MU-FOED02	UCNE Devices	Pair of 24 Vdc UCNEs for MU-FOMK21
MU-NTAP02	UCN 2-Drop Taps	Two 2-drop taps with mounting/grounding hardware
MU-NTAP04	UCN 4-Drop Taps	Two 4-drop taps with mounting/grounding hardware
MU-TMCN02	FTA Mounting Channel	Long (36-inch) narrow FTA Mounting Channel with cable shield ground bar

1.2.1 Overview, Continued

Hardware model numbers, continued

Table 1-1 Remote Hardened Controller Enclosure Assembly Model Numbers, Continued

Model Number	Component	Description
MU-TMCN02	FTA Mounting Channel	Long (36-inch) narrow FTA Mounting Channel with cable shield ground bar
MU-TMCN22	FTA Mounting Channel	Short (24-inch) narrow FTA Mounting Channel with cable shield ground bar
MU-RCHT01	Cabinet Heater Kit	120 Vac 800 W cabinet heater for “ordinary” locations
MU-RCHT02	Cabinet Heater Kit	240 Vac 800 W cabinet heater for “ordinary” locations
MC-HPFH01	Left 7-Slot Card File	Left 7-Slot HPMM or IOP card file – non-CE Compliant
MC-HPFH03	Left 7-Slot Card File	Left 7-Slot HPMM or IOP card file – CE Compliant
MC-HPFI03	Left 7-Slot Card File	Left 7-Slot IOP card file – CE Compliant
MC-HPFH11	Right 7-Slot Card file	Right 7-Slot HPMM or IOP card file – non-CE Compliant
MC-HPFH13	Right 7-Slot Card file	Right 7-Slot HPMM or IOP card file – CE Compliant
MC-HPFI13	Right 7-Slot Card File	Right 7-Slot IOP card file – CE Compliant
MC-HPFX02	15-Slot Card file	15-Slot HPMM or IOP card file – non-CE Compliant
MC-HPFX03	15-Slot Card file	15-Slot HPMM or IOP card file – CE Compliant
MC-HPFI13	15-Slot Card file	15-Slot IOP card file – CE Compliant
MC-HPMS01	Nonredundant HPMM Card Set	Nonredundant HPMM card set
MC-HPMR01	Redundant HPMM Card Set	Redundant HPMM card set with redundancy cable.

TotalPlant Configurator

A **TotalPlant** Configurator (TPC) is used to configure the hardware that can be installed in an enclosure. The expected environmental conditions at the installation site is entered into the TPC by qualified Honeywell personnel.

Using thermal data, which is dependent upon the site’s environment and the enclosure’s suggested hardware configuration, the TPC calculates the temperature rise in the enclosure, and warns if the current requirements of the IOPs and FTAs exceed the capacity of the Power System that is selected. The TPC is based on several premises.

- When the equipment is mounted outdoors, the use of a sun/rain shield is required. The shield must totally protect the unit from direct sunlight.

1.2.1 Overview, Continued

- All thermal calculations and predictions are based on relatively calm air. If the enclosure is mounted outside where significant winds are possible, particularly at cold temperatures, the enclosure must be protected.

Power System selection

The Power System's Power Supply Module provides the required card file and fan assembly voltages. The Power System has single or redundant Power Supply Modules. Depending upon the Power System that is installed in the enclosure, 8, 16, or 20 amperes at 24 Vdc is provided to power single or redundant HPMM card sets, IOPs, FTAs, an IOLE, or a UCNE. Table 1-1 lists the Power Systems that can be installed in the RHC enclosure.

Enclosure complexing

Enclosures can be complexed together at a single remote site. However, there must be adequate spacing between the enclosures to ensure sufficient heat dissipation. Each enclosure must include a Power System. Depending upon the hardware configuration, communication between the enclosures is provided by, I/O Link Interface cables or UCN coaxial cables, or both.

I/O Link Extender

If the remote RHC enclosure contains only IOPs and FTAs, a fiber optic data link of up to 8 kilometers in length can be provided by an I/O Link Extender between the remote I/O and the centrally-located HPMM.

UCN Extender

If the remote RHC enclosure contains single or redundant HPMMs, a fiber optic data link of up to 2 kilometers in length can be provided by an I/O Link Extender between the centrally-located Network Interface Module (NIM) and the remote HPMM(s) can be provided by a Universal Control Network Extender (UCNE).

Corrosion protected hardware

The IOPs, card files, FTAs, and Power Systems are corrosion protected.

NEMA 4X integrity

Conduit connections to the enclosure, as well as other enclosure interface connections, must be provided with gaskets or seals that ensure the NEMA 4X integrity of the enclosure. A rain/sun shield is required when the enclosure is mounted where it can be exposed to the weather.

1.2.1 Overview, Continued

Exterior enclosure size	The exterior size of the enclosure is 2315.8 millimeters high, 799.3 millimeters wide, and 809 millimeters deep (91.3 x 31.47 x 31.85 inches). The height dimension does include the enclosure's support legs, but does not include the enclosure's two lifting eyes, which are located on the top of the enclosure. The two lifting eyes can be removed and the opening sealed with the two plugs that are included with enclosure.
Interior enclosure size	The interior size of the enclosure is 2110.7 millimeters high, 794.0 millimeters wide, 750.0 millimeters deep (83.10 x 31.26 x 29.53 inches).
Weight	The approximate weight of a fully complemented enclosure that contains a standard Power System with redundant Power Supply Modules, a backup battery assembly, two fully populated 15-Slot card files, and eight FTA Mounting Channels that are fully populated is approximately 1424 kilograms (645 pounds). The basic enclosure that contains only two dual fan assemblies, air plenum, and hardware mounting rails is approximately 1137 kilograms (515 pounds).
Low temperature operation	The assemblies in the enclosure are designed to withstand exterior ambient temperatures down to -10°C without auxiliary heating. However, if power is first applied to the enclosure at an ambient temperature near -10°C, a 1-hour warm up period is required before operating the equipment. This assumes a typical complement of equipment is installed in the enclosure (i.e., a minimum of 270 watts of dissipated power). If a power failure should occur at lower temperatures, it should be recognized that the battery capacity will be reduced. The TotalPlant Configurator is used to determine the diminished battery capacity. Diminished capacity is relative to the capacity that would exist at room temperature. For temperatures below -10°C, a heater is required. For “ordinary” environments, one of the optional heaters that are listed in Table 1-1 can be used. Use of a heater will reduce the required warm-up time at lower temperatures in the event of a system shutdown. For Class I, Div 2, Groups A,B,C, and D locations, heat will have to be provided on site by steam tracers or another approved means.

1.2.1 Overview, Continued

High temperature operation

It should be recognized that there may be a combination of conditions that will not allow the enclosure to be installed in an ambient environment of 60°C. For instance, if the combination of HPMM card sets, FTAs, IOPs, and the Power System is such that it generates over 270 watts of heat, the enclosure's internal temperature will rise above the permissible 70°C internal ambient temperature. A condition such as this is easily determined by the **TotalPlant** Configurator. In this case, the **TotalPlant** Configurator will probably recommend that some of the hardware be removed and installed in another enclosure, or that an auxiliary cooling unit (heat exchanger, vortex cooling, and so on) is required.

It is also not recommended to have backup battery where the average ambient temperature is significantly over 40°C. With an average ambient temperature of 60°C, the battery's service life is reduced to 20% of its value at 40°C, which is already reduced to 25% of its value at room temperature. When a backup battery that has a theoretical 60-month life at room temperature is exposed to an average ambient temperature of 60°C, its theoretical life is approximately 3 months.

CE Compliance

Though some of the enclosure components are CE Compliant, the enclosure as a whole is presently not CE Compliant.

1.2.2 Basic Enclosure Description

Introduction

The basic enclosure is a rectangular-shaped stainless steel enclosure with front and rear access doors that are hinged on the left side. The door is secured by three latches that are controlled by a lockable flush handle. The enclosure's construction complies with the National Electrical Manufacturers Association (NEMA) 4X classification. The NEMA 4X classification is similar to the International Electrotechnical Commission (IEC) classification IP56.

1.2.2 Basic Enclosure Description, Continued

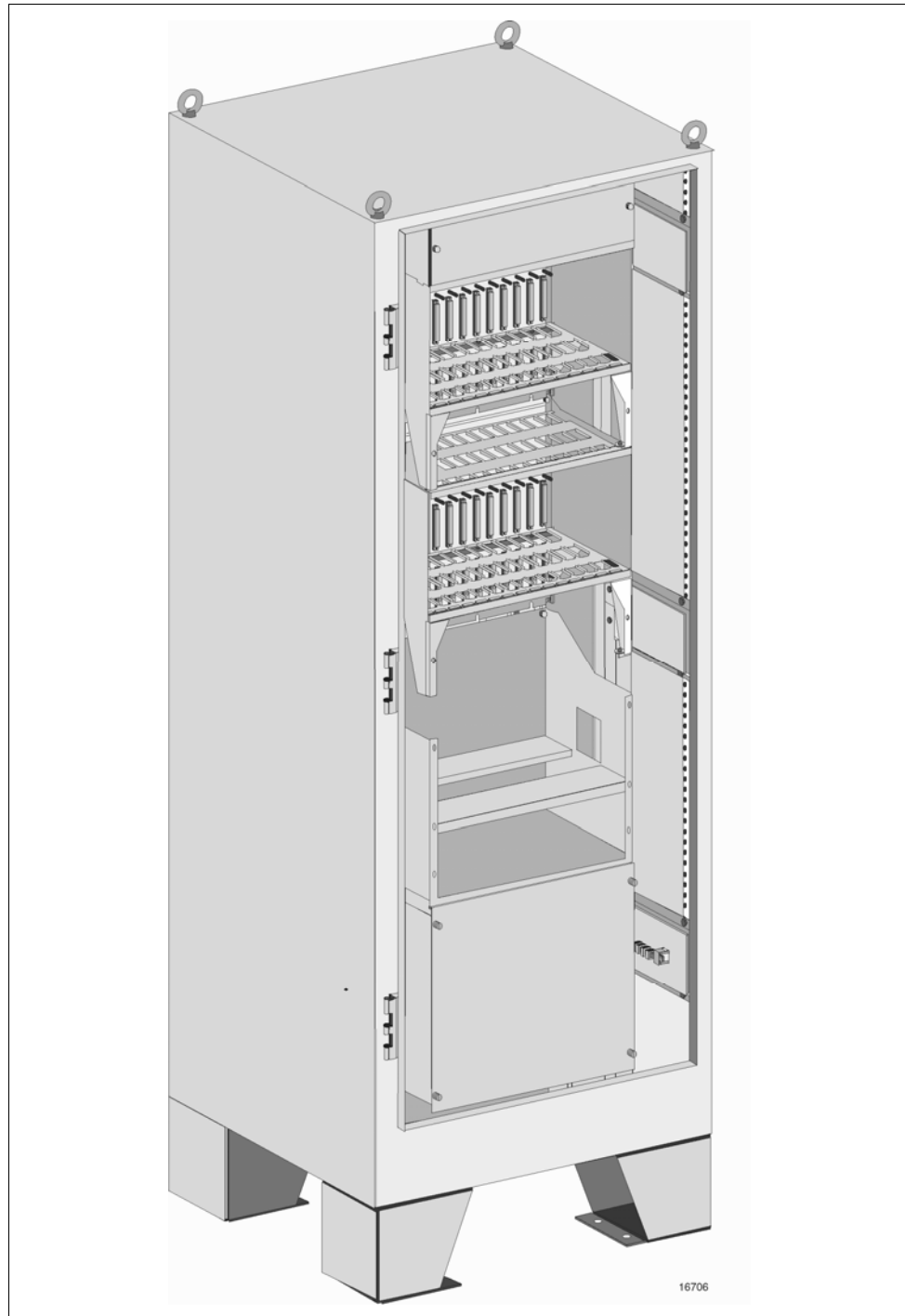
Indoor/outdoor use	The NEMA 4X enclosure is intended for indoor or outdoor use to provide protection against corrosion, wind-blown dust and rain, splashing water, hose-directed water, and external icing. Internal condensation and corrosion protection is provided for the electrical components inside the enclosure.
Door integrity	Each enclosure door has a flush key locking door handle that controls a three-point latch. A multifold gutter around the door restricts the entry of water. A special elastic foam gasket around the opening ensures the sealed integrity of the enclosure.
Door security	A key that is supplied by the manufacturer of the enclosure is required for entry into the enclosure.
Door grounding	A grounding strap that is attached to studs inside the doors and inside the enclosure body ensures proper common ground between the enclosure body and the doors.
Cable entry	The enclosure does not have a gland plate for cable entry. Cable entry from the side, top, or the bottom of the enclosure is determined by the user and access holes must be provided by the user on site. Top entry is not recommended because of the increased susceptibility to water entering the enclosure. The allowable cable entry areas are identified in Section 3, <i>Installation</i> , in this publication. The conduit fittings and conduit must maintain the NEMA 4X integrity of the enclosure.
Basic enclosure interior hardware	The interior of the basic enclosure (model MU-RHCAB1 or MU-RHCAB2) consists of vertical rails to mount the card files, FTA Mounting Channels, a Universal Control Network Extender, and a Power System, upper and lower fan assemblies that sense fan rotation failure, and an air plenum. 7-Slot or 15-Slot card files, a Power System, an optional electric heater, and an optional UCNE are mounted at the front of the enclosure, and FTA Mounting Channels are mounted at the rear of the enclosure. A center horizontal rail at each side of the enclosure has tapped holes that will accept self-threading screws that attach the brackets that are included in a UCN drop tap kit.

1.2.2 Basic Enclosure Description, Continued

Front interior view of RHC enclosure

Figure 1-1 is a front interior view of a Remote Hardened Controller enclosure that contains two card files and a tray for the 20-ampere Power System.

Figure 1-1 Front Interior View of RHC Enclosure

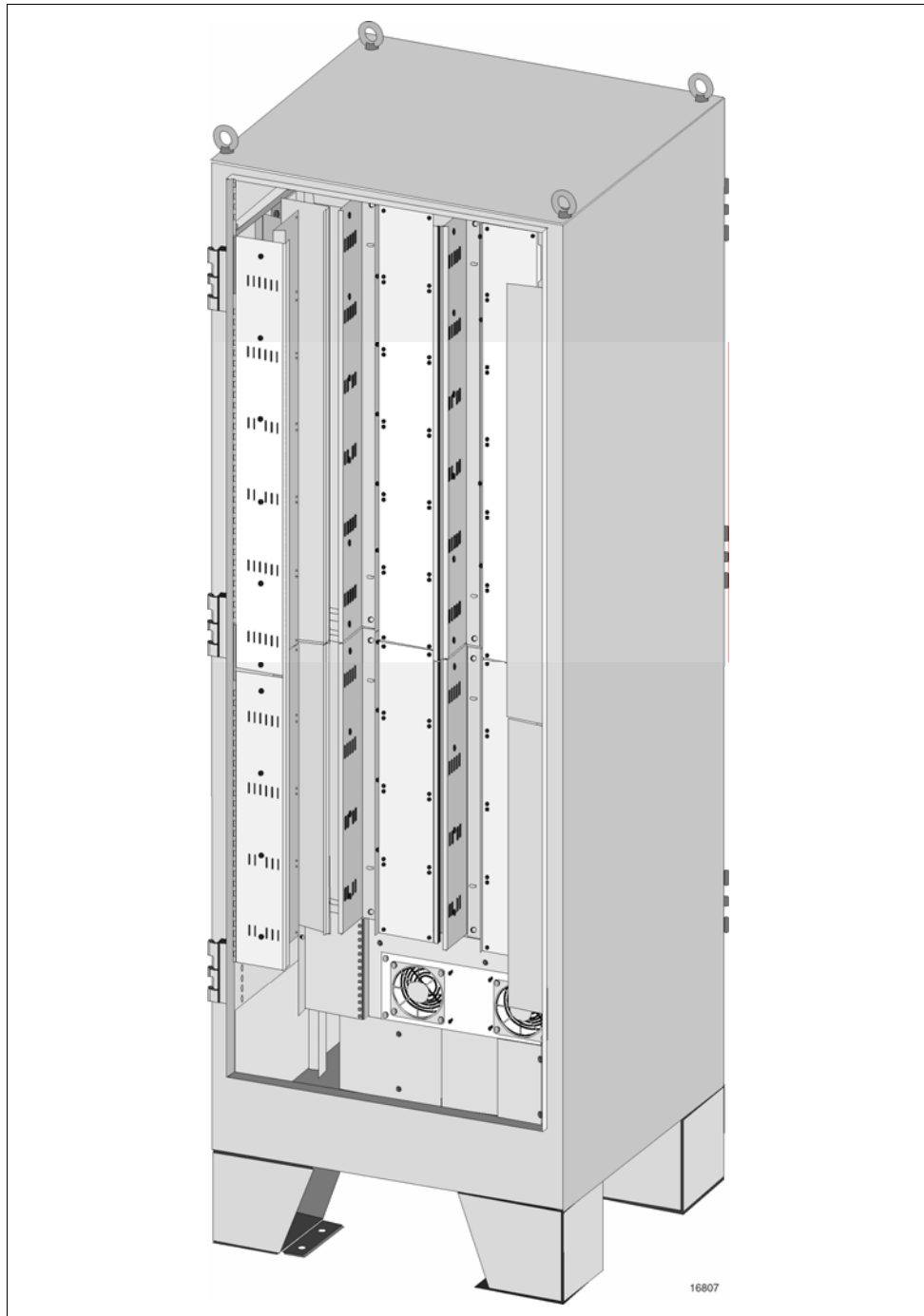


1.2.2 Basic Enclosure Description, Continued

Rear interior view of RHC enclosure

Figure 1-2 is a rear interior view of a Remote Hardened Controller enclosure that contains FTA Mounting Channels.

Figure 1-2 Rear Interior View of RHC Enclosure



1.2.3 Basic Enclosure Option Descriptions

Introduction

A hardware configuration that is based on the following assemblies and kits determines the basic enclosure's functionality.

The assemblies are selected by model number and entered into the **TotalPlant** Configurator. Except for IOPs and FTAs, the available assemblies and kits are listed in Table 1-1. Detailed physical and functional descriptions of the assemblies can be found in the UCN and HPM publications that are referenced in this manual.

FTA Mounting Channels

The rear of the enclosure will accept a pair of FTA Mounting Channels at each side of the enclosure and two pairs of FTA Mounting Channels at the center of the enclosure for a total of eight FTA Mounting Channels. A pair of FTA Mounting Channels is defined as an FTA Mounting Channel that is approximately 36 inches long and an FTA Mounting Channel that is approximately 24 inches long.

A full complement of eight FTA Mounting Channels will accommodate 20 "B" size FTAs or 40 "A" size FTAs.

Alternately, three pairs of FTA Mounting Channels can be mounted across the rear of the enclosure.

Standard Power System

The standard Power System (model MC-PSSX04 or MC-PSRX04) that mounts in the enclosure is identical to the Power System that mounts in a standard HPM cabinet. The Power System provides 20 amperes of 24 Vdc power for HPMM cards, IOPs, and FTAs, 3.6 Vdc for HPMM and IOP memory circuits, and 6 Vac for operation of LLAI IOP line frequency noise cancellation clock circuitry. Three rechargeable C-size NiCad batteries provide backup for the HPMM and IOP CMOS memory circuits.

Two models are available. One model has a single Power Supply Module, and the other model has redundant Power Supply Modules. Both models accept 120 Vac or 240 Vac input power.

Backup battery pack option

A backup battery pack assembly (MU-RBMK01) supports the standard Power System when there is an ac power failure. Depending upon the ambient external temperature, the 48 volt battery pack provides backup for the Power System's 24 Vdc for up to 20 minutes at full load.

The backup battery pack enclosure mounts beneath the front of the enclosure between the mounting legs.

1.2.3 Basic Enclosure Option Descriptions, Continued

AC Only Power System

Depending upon the model that is selected, the AC Only Power System has 8- or 16-ampere nonredundant or redundant Power Supply Modules, but it cannot be supported by the standard Power System's optional 48 Vdc backup battery pack. The AC Only Power System has nonrechargeable alkaline battery CMOS memory backup instead of rechargeable NiCad CMOS memory backup.

Eight models of the AC Only Power System are available. Four models require 120 Vac input power, and the other four models require 240 Vac input power.

Card files

Six types of card files are available for use in the enclosure with their model numbers listed in Table 1-1. Both Non-CE Compliant and CE Compliant models are available. The six types of card files are:

- Left 7-Slot HPMM card file (Slots 1-7)
 - Left 7-Slot IOP card file (Slots 1-7)
 - Right 7-Slot HPMM card file (Slots 9-15)
 - Right 7-Slot IOP card file (Slots 9-15)
 - 15-Slot HPMM card file (Slots 1-15)
 - 15-Slot IOP card file (Slots 1-15)
-

The card files are the same card files that are used in an HPM cabinet. Any combination of card files can be installed in the enclosure, except a Right 7-Slot card file cannot be used unless a Left 7-Slot card file is present. Left and Right 7-Slot card files are generally mounted side by side and populated with redundant HPMM card sets. Two 15-Slot card files, one above the other, can also be populated with redundant HPMM card sets. This configuration is acceptable only when the outside ambient temperature is 40°C or less. The card files are mounted above the Power System.

Card file slots that do not contain HPMM card sets, accept IOPs with their associated FTAs mounted on FTA Mounting Channels at the rear of the enclosure. Power to the FTA is provided by the associated IOP through its IOP to FTA cable.

1.2.3 Basic Enclosure Option Descriptions, Continued

IOPs	The same complement of IOPs that are used in an HPM cabinet can be installed in an RHC enclosure. Lists and descriptions of the IOPs with their model numbers can be found in the <i>Process Manager I/O Specification and Technical Data</i> and <i>Process Manager I/O Installation</i> manuals.
FTAs	The same complement of FTAs that are used in an HPM cabinet can be installed in an RHC enclosure. Lists and descriptions of the FTAs with their model numbers can be found in the <i>Process Manager I/O Specification and Technical Data</i> and <i>Process Manager I/O Installation</i> manuals.
I/O Link Extender	When RHC enclosures are complexed and the cable-distance between the enclosures exceeds the maximum allowable length of an I/O Link Interface cable pair, or the HPMM(s) is not located at the remote site, an I/O Link Extender fiber optic link can be used to “extend” the I/O Link Interface. Two I/O Link Extender models are available. They “extend” the link up to either 1.3 or 8 kilometers. The hardware consists of a pair of I/O Link Extender cards and Fiber Optic Couplers that occupy two card file slots at each end of the fiber optic link.
UCN Extender	A Universal Control Network Extender (UCNE) “extends” the UCN through a fiber optic link, rather than through coaxial cable. When the Network Interface Module (NIM) and other UCN nodes are centrally located and an HPM is installed in an RHC enclosure at a remote location at a distance that does not exceed 2 kilometers, a UCNE can provide a link to the centrally located UCN trunk segment. A UCNE mounting kit (model MU-FOMK21) is required to mount the two UCNEs (model MU-FOED02) that interface the fiber optic cables and the RHC enclosure’s internal UCN coaxial cables.

Section 2 – Planning

2.1 Overview

Introduction

The Remote Hardened Controller (RHC) enclosure has four floor stands that can be attached to a concrete pad or some other solid surface to ensure that no damage to the enclosure is incurred from high winds.

Power and signal cable entry into the enclosure can be through the top, side, and bottom surfaces of the enclosure. Top entry is not recommended because of the increased susceptibility to water entry into the enclosure. Cable entry holes must be cut into the enclosure by the user after determining the location and number of holes. Honeywell provides a diagram of the areas where the entry holes can be cut, as illustrated in Figure 3-2. The conduit fittings and conduit that is attached must maintain the NEMA 4X integrity of the enclosure.

Conduits that offer protection and are large enough to except power, FTA field wiring, coaxial, fiber optic, and I/O Link Interface cables with their connectors must be provided.

A source of adequate power and the correct voltage, optionally redundant, for the enclosure's Power System must be provided.

2.2 Guidelines

Transport the enclosure

The enclosure must be transported in an enclosed shock-mounted air van. Precautions must be taken not to exceed vibrational sinusoidal loads of 0.5 Gs. It must be transported in an upright position.

Weather protection

When installing the enclosure outside where it is exposed to natural elements, a sun shield or rain canopy is mandatory.

Clearance space

A space clearance of at least 91.5 centimeters (36.0 inches) must be maintained around the enclosure to ensure proper heat dissipation. See subsection 2.9, *Enclosure Complexing*, if it is necessary to place the enclosures closer together. Do not place any items on top of the enclosure, or obstruct free air flow around the enclosure in any way.

2.2 Guidelines, Continued

Field servicing	Field servicing of the enclosure during rain and/or high humidity must be avoided. If unavoidable, precautions must be taken not to wet the interior of the enclosure. When servicing is complete, the interior walls of the enclosure must be wiped with a dry cloth before closing and latching the door.
------------------------	---

2.3 AC Power Requirements

2.3.1 Overview

Scope	This section will guide the user through the process of planning the ac power requirements for a TotalPlant Solution (TPS) system installation that may have Local Control Network (LCN), Data Hiway (DH), Process Manager (PM), Advanced Process Manager (APM), High-Performance Process Manager (HPM), and Logic Manager (LM) equipment.
--------------	---

Introduction	The Remote Hardened Controller must be provided with ac power that is within certain limits, and the system electrical equipment must be properly grounded. The Remote Hardened Controller's performance and safety is directly related to the quality of the power and grounding system that is provided by the user. Honeywell provides the portion of the electrical power and grounding equipment that is contained within the RHC enclosure.
---------------------	---

2.3.1 Overview, Continued

Analyze before planning

When planning the power system, carefully analyze the present requirements, together with any expansions that are planned for the future and anticipate unplanned expansion. It is always easier and less expensive to initially install power systems to serve any future needs, than to retrofit cabling trays, new distribution panels, etc., in the future. Technical assistance is available from your Honeywell Project Engineer or Project Manager.

The electrical power requirements for each system component are listed in the *Local Control Network Planning*, *Process Manager/Advanced Process Manager Planning*, *High-Performance Process Manager Planning*, *Logic Manager Planning*, and *Data Hiway Planning* manuals. To determine the total KVA and Btu/hr, use the installation specifications in the manuals.

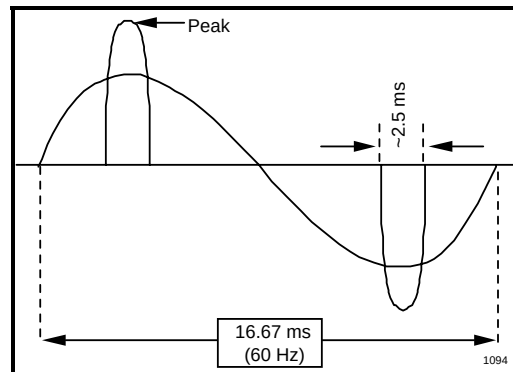
2.3.1 Overview, Continued

Background

Note that the electrical power (amperage) requirements that are specified in the *Local Control Network Planning*, *Process Manager/Advanced Process Manager Planning*, *High-Performance Process Manager Planning*, *Logic Manager Planning*, and *Data Hiway Planning* manuals for each of the TPS system components are given in terms of RMS, peak, and watts. It is important to understand how the values were measured and the measuring device that was used.

Most ac-type volt/ampmeters use a rectification and averaging technique to measure current, but are scaled to indicate RMS, based on an assumed sinusoidal input.

For the TPS system, the majority of the loads are of the rectifier/capacitor-input type, where current flows on the peaks of the line voltage, as illustrated.



Higher peak currents

The current drawn by the input is far from sinusoidal and contains large amounts of odd harmonics. As a result, the peak currents are much higher than those produced by a linear (resistive or inductive) load.

This relationship between the peak value and the RMS, or effective value of the current, is specified as the "crest" factor, which is 1.41 for a resistive load (sinusoidal current).

Large errors

Using an average reading instrument that is scaled for sinusoidal RMS values to measure this type of load can result in errors of 50% or more; that is, the reading is half of the RMS effective value of the current.

The RMS (effective) value of the current determines the heat generated as a result of resistive losses in the TPS system. The peak current, however, determines the voltage losses in the power source and the distribution system; therefore, when the current waveform is distorted and the source impedance is not low enough, there is an appreciable amount of distortion. The fuse or circuit breaker ratings, however, should be based on RMS values (taking into account possible inrush current).

2.3.2 AC Power System Design

Overview

Strive to design the power distribution system for no more than 5% power loss.

- Transformer wire losses $\leq 3\%$
- Distribution system $\leq 2\%$

Guidelines for power system components

Breaker sizing and wire sizing should be done according to the lowest acceptable voltage (use maximum current listed in specification table) because the constant current outputs will draw more input current when the line voltage drops to the lowest acceptable voltage.

Table 2-1 lists the ac power system guidelines.

2.3.2 AC Power System Design, Continued

Guidelines for power system components, continued

Table 2-1 AC Power System Guidelines

Circuit Breakers:	<u>Inrush current</u>—Breakers are selected on the basis of RMS current. • LCN systems have soft start circuitry to limit turn-on current to the rated, full-load, running current. • PM, APM, and HPM systems do have soft start circuitry, but the inrush current is 35 amps. • Data Hiway systems have CVT input transformers that have a one cycle current inrush of up to 30 times the running current. <u>Guideline</u>—Startup philosophy will determine the selection of delay curve type of breaker. After a total power failure and power returns, all power supplies will start simultaneously and can cause a significant breaker overload. • Breakers with only CVT power supplies should have the 25x delay curve. • Breakers with 50% CVT loads can have medium (6x) delay curves. • Breakers for the PM, APM, and HPM should have medium (6x) delay curves. <u>Sizing</u>—Use the typical RMS currents from the planning manuals for sizing breakers. The typical RMS values do not include the increase in current caused by low voltages (102/204 Vac.) • Each cabinet must have its own dedicated breaker so that power can be removed from a single cabinet (or power strip in an equipment cabinet) separately. • Because each cabinet or power strip is limited to 15 amperes by the breaker in the cabinet, a larger breaker in the distribution box is not required. • The purpose of the breakers in the distribution panel is twofold: – Protect the power wiring from damage. – Allow for complete removal of power to the station or cabinet.
-------------------	---

2.3.2 AC Power System Design, Continued

Guidelines for power system components, continued

Table 2-1 AC Power System Guidelines, continued

Distribution Wiring:	Use size 12 AWG (2.5 mm²) wire from the Universal Station or cabinet to the distribution panel if: • the distance from the distribution panel to the Universal Station or cabinet is 75 feet (22 meters) or less • the total current in the cabinet is 10 amps or less If the distance is greater than 75 feet or if the current is larger than 10 amps, use Table 3-2, 3-3, or Table 3-4. For wire sizes larger than 12 AWG (2.5 mm²), special procedures must be followed to reduce the wire size to 12 AWG as it enters the Universal Station, LCN equipment cabinet, or PM/APM/HPM cabinet (the power terminal will not accept wire larger than 12 AWG or 2.5 mm²).
Transformer Sizing:	Transformers should be sized using typical RMS values. When a transformer is used to directly supply ac power to the TPS equipment (LCN, PM, APM, HPM, LM, or PC-based systems), an Underwriter's Laboratory (UL) rated transformer with a K factor of 9 or better is recommended with 5% taps to provide an additional 5% output voltage on a contingency basis. The resistance rating of the transformer should be 3% or better and the total impedance should be less than 5%. Low temperature-rise models or oversized units may be used to get the equivalent source resistance and to protect against additional temperature rise due to the harmonic eddy current content that is defined by the K = 9 rating. The 3% resistance is required to minimize the "flat-top" voltage effect. The K factor-rated transformers are made by many vendors and have a premium cost factor of from 10% (for transformers in the 15 KVA range) to 100% (for 500 KVA). K = 9 transformers will have longer life. Transformers for use in systems that have only Data Hiway equipment can be normal-rated transformers. Systems having a mix of Data Hiway and LCN, PM, APM, HPM, LM, and personal computer equipment should have transformers with a K factor = 9.

2.3.2 AC Power System Design, Continued

Guidelines for power system components, continued

Table 2-1 AC Power System Guidelines, continued

UPS Systems:	Note: The following procedure is for sizing Uninterruptible Power Systems. 1. Add up the individual typical RMS currents for the TDC 3000^X equipment chosen. 2. Multiply the total currents by 120 or 240 volts. 3. Check that the UPS equipment can handle a crest factor of at least 3.0. The steps do not account for any overhead from other equipment or for possible expansions. The values are typical and do not include any margins.
--------------	---

Wire size table

Select the wire size from Table 2-2 according to the load current, wire length, and percentage of voltage drop that is allowed for 120 Vac. The table determines the wire size based on the RMS current.

Table 2-2 Copper Wire Size Based on RMS Current Requirement for 120 Vac

Amp s	1% Voltage Drop						2% Voltage Drop						KVA
	Cable Feet						Cable Feet						
	25	50	75	100	200	500	25	50	75	100	200	500	
5	12	12	12	10	8	4	12	12	12	12	10	6	0.6
10	12	10	8	8	4	1	12	12	12	10	8	4	1.2
15	12	8	6	6	2	2/0	12	12	10	8	6	2	1.8
20	10	8	6	4	2	3/0	12	10	8	8	4	1	2.4
25	10	6	4	4	1	4/0	12	10	8	6	4	0	3.0
30	8	6	4	2	0	–	10	8	6	6	6	2/0	3.6
40	8	4	2	2	2/0	–	8	8	6	4	2	3/0	4.8
50	6	4	2	1	3/0	–	8	6	4	4	1	4/0	6.0
60	6	2	1	0	4/0	–	6	6	4	2	0	–	7.2
80	4	2	0	2/0	–	–	4	4	2	2	2/0	–	9.6
100	2	1	2/0	3/0	–	–	2	2	2	1	3/0	–	12
125	1	0	3/0	4/0	–	–	1	1	1	0	4/0	–	15

2.3.2 AC Power System Design, Continued

Wire size table for 240 Vac

Select the wire size from Table 2-3 according to load current and length in meters for a 2% maximum voltage drop for 240 Vac.

Table 2-3 Maximum Cable Length in Meters for 240 Vac

Cable Size (mm ²)	Current (Amperes)				
	5	10	15	20	30
1,5	38	19	—	—	—
2,5	63	32	21	—	—
4	100	50	33.3	25	—
6	150	75	50	37.5	25
10	252	126	84	63	42
16	400	200	133	100	66.6

Wire size table for 120 Vac

Select the wire size from Table 2-4 according to load current and length in meters for a 2% maximum voltage drop for 120 Vac.

For a 1% maximum voltage drop, allow half of the cable length in Table 2-4.

Table 2-4 Maximum Cable Length in Meters for 120 Vac

Cable Size (mm ²)	Current (Amperes)				
	5	10	15	20	30
1,5	19	9.5	—	—	—
2,5	31.5	16	10.5	—	—
4	50	25	16.6	12.5	—
6	75	37.5	25	18.7	12.5
10	126	63	42	32	21
16	200	100	66	50	33.3

2.3.3 AC Power Distribution Guidelines

Conduit

Metal conduits are highly recommended for all power cable runs. The conduit must be grounded at the power distribution box and at the system equipment power entry box. The case of the equipment power entry box and branch-circuit distribution panel must be grounded (bonded) to building steel.

The IEC allows cable trays for power cables when it is desirable.

ATTENTION

- To avoid electrical noise problems in the system, observe the power cable(s) to signal cable(s) separation rules that are discussed in the *Data Hiway Planning*, *Process Manager/Advanced Process Manager Planning*, and *High-Performance Process Manager Planning* manuals.

- **All ac power wiring must be three-conductor with a fire-resistant jacket for the following connections.**
 - Hot
 - Neutral
 - Safety Ground

Distribution box sizing

The size or capacity of the system power distribution box or dedicated branch circuit(s) depends on the size of the system. The panel must contain a Safety Ground (equipment) bus bar to which the ground (green wire or yellow green) from the system cabinets, consoles, and device receptacles can be connected. The Safety Ground bus bar must then be connected to the ac Safety Ground rod through an insulated cable specified by local code requirements. Refer to the subsection 2.4 in this publication.

Power entry boxes

Each TPS product has its own power entry scheme. Refer to the **TotalPlant** *System Planning* manual.

2.3.3 AC Power Distribution Guidelines, Continued

PM, APM, and HPM cabinet power distribution

Process Manager (PM), Advanced Process Manager (APM), and High-Performance Process Manager (HPM) equipment cabinets do not have ac breaker boxes. The ac power connects directly to the system power distribution panel.

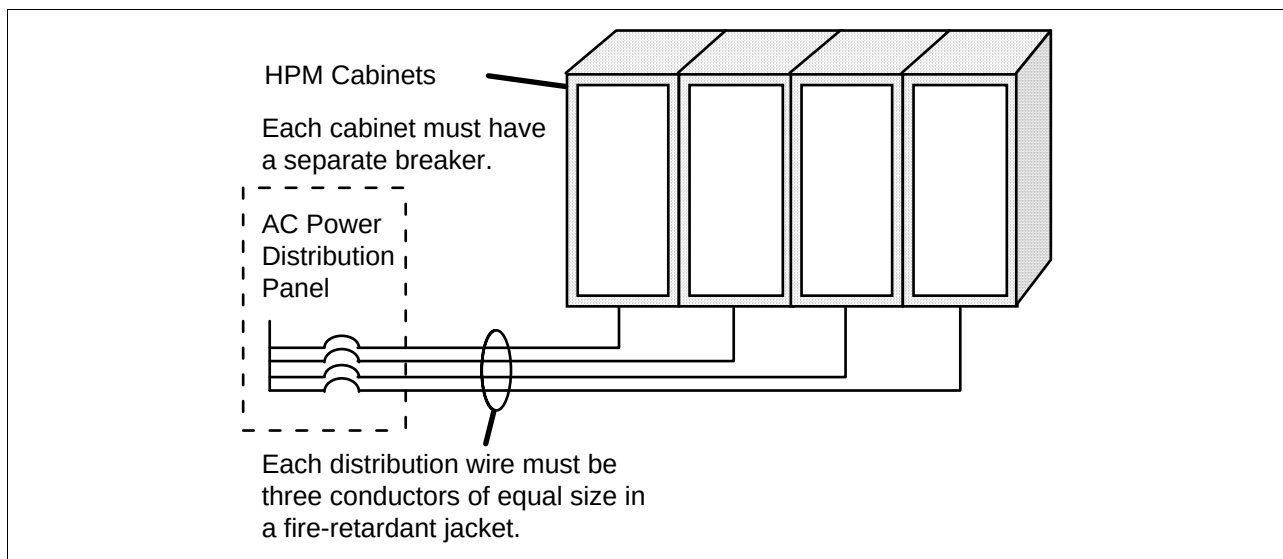
PM, APM, and HPM cabinet power distribution

Process Manager (PM), Advanced Process Manager (APM), and High-Performance Process Manager (HPM) equipment cabinets do not have ac breaker boxes. The ac power connects directly to the system power distribution panel.

Single source power distribution

Figure 2-1 illustrates typical power distribution to an HPM cabinet complex from a single power source.

Figure 2-1 HPM AC Power Distribution—Single Power Source



2.3.3 AC Power Distribution Guidelines, Continued

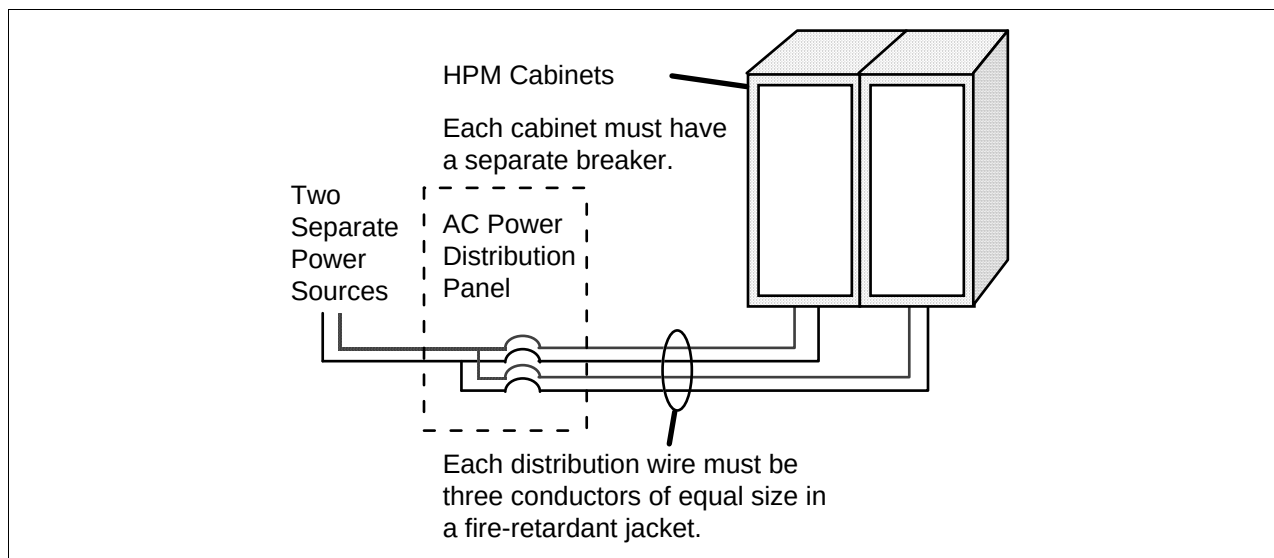
Redundant power sources

PM, APM, and HPM equipment cabinets can be connected to two power sources. Each source must be separately wired. This provides automatic power backup. **An individual 15 A breaker is required for each power source.**

Dual source power distribution

Figure 2-2 illustrates typical power distribution to an HPM cabinet complex from two power sources.

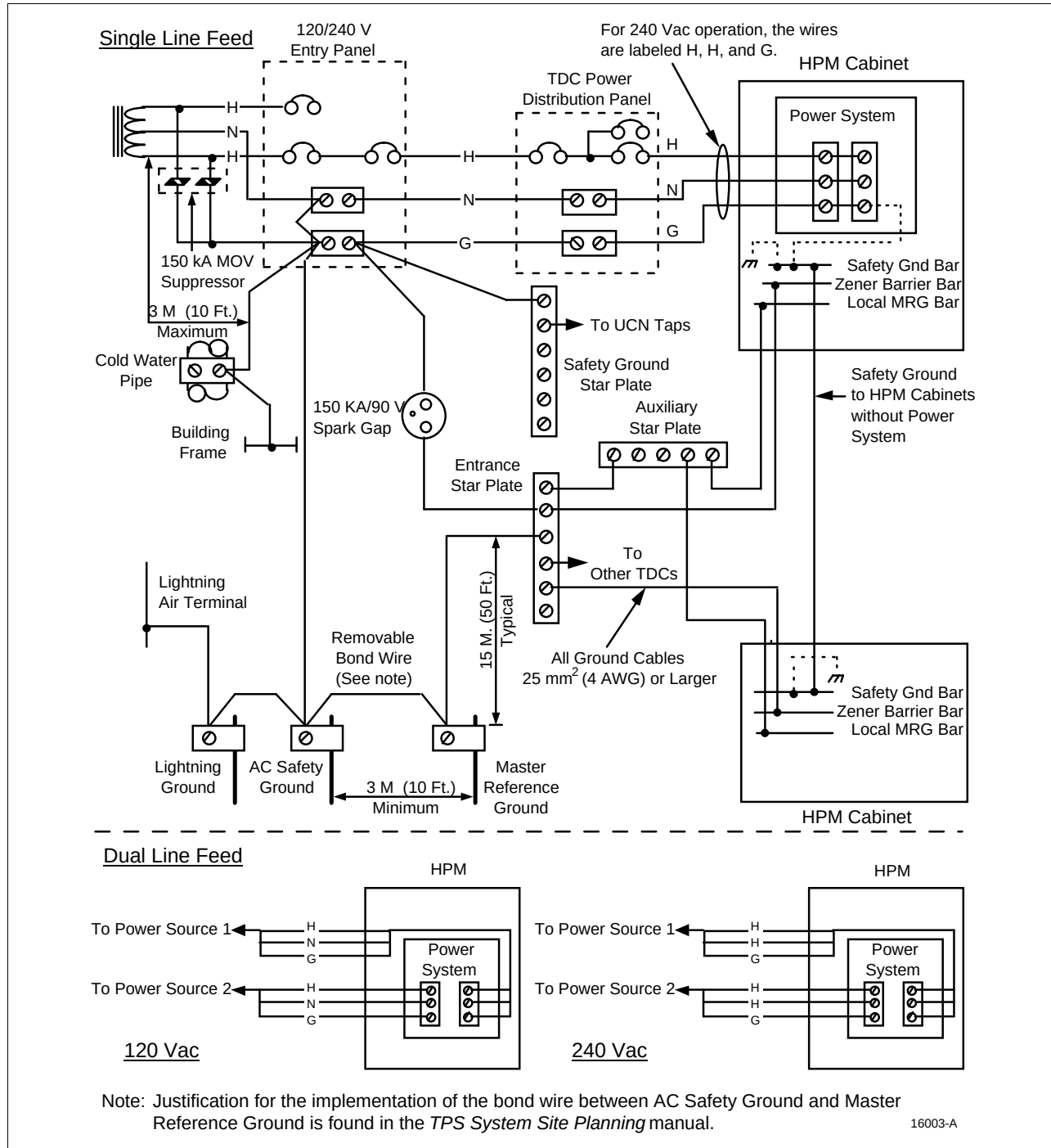
Figure 2-2 HPM AC Power Distribution—Separate Power Sources



2.3.3 AC Power Distribution; Guidelines, Continued

AC power and ground Figure 2-3 illustrates the ac power and ground connections for a typical multi-ground HPM installation that includes Master Reference Ground (MRG). The ground system is non-CE Compliant.

Figure 2-3 Subsystem AC Power and Ground Connections—Multi-Ground System



2.4 Ground Requirements

2.4.1 System Grounds

Required grounds

Grounding of electronics can make the difference between a system that is unusable and one that performs well. The required grounds, their connections, and design are discussed below.

TPS systems require the use three ground systems.

- Safety Ground
- Lightning Ground
- Master Reference Ground

AC Safety Ground

AC Safety Ground interconnects all metallic structures and enclosures in an area. This minimizes the voltage that people can come into contact with. AC Safety Ground is used to ground all power systems for lightning. It is also used to ground the UCN coaxial cable shields at every tap.

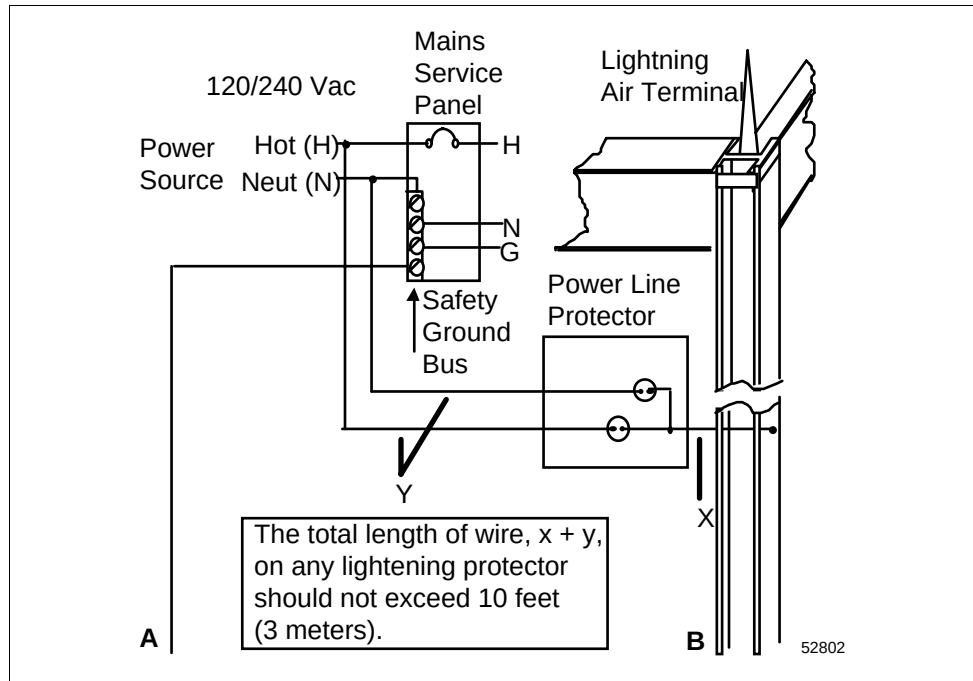
For each building power entry, two independent ac Safety Ground systems with two electrode cables are preferred. Examples of such independent ground systems would be a driven electrode and a metal water pipe as shown in Figures 2-4 and 2-5.

2.4.1 System Grounds, Continued

TPS system ground system – upper view

Figure 2-4 illustrates an upper view of a TPS system ground system.

Figure 2-4 TPS System Ground System—Upper View

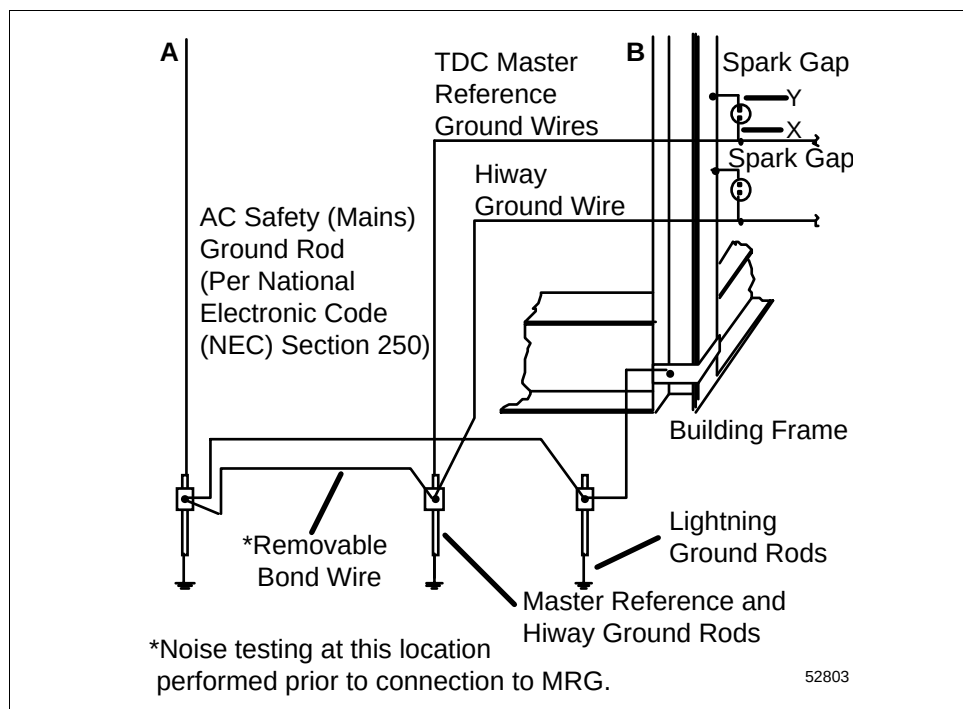


2.4.1 System Grounds, Continued

TPS system ground system – lower view

Figure 2-5 illustrates a lower view of a TPS system ground system.

Figure 2-5 TPS System Ground System—Lower View



Electrode system requirements

Safety grounding requires electrode systems that provide grounding resistances between 0.1 and 5.0 ohms. The required performance varies with the circumstances described as follows:

- When a facility has neither zener barriers nor lightning problems, ac Safety Ground has only to meet the minimum grounding requirements specified in the local electrical code. Usually, such codes require a grounding resistance of 5 ohms maximum to true earth.
- When a facility uses zener barriers for intrinsic safety, ac Safety Ground should be less than 0.1 ohms and Master Reference Ground less than 1.0 ohms to true earth. Refer to the Lightning Ground discussion that follows.

2.4.1 System Grounds, Continued

Lightning Ground

The purpose of lightning ground is to safely disperse a lightning charge that may be picked up by the facility metallic structure and electrical system. Lightning ground is frequently integrated with ac Safety Ground as shown in Figure 2-4 and Figure 2-5.

Lightning grounds must conform to applicable codes such as NFPA No. 78 (ANSI), IEEE Std 142-1972, Code LP1-175, and other local codes. When a facility is subject to lightning, each lightning rod should be grounded at the base with a maximum resistance of 0.1 ohms to earth.

The TPS system lightning protection philosophy is that each lightning rod should be grounded with a resistance to earth of less than 0.1 ohms. A 100 kA lightning strike into a 0.1 ohm ground generates 10 kV. Experience has shown that 10 kV does not cause sparking inside of wiring trays or across terminal blocks. Without sparking and the Electromagnetic Interference (EMI) it generates, a TPS system performs well during a lightning strike.

2.4.1 System Grounds, Continued

Master Reference Ground properties

Master Reference Ground (MRG) is an isolated ground electrode array that references Honeywell TPS system electronics and the process signals as illustrated in Figure 2-5.

An isolated MRG assures stable process signals to the Process Manager, Advanced Process Manager, or High-Performance Process Manager by divorcing signal ground from ac Safety Ground. An MRG installation can have the following properties:

- An MRG is an isolated ground electrode system reserved for Honeywell TPS systems.
 - An MRG usually has a maximum resistance through the earth to ac Safety Ground of 5 ohms.
 - When the MRG is grounding zener barriers, it must have a maximum resistance through the earth to ac Safety Ground of less than 1.0 ohm.
 - Use a minimum separation of 3 m (10 feet) between the MRG electrodes and any grounded metal such as pipes or a grounding grid or mat.
 - MRG electrodes usually perform well when placed immediately next to building footings with electrically floating reinforcing bar.
 - TPS system electronics are to be no further than 152.4 cable-meters (500 feet) from its MRG electrode.
 - An MRG electrode conductor is typically 11.8 mm (4/0 AWG), is 13 m (50 feet) long, and terminates at a star plate upon entry into a building.
 - An MRG entry may require a spark gap to ac Safety Ground to control arcing during a lightning strike. See the previous discussion on lightning in this subsection for more information.
 - Large TPS system can use MRG star plates that feed other MRG star plates.
 - An MRG star plate is insulated from the building structure and concrete, and is capable of accepting up to three 5.3 mm (4 AWG) local Master Reference Ground conductors from each TPS system cabinet or complex in the TPS system.
 - Radial wiring to a star plate is preferred to daisy-chaining from cabinet-to-cabinet.
-

2.4.2 Star Plate

Description

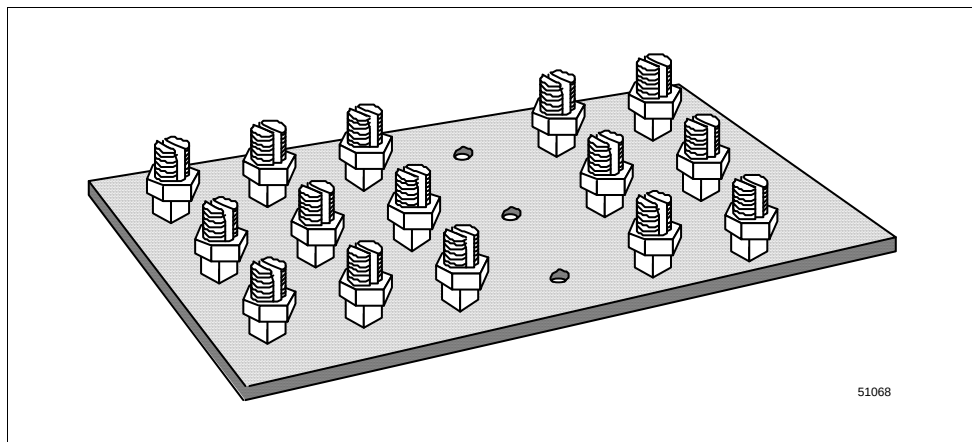
A star plate is a copper or brass plate that provides for the connection of ground wires at a common point. Following are the required attributes of a star plate.

- **Must be completely physically isolated from all conductive material except those ground wires that are to be connected.**
- Must be heavy enough (thick enough) to resist damage from being stepped on or a heavy blow. A minimum of 6.35 mm (1/4 inch) thick.
- Have more fastening studs (or stud holes) on the plate than the ground system wiring requires. **Star plates cannot be replaced with any of the attached system hardware in operation.**
- Each fastening lug must be capable of retaining a single 5.2 mm (#4/0 gauge) insulated ground wire. **One, and only one, wire may be terminated in each lug.**
- Must be located where it can be readily inspected.
- Must be kept dry.

Star plate example

Figure 2-6 is an example of a star plate.

Figure 2-6 Example of a Star Plate



2.4.2 Star Plate, Continued

Application rules

Star plates can be used when:

- the total distance from the stake to the farthest end point to be grounded is less than 152.4 meters (500 feet). Refer to the MRG grounding requirements shown in Figure 2-5 and described in subsection 2.4.1.
- each lug has one, and only one, #4/0 copper wire terminated in it.
- **only ground wires from Honeywell TPS system equipment are to be connected to the star plate.**

Star plates for Safety Ground systems, MRG systems, and zener barrier grounds systems must be kept separate until they reach their respective ground electrode.

2.4.3 PM/APM/HPM Grounding

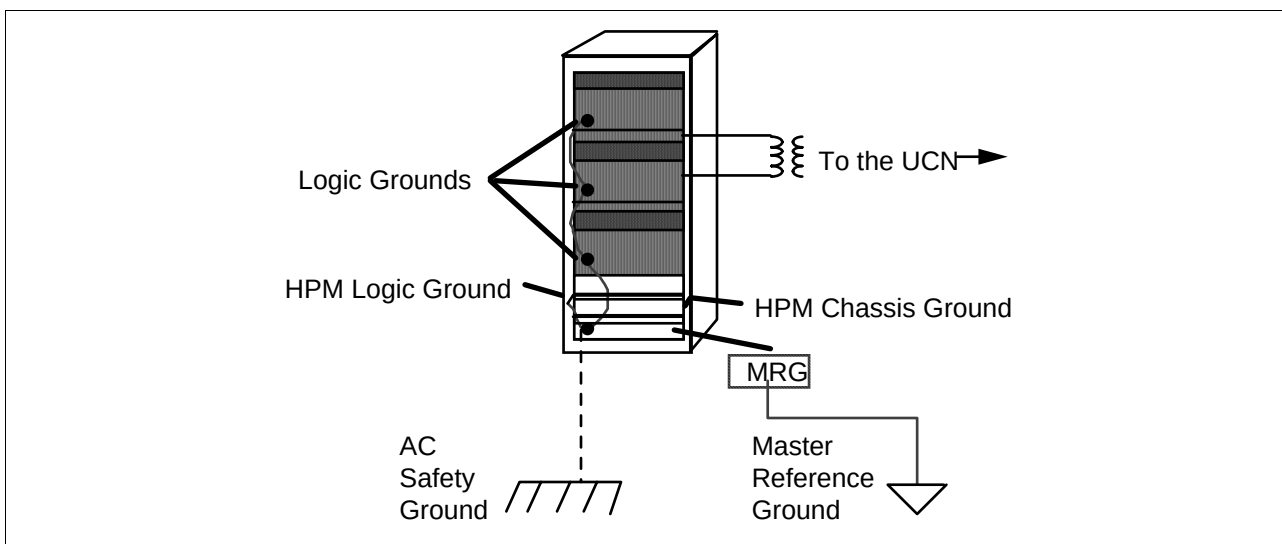
Introduction

The PM, APM, and HPM are grounded as illustrated in Figures 2-7 and 2-8. Each component of the TPS system is individually grounded and communication between each component and the Local Control Network (LCN) is transformer coupled.

HPM cabinet grounding

Figure 2-7 illustrates HPM cabinet grounding.

Figure 2-7 HPM Cabinet Grounding

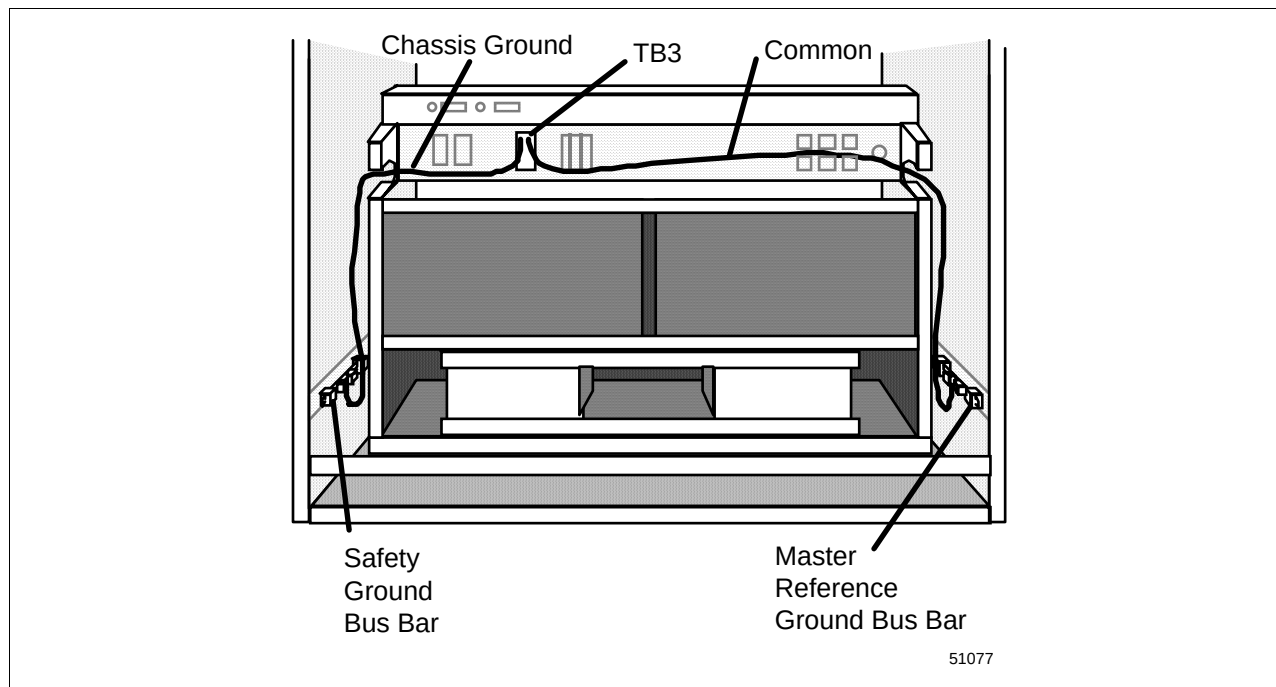


2.4.3 PM/APM/HPM Grounding, Continued

HPM cabinet ground wiring

Figure 2-8 illustrates HPM cabinet ground wiring.

Figure 2-8 HPM Cabinet Ground Wiring (Factory Provided)



2.4.3 PM/APM/HPM Grounding, Continued

PM/APM/HPM component grounding

PM/APM/HPM FTA signals that are not grounded at the process end must have the cable shields for the process-connected wiring connected to ground as follows:

- Analog Input FTAs and Analog Output FTAs must be connected to MRG.
- 3-30 Vdc Digital Input FTAs must be connected to MRG.
- 30-200 Vdc Digital Input FTAs that have safety grounds or shields must be connected to Safety Ground.
- Ac Digital Output FTAs that have safety grounds or shields must be connected to Safety Ground.

The PM/APM/HPM cabinets have a separate bus bar for Safety Ground, MRG, and an optional zener barrier ground bus bar that is located near the bottom of the cabinet. Each bus bar is connected to the appropriate ground system star plate (one for Safety Ground, one for MRG, and one for zener barrier ground). Refer to the *Process Manager/Advanced Process Manager Planning*, *High-Performance Process Manager Installation*, or *Process Manager I/O Installation* manual for additional PM, APM, or HPM grounding information.

2.5 Hardware Selection

Introduction

Additional hardware must be ordered or procured for the basic Remote Hardened Controller enclosure(s). Depending upon the application, the following assemblies and cables must be considered.

- Power System
- Power System Backup Battery
- Card file(s)
- HPMM card set(s)
- IOPs
- FTAs
- FTA Mounting Channels
- I/O Link Extender (Standard or Long Distance)
- Universal Control Network Extender
- 24 Vdc Power cables
- IOP to FTA cables
- I/O Link Interface cables
- Fiber optic cables
- Enclosure electrical heater (Internal)

Fiber optic cables

All the assemblies that are listed can be provided by Honeywell except the fiber optic cables that support I/O Link Interface Extender and Universal Control Network Extender hardware. Fiber optic cables must be installed by a qualified contractor.

2.5.1 Power System Selection

Introduction

Two types of Power Systems, standard and AC Only, are available. The Power System mounts in the second position from the bottom of the enclosure. The available Power Systems are listed by model number in Table 1-1.

Standard Power System

There are two models of the standard Power System. They differ only in the number of Power Supply Modules that they have, single or dual. Dual Power Supply Modules provide power redundancy.

Both models of the Power System provide 20 amperes of 24 Vdc power and accept either 120 Vac or 240 Vac source power.

The standard Power System can be supported by a battery backup assembly that mounts beneath the front of the enclosure. Should power fail, the backup time is dependent upon the battery load and the ambient temperature.

AC Only Power System

There are eight models of the AC Only Power System. They differ in the number of Power Supply Modules that they have, single or dual, the input voltage requirement, 120 Vac or 240 Vac, and the 24 Vdc output current, 8 amperes or 16 amperes, that the Power System provides. Dual Power Supply Modules provide power redundancy.

The AC Only Power System cannot be supported by a battery backup assembly.

2.5.2 Card File Selection

Introduction

Three types of card files can be mounted in the Remote Hardened Controller enclosure. The available card files are listed by model number in Table 1-1.

- Left 7-Slot card file
- Right 7-Slot card file
- 15-Slot card file

The card slots in the Left 7-Slot card file are designated slots 1 through 7, and the card slots in the Right card file are designated slots 9 through 15. There is no slot 8 in either card file. Card slots in the 15-Slot card file are designated slots 1 through 15.

Two tiers of card files

The enclosure will accept two tiers of card files in the front upper portion of the enclosure. Left and Right card files are mounted side by side on horizontal rails.

Card file combinations

Any combination of card file types can be installed. However, a Right 7-Slot card file must not be installed in the enclosure unless the intent is to provide HPMM redundancy. Left 7-Slot and Right 7-Slot card files can provide HPMM redundancy when populated with HPMM card sets.

Two 15-Slot card files can also provide HPMM redundancy when populated with HPMM card sets. In some applications, the 15-Slot card file may be populated with 15 IOPs.

When environmental conditions exist such that the maximum average ambient temperature is anticipated to be above 40°C and the intent is to provide HPMM redundancy, two 7-Slot card files, Left and Right, must be used in the uppermost card file position.

2.5.3 HPMM Card Set Selection

Introduction

The HPMM card set can be installed in any of three types of card files. The set consists of two cards and a module, and are installed in the card file's two leftmost card slot positions. IOPs occupy the remaining card slot positions in the card file. The HPMM card set(s) are listed by model number in Table 1-1. When installing redundant HPMMs, a redundancy cable is provided with the card sets.

HPMM card types

The three types of cards that constitute an HPMM are:

- High-Performance Comm/Control card
- HPM/UCN Interface module
- High-Performance I/O Link card

The High-Performance Comm/Control card and HPM/UCN Interface module occupy the same leftmost card slot position. The module is installed in the connector below the High-Performance Comm/Control card. The High-Performance I/O Link card is installed in the adjacent card slot.

2.5.4 IOP Selection

Introduction

Input/Output Processor (IOP) card selection is dependent upon the user's process that is being controlled and/or monitored. IOPs can be installed in any card file card slot that is not occupied by an HPMM card set.

Available IOPs

Based on their functionality, there are essentially five types of IOPs.

- Analog Input
- Analog Output
- Digital Input
- Digital Output
- Special purpose

All IOPs are corrosion protected.

2.5.4 IOP Selection, Continued

Description and model numbers

Refer to the *Process Manager I/O Specification and Technical Data* or *Process Manager I/O Installation* manual for descriptions and model numbers of the available IOPs. The model number for a corrosion protected IOP is prefixed with “MC.”

2.5.5 FTA Selection

Introduction

Field Termination Assembly (FTA) selection is dependent upon the user’s process that is being controlled and/or monitored. FTAs are installed on FTA Mounting Channels at the rear of the Remote Hardened Controller enclosure.

Available FTAs

Based on their functionality, there are essentially five types of FTAs.

- Analog Input
- Analog Output
- Digital Input
- Digital Output
- Special purpose

All FTAs are available corrosion protected.

FTA sizes

There are three standard FTA sizes. Size refers to the length of the FTA. The width of all three standard FTA sizes is the same and matches the mounting surface of the FTA Mounting Channel. The three sizes and their approximate lengths are:

- “A” size – 6 inches
- “B” size – 12 inches
- “C” size – 18 inches

Standard FTA sizes exist so that the number of FTAs that can be mounted on a given number of FTA Mounting Channels can be calculated. The FTA Mounting Channels are 24 inches and 36 inches in length.

Description and model numbers

Refer to the *Process Manager I/O Specification and Technical Data* or *Process Manager I/O Installation* manual for descriptions and model numbers of the available FTAs. The model number for a corrosion protected FTA is prefixed with “MC.”

2.5.6 FTA Mounting Channel Selection

Introduction	FTA Mounting Channels are available in two lengths, approximately 24 inches and 36 inches. Only narrow width FTA Mounting Channels are available for the Remote Hardened Controller enclosure.
Model numbers	The two types of FTA Mounting Channels are listed by model number in Table 1-1.

2.5.7 I/O Link Extender Selection

Introduction	The I/O Link Extender is used to extend the HPM's I/O Link Interface through fiber optic cable. If a portion of an HPM's I/O is installed in an enclosure (cabinet) other than the enclosure that the HPM is resident in, and the distance between the enclosures exceeds the allowable segment length of an I/O Link Interface cable, the use of an I/O Link Extender can be considered. A short or long distance I/O Link Extender (1.3 or 8 kilometers) is available.
Model numbers	The short or long distance I/O Link Extender components are listed by model number in Table 1-1.

2.5.8 Universal Control Network Extender Selection

Introduction	The Universal Control Network Extender (UCNE) is used to extend the Universal Control Network through fiber optic cable that can be up to 2 kilometers in length. The UCNE increases the end-to-end distance of a single UCN without resorting to special, wide-diameter coaxial trunk cables and increases the UCN signal quality at the ends of the UCN that would otherwise be connected by long coaxial trunk cables. The UCNE provides ground isolation between UCN trunk cable segments.
Model numbers	The Universal Control Network Extender components are listed by model number in Table 1-1.

2.6 Cabling Selection

2.6.1 24 Vdc Power Cable Selection

Introduction	Power cables are required to provide 24 Vdc power from the Power System to the card file(s) and Power Distribution Assembly(ies).
Model numbers	Refer to the <i>Process Manager I/O Specification and Technical Data</i> or <i>Process Manager I/O Installation</i> manual for descriptions and model numbers of the available power cables.

2.6.2 IOP to FTA Cable Selection

Introduction	IOP to FTA cables are required between the IOP's card file connector and its associated FTA.
Model numbers	Refer to the <i>Process Manager I/O Specification and Technical Data</i> or <i>Process Manager I/O Installation</i> manual for descriptions and the model numbers of the available IOP to FTA cables.

2.6.3 I/O Link Interface Cable Selection

Introduction	When more than one card file exists in an enclosure, I/O Link Interface cables are required between the card files for communication between the HPMM(s) and IOPs. The cable's connectors daisy-link the card files. Cables with up to six connectors can be selected, but the maximum number required is four.
Model numbers	Refer to the <i>Process Manager I/O Specification and Technical Data</i> or <i>Process Manager I/O Installation</i> manual for the descriptions and the model numbers of the available I/O Link Interface cables.

2.6.4 I/O Link Extender Cable Selection

Introduction	Fiber optic cables are required for I/O Link Extenders. The cables should be installed by qualified contractors.
Fiber optic cable installation	Refer to the <i>High-Performance Process Manager Planning</i> and <i>Process Manager I/O Installation</i> manuals for I/O Link Extender fiber optic cable selection information.

2.6.5 UCN Cable Selection

Introduction	When single or redundant HPMMs are installed in an enclosure, the connection to the Universal Control Network must be provided by either coaxial cables or a Universal Control Network Extender fiber optic link. If the connection is through coaxial cable, a pair of RG11-type trunk cables, cable A and cable B, must be brought into the enclosure and connected to UCN drop taps. One UCN drop tap is required for each cable.
UCN drop tap kit	The model MU-NTAP02 and MU-NTAP04 kits include a pair of two-port or four-port UCN drop taps, enclosure mounting hardware, and instructions. Additional information can be found in the <i>High-Performance Process Manager Installation</i> manual.
Drop cables	RG6-type drop cables must be installed between the UCN drop tap and the HPM UCN Interface module in each HPMM card set. Various cable lengths are available, but 2-meter cables are recommended. Refer to the <i>High-Performance Process Manager Service</i> manual for cable selection information.

2.6.6 UCN Extender Cable Selection

Introduction

When single or redundant HPMMs are installed in an enclosure, the connection to the Universal Control Network must be provided by either coaxial cables or a Universal Control Network Extender fiber optic link.

If the connection is through a Universal Control Network Extender fiber optic link, a UCNE kit must be installed in the enclosure and fiber optic cables must be installed between the UCNE devices at each end of the link.

UCNE kits

A model MU-FOMK21 UCNE mounting kit must be installed in the enclosure and populated with a pair of 24 Vdc UCNEs which are included in the model MU-FOED02 UCNE kit. Additional information can be found in the *Universal Control Network Installation* manual.

UCN drop tap kits

The model MU-NTAP02 or MU-NTAP04 kit is also required and includes a pair of two-port or four-port UCN drop taps, enclosure mounting hardware, and instructions. The model MU-NTAP02 kit is required for a single HPMM, and the model MU-NTAP04 kit is required for redundant HPMMs. Additional information can be found in the *High-Performance Process Manager Installation* manual.

UCN drop cables

RG6-type drop cables must be installed between the UCN drop tap and the HPM UCN Interface module in each HPMM card set, and between the UCN drop tap and the UCNE. Various cable lengths are available, but 2-meter cables are recommended. Refer to the *High-Performance Process Manager Service* manual for cable selection information.

Fiber optic cables

Fiber optic cables are required for Universal Control Network Extenders. The cables should be installed by qualified contractors.

Fiber optic cable installation

Refer to the *Universal Control Network Planning* and *Universal Control Network Installation* manuals for Universal Control Network Extender fiber optic cable selection information.

2.7 Conduit Selection

Introduction

In order to maintain the NEMA 4X rating of a Remote Controller enclosure, all cable entries into the enclosure must be through conduit fittings that are certified to meet NEMA 4X specifications.

Recommended vendor

Bridgeport Fitting, Inc., located in Stratford, Connecticut, manufactures a line of fittings that are certified to meet NEMA 4 watertight and NEMA 4X corrosion resistant specifications.

In addition, Hoffman Engineering Company, the manufacturer of the RHC enclosure, offers stainless steel NEMA 4X rated wireways, that are ideal for making power and I/O Link Interface connections between complexed enclosures. There are 4-inch x 4-inch and 6-inch x 6-inch cross sections available in lengths of 12, 24, 36, 48, 60, and 120 inches with the appropriate fittings, connectors, elbows, etc.

Conduit fitting list

Table 2-5 lists the part numbers and size of fittings that are manufactured by Bridgeport Fitting, Inc. The fittings are supplied with a bonding screw that is required for Remote Hardened Controller installations.

Table 2-5 Conduit Fittings – Bridgeport Fittings, Inc.

Size (Inches)	Cutout Size (Inches)	Part Number
0.5	0.875	161-DC
0.75	1.109	162-DC
1.0	1.375	163-DC
1.25	1.734	164-DC
1.5	1.984	165-DC
2.0	2.469	166-DC
2.5	2.969	167-DC
3.0	3.594	168-DC
3.5	4.125	169-DC
4.0	4.641	170-DC

2.7 Conduit Selection, Continued

Type of conduit recommended

It is generally recommended that the conduit used be a rigid service-entrance type conduit that is galvanized and threaded.

Fitting and conduit size selection

The size of the fitting and conduit is determined by the size of the largest cable connector and the number of cables that must be routed through the conduit because cables normally have connectors attached to them.

In some cases, a cable connector can be removed and reinstalled or installed after the cable is routed through the conduit. A smaller fitting and conduit can then be used.

2.8 Heater Selection

Introduction

When the exterior ambient temperature approaches -10°C or lower, a thermostatically controlled 800-watt electric heater can be mounted inside the enclosure to maintain an acceptable operating temperature with the assumption that the RHC will be mounted in an “ordinary” location. Two heater models are available. The heaters are not approved for a Class I, Division 2 location.

Two models available

The model MU-RCHT01 heater kit requires 120 Vac to operate and the model MU-RCHT02 heater kit requires 240 Vac. The heater specifications are listed in Table 2-6.

Table 2-6 Model MU-RCHT01/MU-RCHT02 Specifications

	Model MU-RCHT01	Model MU-RCHT02
Input Voltage	115 Vac	230 Vac
Input Frequency	50/60 Cycles	50/60 Cycles
Input Current	6.5 Amperes	3.3 Amperes
Wattage	800 Watts	800 Watts
Thermostat Control Range	-18°C to 38°C (0°F to 100°F)	

2.8 Heater Selection, Continued

Mounting

The heater is mounted on the left side of a mounting plate that is located at the front of the enclosure, near the bottom. The heater is regulated by a thermostat, which is an integral part of the heater, and protected by a thermal switch, should the thermostat “stick” in the on position.

CAUTION—Power for the heater must be provided by a circuit breaker that is separate from the circuit breaker that provides power to the RHC Power System. Use a separate 15 amp circuit breaker for both 120 Vac and 240 Vac operation and 14-gauge wire for the connections.

2.9 Enclosure Complexing

2.9.1 Thermal Considerations

Introduction

TotalPlant Configurator (TPC) Release 3.3 and later releases have the capability to predict the maximum internal temperatures of the RHC enclosure, which are based on the maximum external environment and electronic assemblies that are installed in the enclosure. These predictions, as of TPC R3.3, are all based on stand-alone enclosures. In other words, no other RHC enclosures, or any other type of heat generating equipment is mounted in close proximity to the RHC enclosure.

2.9.1 Thermal Considerations, Continued

Stand-alone RHC enclosure internal temperature formula

For a stand-alone RHC enclosure, the cooling is accomplished by free convection and radiation. The TPC predicts the internal temperature of the RHC enclosure by using the following formula.

$$y = .037x + b$$

Where: b = the ambient temperature surrounding the enclosure

x = the power dissipated

y = the temperature of the air stream inside the enclosure

$.037$ = the temperature increase in °C per watt of dissipated power

This expression was determined experimentally. While it is relatively easy to develop an expression for a stand-alone RHC enclosure by experimental means, it is more difficult to do a similar experiment with complexed enclosures. Therefore, the predictions are based on theory.

2.9.1 Thermal Considerations, Continued

Thermal analysis for an enclosure complex

A thorough thermal analysis was conducted, and the results are presented in Table 2-7 with two examples following the table.

Table 2-7 Enclosure Complexing Thermal Analysis

Complex Geometry		Two Adjacent RHC Cabinets		RHC Cabinet with Adjacent RHC Cabinets	
Separation (Feet)	View Factor	Percent of Reduced Radiated Heat from Cabinet	Multiplication Factor Applied to Heat Rise for Stand-Alone Cabinet	Percent of Reduced Radiated Heat from Cabinet	Multiplication Factor Applied to Heat Rise for Stand-Alone Cabinet
0.0	1.000	22	1.22	44	1.44
1.0	0.694	6	1.06	12	1.12
2.0	0.500	4	1.04	8	1.08

For 36 inches and over, it is reasonable to assume that the temperature increase is small and can be ignored.

Two enclosure complex example

An RHC enclosure is entered into the **TotalPlant** Configurator (TPC). The controller consists of a redundant HPM, 20 amp power supply, UCNE and an assortment of IOPs and FTAs. The project engineer on the job determined that the maximum average ambient temperature was 40°C. The TPC determines that there will be 345.5 watts dissipated for the chosen equipment and there will be a thermal rise of 12.8°C, bringing the internal enclosure temperature to 52.8°C. However, this particular RHC enclosure will be spaced 1 foot from another RHC enclosure with a similar thermal load, thereby reducing the heat exchange at one side of the enclosure. Table 2-7, indicates that a multiplication factor of 1.06 is appropriate for this configuration. Therefore, the internal temperature rise in the RHC enclosure will be:

Stand-Alone Cabinet Temperature Rise x Multiplication Factor =
12.8°C x 1.06 = 13.6°C

This raises the internal enclosure temperature to 40°C + 13.6°C = 53.6°C.

2.9.1 Thermal Considerations, Continued

Three enclosure complex example

An RHC enclosure with the same electronic assemblies that were described in the previous example is placed between two RHC enclosures with 1-foot spacing on both sides and are operating at approximately the same internal temperature.

Stand-Alone Cabinet Temperature Rise x Multiplication Factor =
 $12.8^{\circ}\text{C} \times 1.12 = 14.3^{\circ}\text{C}$

This raises the internal enclosure temperature to $40^{\circ}\text{C} + 14.3^{\circ}\text{C} = 54.3^{\circ}\text{C}$.

Cabling considerations

To maintain the NEMA 4X integrity of the system, all entrances to and exits from the RHC enclosure must be through approved NEMA 4X conduit, wireways, fittings, etc., or approved alternate methods. Conduit selection has already been addressed in subsection 2.7. In addition, Hoffman, the manufacturer of the RHC enclosure, offers stainless steel NEMA 4X rated wireways, that are ideal for making power and I/O Link connections between complexed enclosures. There are 4-inch x 4-inch and 6-inch x 6-inch cross sections available in lengths of 12, 24, 36, 48, 60, and 120 inches with appropriate fittings, connectors, elbows, and so on.

A balance needs to be maintained between the need to separate RHC enclosures to meet thermal requirements and spacing them close enough to avoid difficulties due to the length or availability of standard cabling. The longest available I/O Link cable is currently a 6-drop I/O Link which is probably sufficient for a 1- to 2-foot separation between RHC enclosures, depending on cable routing and whether feed-through wireways are used between the sides of the enclosures.

Section 3 – Installation

3.1 Overview

Introduction	Typically, a Remote Hardened Controller (RHC) enclosure arrives from the factory fully assembled and properly configured. Generally, the only required on site task is cutting cable entry holes in the enclosure for the cables that must connect to the enclosure's internal electrical components.
External cable entry	A Remote Hardened Controller enclosure can be mounted in its planned location before preparing the enclosure for external cable entry if conduit entry hole alignment with another enclosure is not a consideration.
Number of entry holes	The number of components that are installed in the enclosure determines the number, size, and location of the entry holes that must be cut in the enclosure.
Entry fittings	A conduit fitting or some type of sealed transit device that is NEMA 4X rated must be used for the entry into the enclosure of ac power cable(s), fiber optic cables, coaxial cables, and field control wiring.

3.2 Installation Considerations

Considerations

Consider the following before you begin a site installation.

- A single ac power source or two independent ac power sources must provide power to the Power System in the enclosure. In either case, the ac input power lines to the Power System's Power Supply Module(s) must be protected by an exterior 15-ampere circuit breaker.
Battery backup for the Power System's Power Supply Module(s), can be provided by an externally mounted Battery Backup assembly. If power stability is a concern, an uninterruptible power source (UPS) should be used to provide the ac power.
 - If there will be a complex of enclosures at the remote site, the I/O Link Interface between the enclosures must be provided by either I/O Link Interface cables or a fiber optic I/O Link Extender. See subsection 2.9, *Enclosure Complexing*, for detailed information.
 - If an I/O Link Interface Extender is installed, the A and B fiber optic cables must be correctly connected to the Fiber Optic Coupler modules. The card file's I/O Link Interface address and the slot location of the I/O Link Extender card determine which fiber optic cable will connect to each Fiber Optic Coupler module.
 - The HPMM(s) must be connected to the Universal Control Network (UCN) through either coaxial cables or a fiber optic Universal Control Network Extender.
 - All card files must be pinned (configured) for their assigned I/O Link Interface address as defined by the High-Performance Process Manager subsystem's configuration. Generally, each card file is properly configured in the factory.
 - HPMM card files must be pinned for their assigned Universal Control Network (UCN) node address. Generally, each card file is properly configured in the factory.
-

3.3 Enclosure Mounting, Continued

Enclosure mounting procedure

Use the procedure in Table 3-1 to mount an enclosure on a concrete pad. It is difficult to prepare a concrete pad with mounting studs that are accurately embedded in the concrete. Therefore, the procedure assumes that mounting screws (bolts) will be used to secure the enclosure after drilling holes and then installing lead sinks in the appropriate locations in the concrete.

Table 3-1 RHC Enclosure Mounting Procedure

Step	Action
1	Using either the template dimensions shown in Figure 3-1 or by positioning the enclosure in the appropriate location on the concrete pad, accurately mark the pad surface for the location of the lead sinks that will be embedded in the concrete. Each of the four floor stands has three mounting holes (11 mm, .44 inches), a center hole and a pair of holes that are located at the edge of the floor stand. Choose the holes that will be used to attach the enclosure to the pad.
2	After marking the locations where the lead sinks will be installed, drill holes in the concrete that are large enough to accommodate the diameter of the lead sinks. The mounting screws (bolts) must be no smaller than 3/8 inches in diameter and 2 inches in length. Use lead sinks that will accommodate the screws (bolts) that you have chosen.
3	Install the lead sinks in the holes.
4	Position the enclosure's mounting holes over the lead sinks and attach the enclosure to the concrete pad with the mounting screws (bolts).
5	Repeat Steps 1 through 4 for any additional enclosures.

3.4 Conduit Installation

Introduction

When multiple Remote Hardened Controller enclosures exist at a site, appropriate sized conduit must be installed between the enclosures to route the I/O Link Interface and/or UCN cables. Conduit for power cables and field device wiring to FTAs must also be a consideration.

Conduit installation procedure

Use the procedure in Table 3-2 to install conduit between Remote Hardened Controller enclosures.

Table 3-2 Multiple Enclosure Conduit Installation Procedure

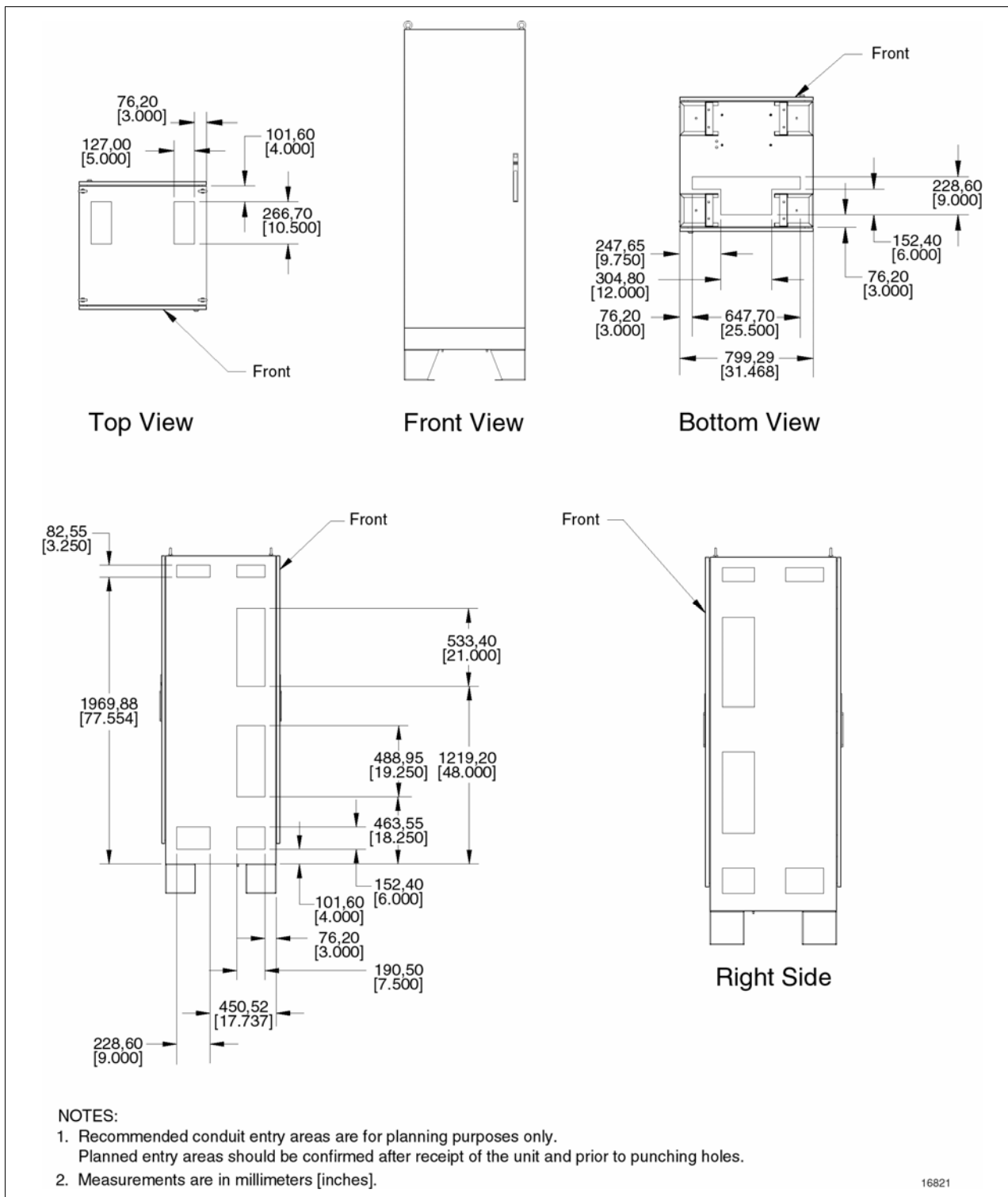
Step	Action
1	Referring to Figure 3-2, which identifies areas where openings can be cut or punched for cable entry into an enclosure through conduit, mark the location(s) on the enclosure that must be cut or punched.
2	Drill or cut holes in the enclosure at the locations that are marked for cable entry. Refer to Table 2-5 for conduit cutout sizes. Be very careful that metal chips or filings are not deposited on any of the electrical components. Use antistatic cloth or similar material to protect them. Remove any metal chips or filings that enter the enclosure with a vacuum cleaner, if necessary.
3	Repeat Steps 1 and 2 for the second enclosure, and any other enclosures.
4	Install the appropriate size of conduit between the Remote Hardened Controller enclosures.

3.4 Conduit Installation, Continued

Enclosure conduit entry areas

Figure 3-2 illustrates the acceptable areas where openings can be cut or punched in the enclosure's outer surface for conduit entry.

Figure 3-2 Acceptable Enclosure Conduit Entry Areas



3.5 AC Power and Ground Connections

3.5.1 Standard Power System

AC Power connections

Figure 3-3 illustrates the ac power and safety ground connections at TB1 and TB2 on the standard Power System's backpanel for the primary and redundant Power Supply Modules when either a single or two independent ac power sources provide power. If there is a single ac power source, TB1 and TB2's terminals must be strapped together.

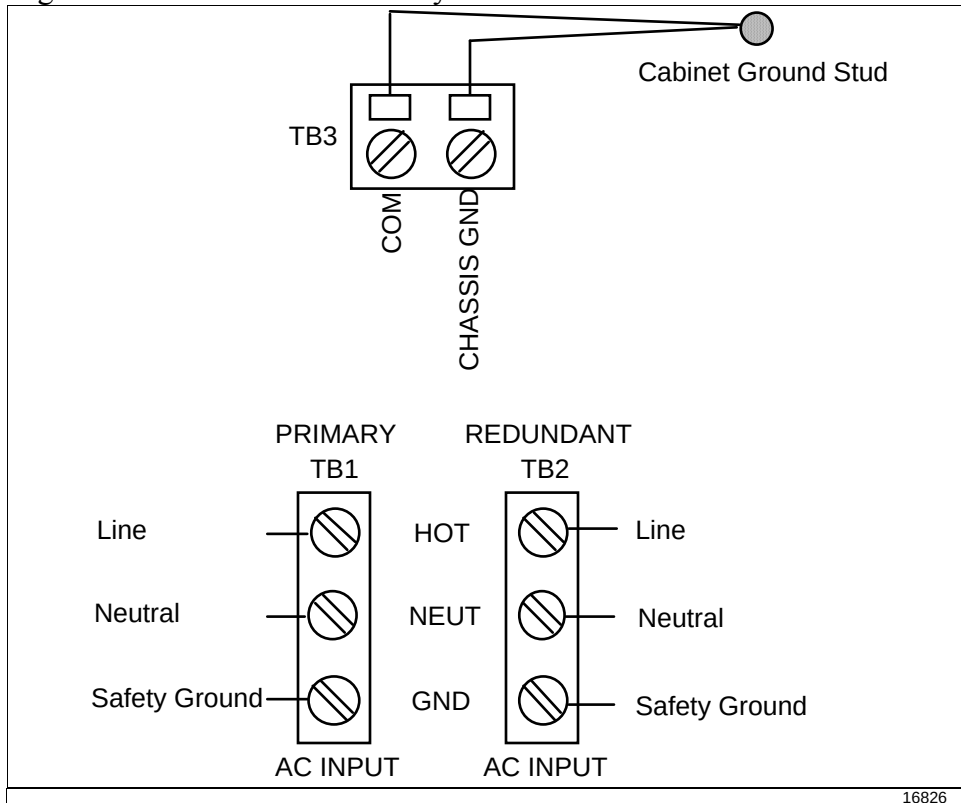
Ground connections

TB3 provides connection terminals for **CHASSIS GND** (Safety Ground) and **COM** (optionally Master Reference Ground). The normal practice is to connect both the **CHASSIS GND** and **COM** terminals to the ground stud inside the enclosure with separate 2.0 mm² (14 AWG) wires. TB3 has compression terminals while TB1 and TB2 have screw terminals. Spade lugs are recommended for the connections. To have access to TB1 and TB2, the Primary (left) Power Supply Module must be removed.

3.5.1 Standard Power System, Continued

Ground connections

Figure 3-3 Standard Power System Power and Ground Connections



3.5.2 AC Only Power System

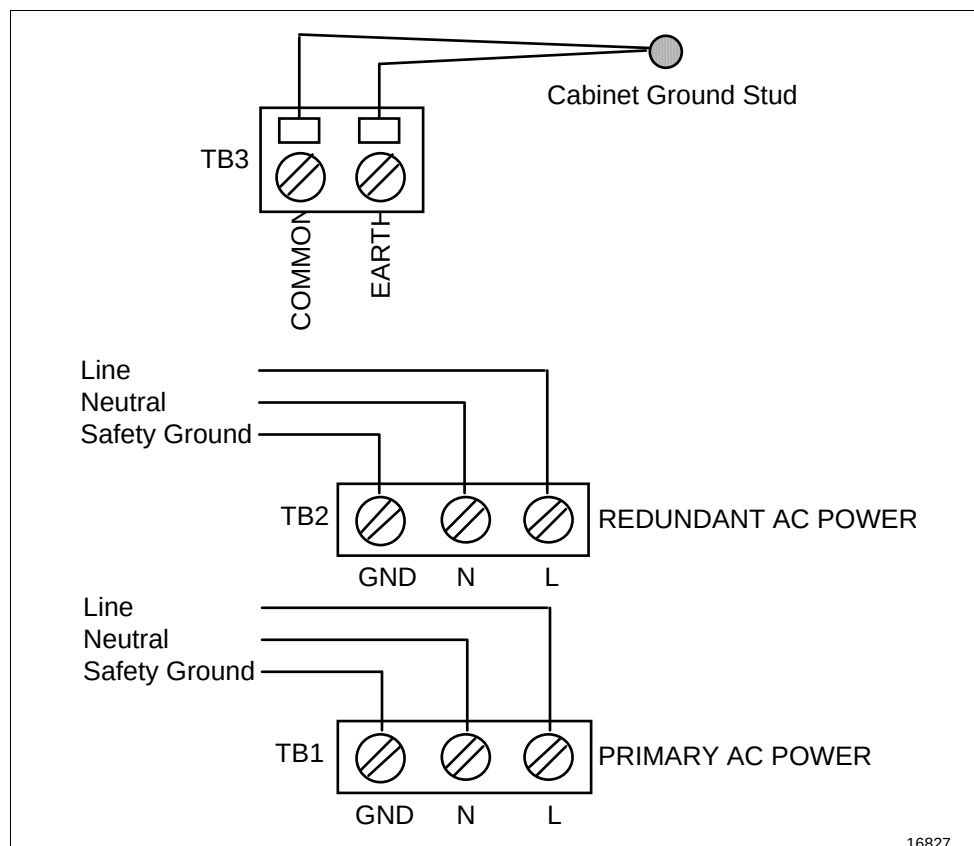
AC Power connections

Figure 3-4 illustrates the ac power and safety ground connections at TB1 and TB2 on the AC Only Power System's AC/DC Distribution Assembly for the primary and redundant Power Supply Modules when either a single or two independent ac power sources provide power. If there is a single ac power source, TB1 and TB2's terminals must be strapped together.

Ground connections

TB4 provides connection terminals for **EARTH** (Safety Ground) and **COMMON** (optionally Master Reference Ground). The normal practice is to connect both the **EARTH** and **COMMON** terminals to the ground stud inside the enclosure with separate 2.0 mm² (14 AWG) wires. TB3 has compression terminals while TB1 and TB2 have screw terminals. Spade lugs are recommended for the connections.

Figure 3-4 AC Only Power System Power and Ground Connections



3.6 Alarm Contacts Connections

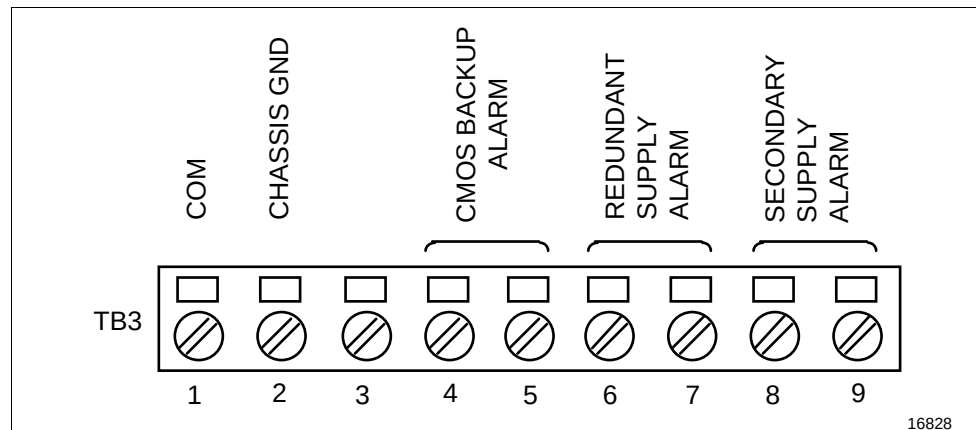
3.6.1 Standard Power System

Alarm contacts

Figure 3-5 illustrates the connections at TB3 on the standard Power System's backpanel for the Power Supply Module alarm annunciator contacts. The contacts are normally closed when the component that is being monitored is functional.

There are alarm contacts for each Power Supply Module, primary and secondary, if present, and the CMOS battery backup assembly. Typically, the contacts are connected to the field terminals of a 24 Vdc Digital Input FTA.

Figure 3-5 Standard Power System Power Supply Module Alarm Annunciator Connections



3.6.2 AC Only Power System

Power Supply Module alarm contacts

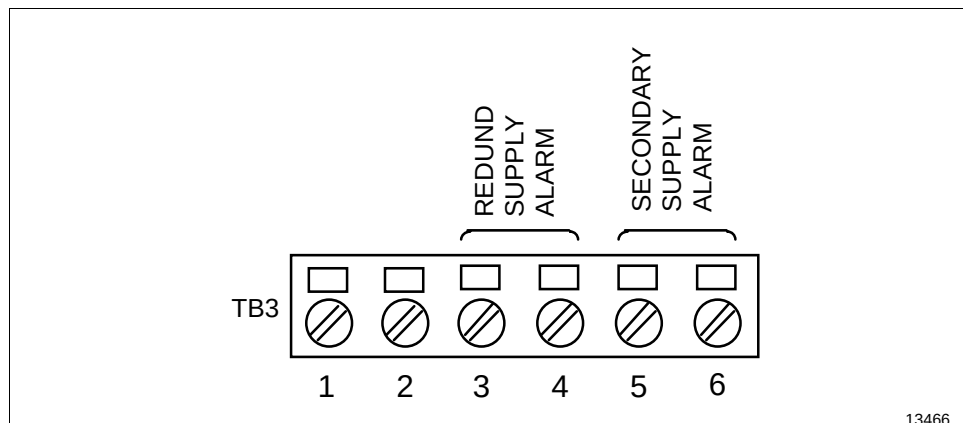
Figure 3-6 illustrates the connections at TB3 on the AC Only Power System's backpanel for the Power Supply Module alarm annunciator contacts. The contacts are normally closed when the Power Supply Module is functional.

There are alarm contacts for each Power Supply Module, primary and secondary, if present.

Typically, the contacts are connected to the field terminals of a 24 Vdc Digital Input FTA.

3.6.2 AC Only Power System, Continued

Figure 3-6 AC Only Power System Power Supply Module Alarm Annunciator Connections



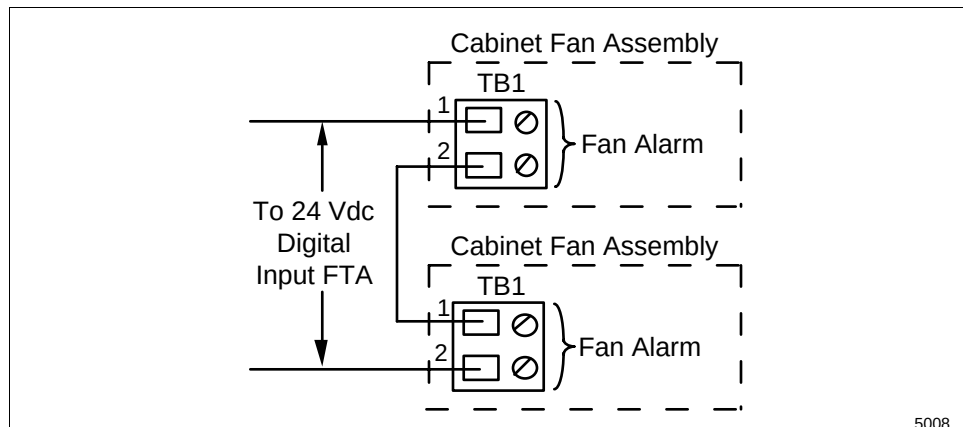
3.6.3 Fan Assembly

Alarm contacts

The fan assembly's alarm contacts are accessible through compression terminals at TB1. Relay K1's normally open contacts are generally connected to a Digital Input FTA as a failure sensing device. The contacts close while both fans are rotating and open when one or both fans are not rotating.

Figure 3-7 illustrates the suggested connections to TB1 on the fan assemblies.

Figure 3-7 Fan Assembly Alarm Annunciator Connections



3.7 I/O Link Interface Cables

Introduction

When a site installation consists of a complex of Remote Hardened Controller enclosures and the distance between them is acceptable, communication between the HPMM(s) and IOP cards is provided by a pair of redundant I/O Link Interface cables.

If the distance between the enclosures is too great, or the HPMM(s) or IOPs are located in another area, communication between the HPMM(s) and IOP cards must be provided by a fiber optic I/O Link Extender.

Interconnections

The I/O Link Interface cables are daisy-chained between card files in the enclosures.

Distance between connectors

The distance between the card file connectors is restricted to no more than 2 meters (6.6 feet).

Cable selection

Sets of I/O Link Interface cables are available in five lengths. The shortest cable, 1.5 meters (5 feet) long, has 2 connectors. The longest cable, 7.5 meters (25 feet) long, has 6 connectors with each connector separated by 1.5 meters (5 feet).

Any of the 5 lengths of cables can be installed between the enclosures in an enclosure complex; however, the end connectors must always be used. The connectors between the end connectors, if any exist, can be left unconnected.

See Table 5-2 in subsection 5.2 for cable selection.

ATTENTION

I/O Link Interface cables must be routed in conduit between enclosures.

Installation

Route the cables between the enclosures and connect them to J23 for Cable A and J24 for Cable B on any 7-Slot card files and J42 for Cable A and J43 for Cable B on any 15-Slot card files in each enclosure. Be sure not to cross the A and B cables.

3.7 I/O Link Interface Cables, Continued

Cable shield grounding

Both I/O Link Interface cables must be grounded at one end. The 7-Slot and 15-Slot card files provide that capability. For consistency, install jumpers between the pins at locations J22 and J29 on the 7-Slot card file backpanel and J44 and J45 on the 15-Slot card file backpanel that contains an HPMM. If redundant HPMMs exist, jumpers must be installed on only one HPMM card file. See Figure 3-15 or 3-16 for the locations of J22, J29, J44, or J45.

3.8 I/O Link Interface Extender

Introduction

The I/O Link Interface Extender fiber optic cable set consists of two cables that are labeled A and B. The pair of I/O Link Interface Extender cards can be installed in any odd- and even-numbered card file slots, but it is recommended that they be installed in the same card file. If both cards are not installed in the same card file, the cards must be installed in an odd- and even-numbered slot in the separate card files, such as slot 3 and slot 4. The I/O Link Interface cables between the card files complete the I/O Link Interface communications path.

I/O Link Extender A and B determination

The slots where the I/O Link Extender cards are installed, together with the assigned I/O Link Interface address of the card file, determines which card slot is I/O Link Extender A and which is I/O Link Extender B. The following rules apply.

Determination rules

If the card file number (1 through 8) and card file's slot number (1 through 7, 9 through 15, or 1 through 15) are both an even or odd number, I/O Link Interface Extender cable A must be connected to the slot's Fiber Optic Coupler module. If both the card file number and card file's slot number are not an even or odd number, I/O Link Interface Extender cable B must be connected to the slot's Fiber Optic Coupler module.

Reference

See the I/O Link Extender section in the *Process Manager/Advanced Process Manager Planning* or *Process Manager I/O Installation* manual for a detailed discussion of the I/O Link Extender.

3.8 I/O Link Interface Extender, Continued

Cable installation

Connect the I/O Link Interface Extender cables to their appropriate Fiber Optic Coupler modules. Be sure the “A” and “B” cables connect to the appropriate Fiber Optic Coupler modules.

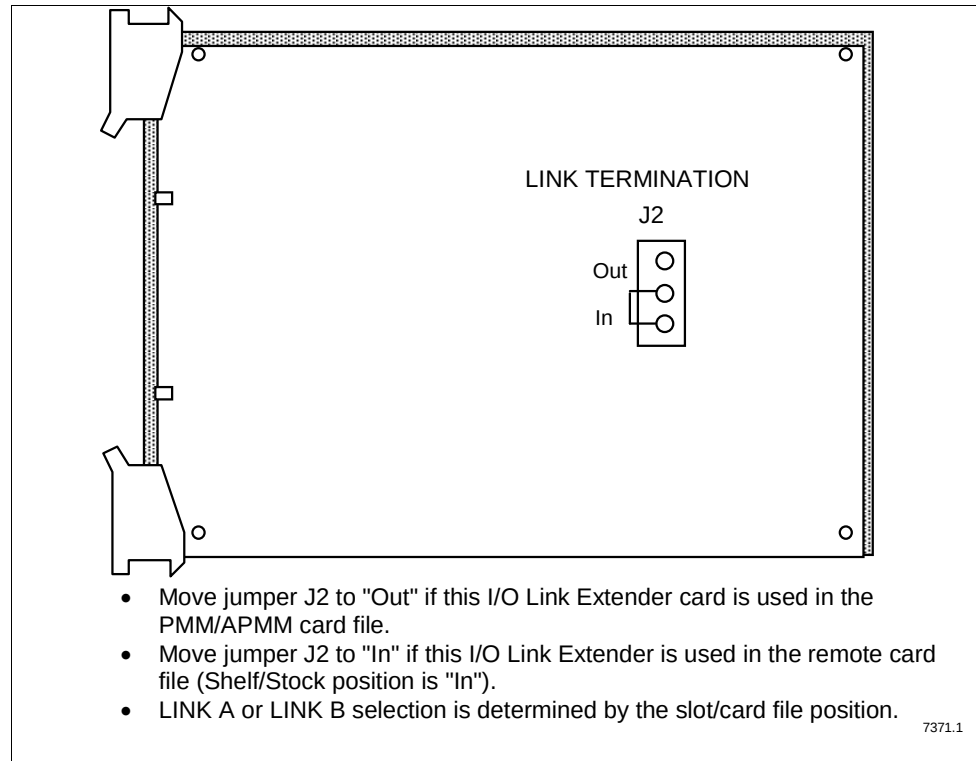
Standard I/O Link Interface Extender termination

Both Link A and Link B of the I/O Link Interface must be terminated at both ends. The HPMM or local IOP card file provides local I/O Link Interface termination. To provide termination at the remote card file, the I/O Link Interface Extender card has jumper selectable terminators available. The **LINK TERMINATION** jumper, **J2**, has two positions, **IN** and **OUT**. At the HPMM or local IOP card file end of the fiber link, the jumper must be placed in the **OUT** position. The opposite is true at the remote end of the fiber link. The jumper must be placed in the **IN** position. See Figure 3-8 for the location of the jumper on the Standard I/O Link Interface Extender card.

3.8 I/O Link Interface Extender, Continued

Standard I/O Link Extender card pinning

Figure 3-8 Standard I/O Link Interface Extender Card Pinning



Long Distance I/O Link Extender link selection

Before installing the Long Distance I/O Link Interface Extender card in its card file slot, it must be configured as Link A or Link B by jumper selection on the card. The jumper location is labeled **A/B LINK SELECTION** and has two positions labeled **LINK A** and **LINK B**. Placing the jumper in the proper position provides the selection. After installation of the link and upon applying power to the equipment, the link selection is indicated by one of two LEDs at the front of the Fiber Optic Coupler module. See Figure 3-11 for the location of the jumper on the Long Distance I/O Link Interface Extender card.

3.8 I/O Link Interface Extender, Continued

Long Distance I/O Link Interface Extender termination

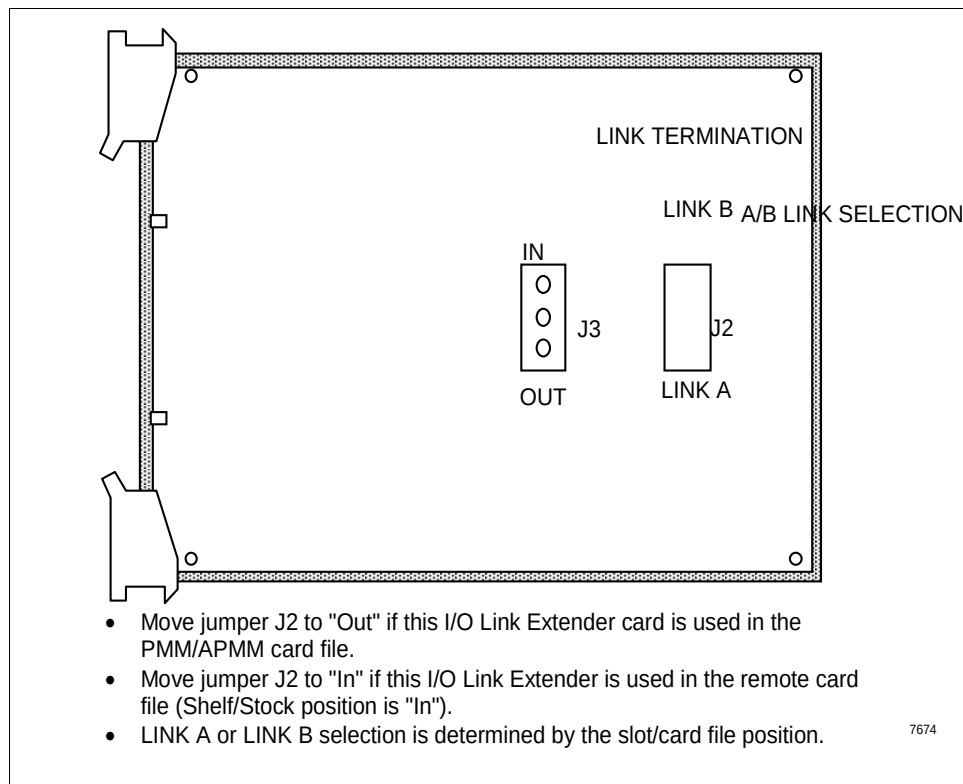
Both Link A and Link B of the I/O Link Interface must be terminated at both ends. The HPMM or local IOP card file provides local I/O Link Interface termination. To provide termination at the remote card file, the I/O Link Interface Extender card has jumper selectable terminators available. The **LINK TERMINATION** jumper, **J2**, has two positions, “**IN**” and “**OUT**.” At the HPMM or local card file end of the fiber optic link, the jumper must be placed in the “**OUT**” position. The opposite is true at the remote end of the fiber optic link. The jumper must be placed in the “**IN**” position. See Figure 3-9 for the location of the jumper on the Long Distance I/O Link Interface Extender card.

3.8 I/O Link Interface Extender, Continued

Long Distance I/O Link Interface Extender card pinning

Figure 3-9 illustrates the pinning options on the Long Distance I/O Link Extender card.

Figure 3-9 Long Distance I/O Link Interface Extender Card Pinning



3.9 Universal Control Network Extender

Introduction

The Universal Control Network Extender (UCNE) extends the Universal Control Network through fiber optic cable that can be up to 2 kilometers in length.

UCNE devices are installed in pairs at each end of a redundant fiber optic link; that is, a UCNE-A and UCNE-B device at one end of the fiber optic link and a UCNE-A and UCNE-B device at the other end of the fiber optic link. Therefore, a total of four UCNE devices are required to implement one redundant fiber optic link.

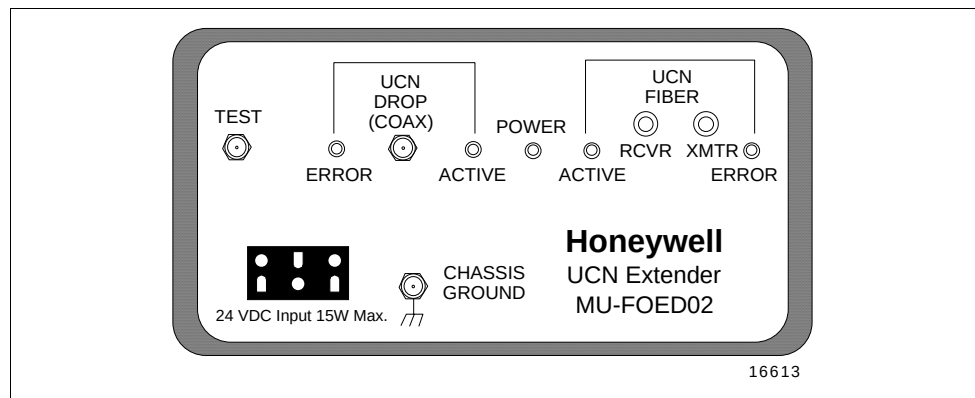
The UCNE device at each end of the fiber optic link must never connect directly to the UCN coaxial trunk cable, but instead must connect to the drop port of a UCN drop tap on that coaxial segment.

Standard coaxial UCN drop cables (RG6-type) are used to connect the UCNE devices to the UCN tap drop ports.

Each UCNE device provides coax-to-fiber optic and fiber optic-to coax cable conversion for the UCN signals.

Figure 3-10 illustrates the front panel of a dc-powered UCNE device.

Figure 3-10 DC-Powered UCNE Device



References

See the Universal Control Network Extender section in the *Universal Control Network Planning* and *Universal Control Network Installation* manuals for a detailed discussion of the Universal Control Network Extender.

3.9 Universal Control Network Extender, Continued

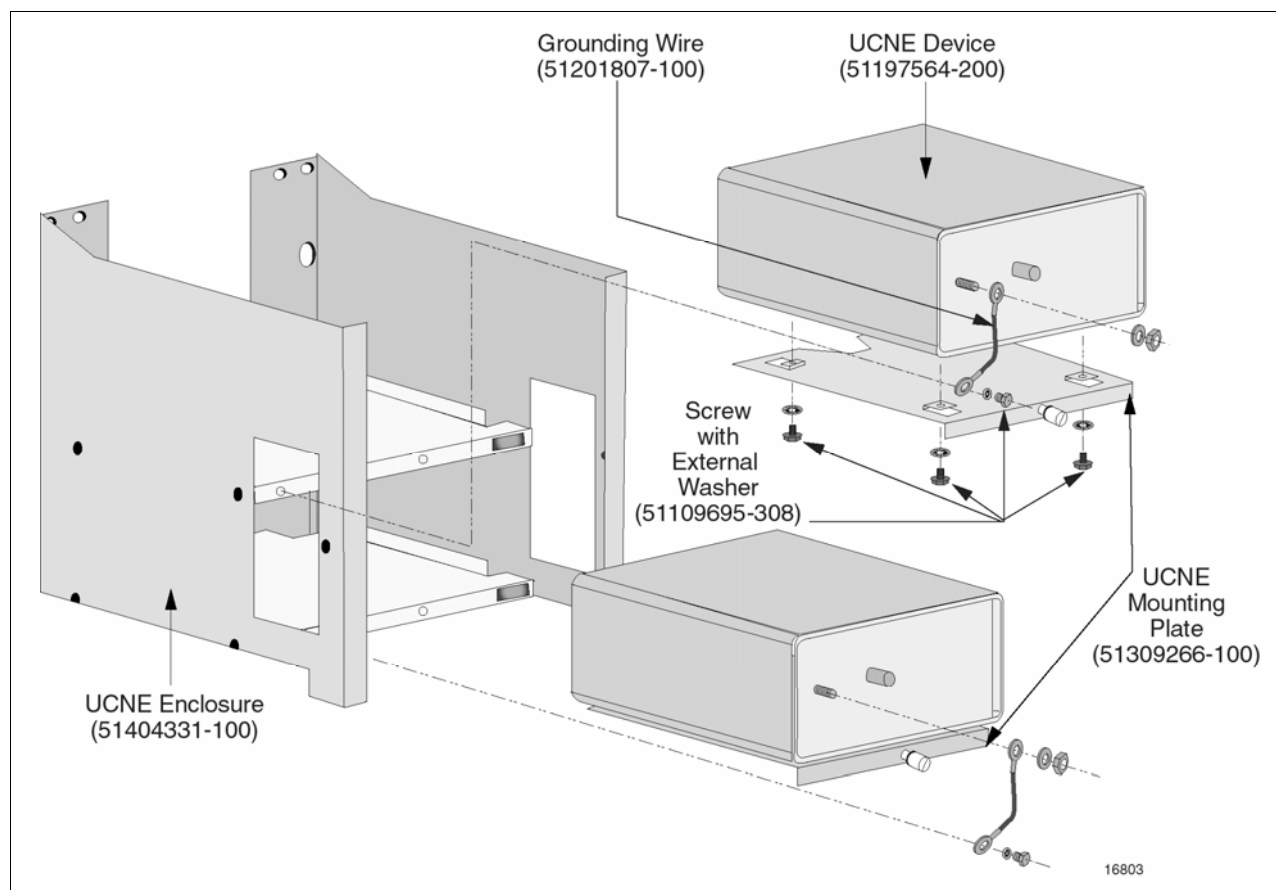
No UCNE configuration pinning or adjustments

The UCNE does not require any configuration pinning or adjustments.

UCNE mounting kit

The UCNE enclosure that is included in the model MU-FOMK21 mounting kit mounts on the right side of the mounting panel that is located at the front bottom of the RHC enclosure with four thread-forming screws. The UCNE enclosure consists of a chassis with two shelves and two trays. A UCNE device is attached to each mounting tray with four screws, and a tray is mounted on each shelf. Figure 3-11 is an exploded view of the installation.

Figure 3-11 Exploded View of UCNE Enclosure Installation



3.9 Universal Control Network Extender, Continued

UCNE enclosure Installation procedure

Use the procedure in Table 3-3 to install a UCNE enclosure and UCNEs. Refer to Figures 3-12 and 3-13 during the procedure.

Table 3-3 UCNE Enclosure Installation Procedure

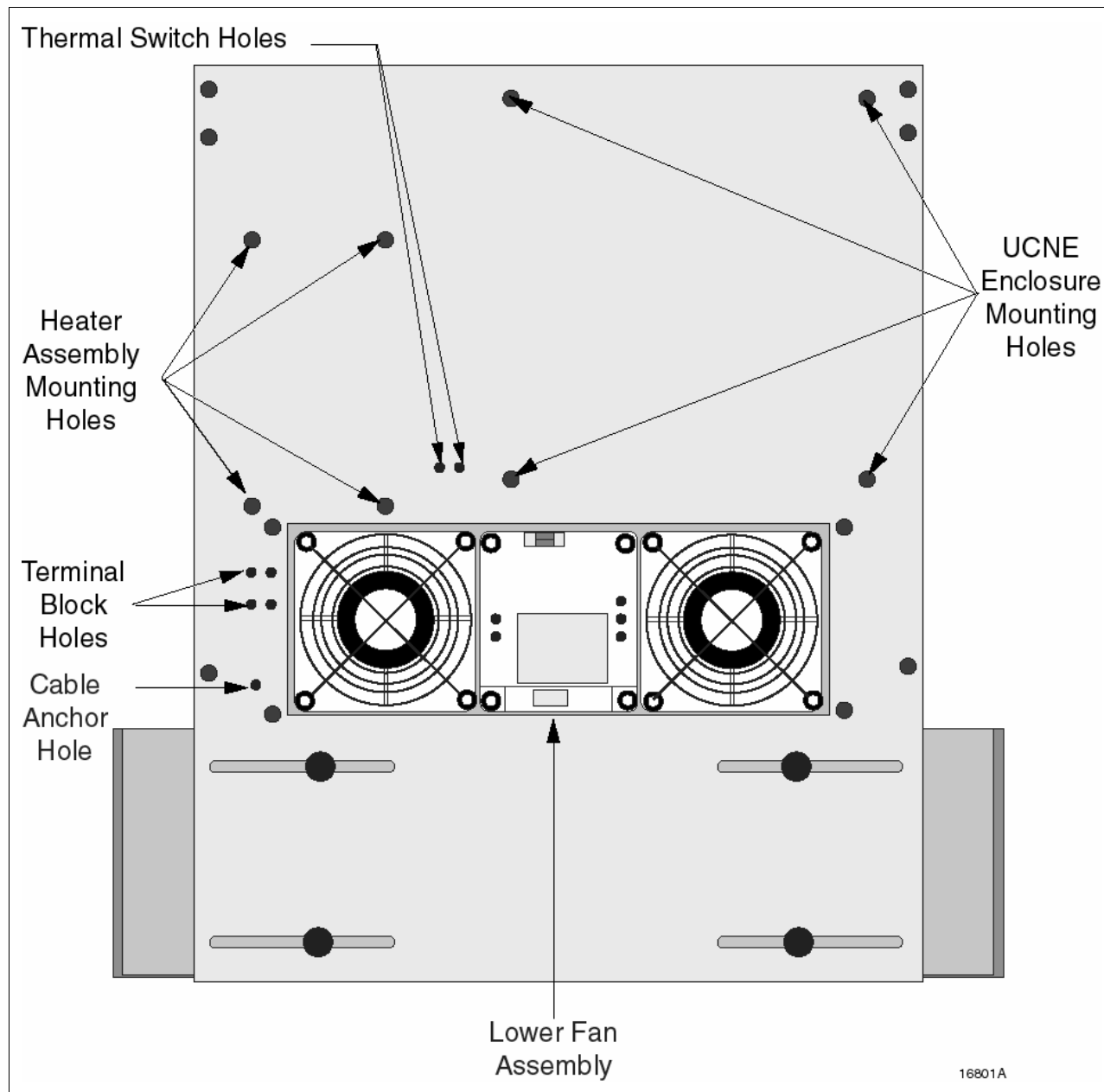
Step	Action
1	Mount the UCNE enclosure on the right side of the mounting panel at the front bottom of the RHC enclosure with four M5 lockwashers and screws. The location of the mounting holes in the mounting panel is illustrated in Figure 3-12. Figure 3-13 illustrates the panel with the UCNE enclosure attached.
2	Attach the pair of UCNE devices to their mounting plates (51309266-100) with four M4 pan-head screws that include a captive external lockwasher (51109695-308).
3	Place one of the UCNE devices on the enclosure's upper shelf and secure it in place with the mounting plate's captive screw.
4	Place the second UCNE device on the enclosure's lower shelf and secure it in place with the mounting plate's captive screw.
5	Connect a grounding wire (51201807-100) between the ground stud on the UCNE's front panel and the enclosure chassis. Use a M4 pan-head screw (51109695-308) to fasten it to the enclosure chassis.
6	Connect a power cable (51201397-002) between a power connector on the Power System's backpanel and the power connector at the front of each UCNE.

3.9 Universal Control Network Extender, Continued

Mounting panel attachment holes

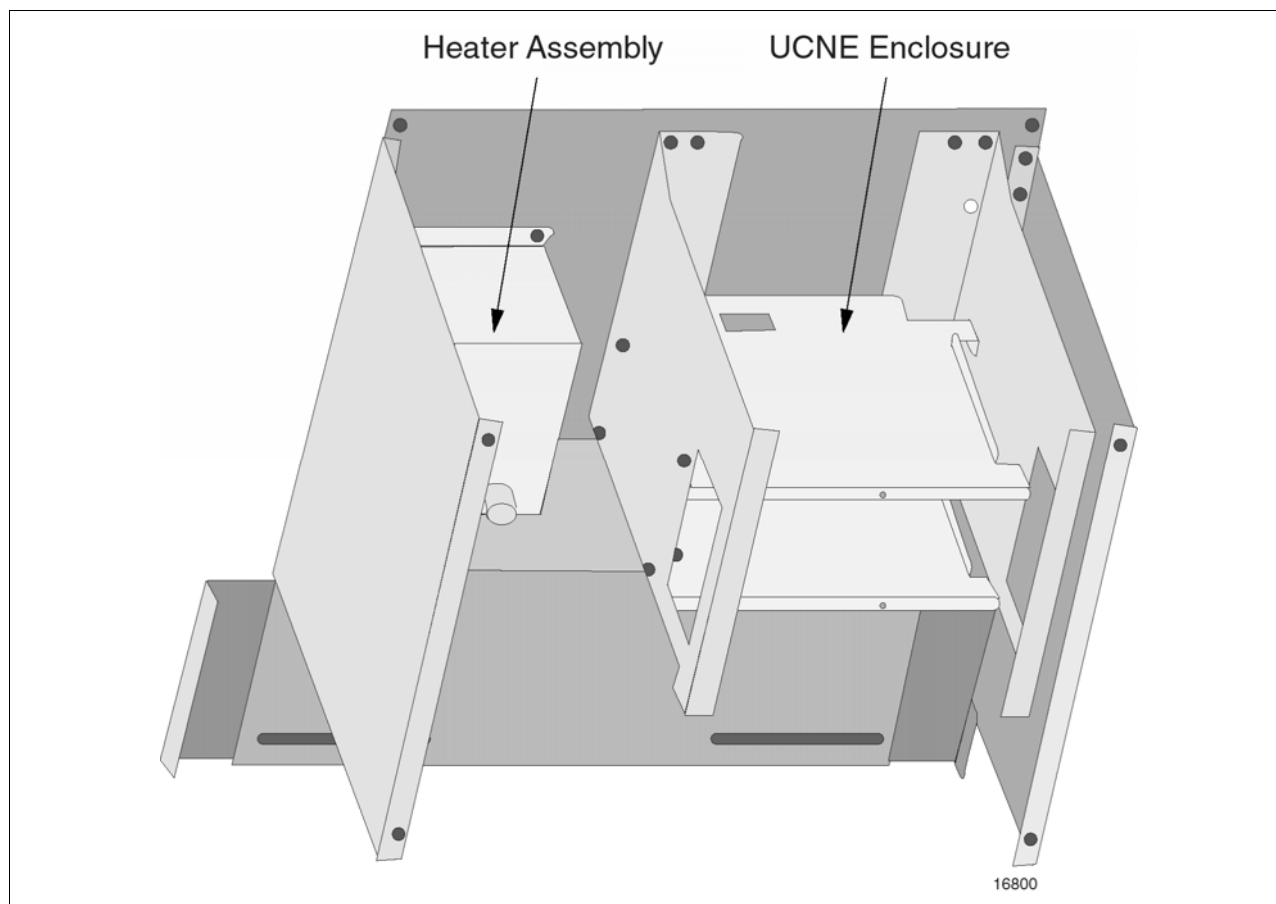
Figure 3-12 illustrates the location of the mounting attachment holes for the UCNE enclosure and heater assembly.

Figure 3-12 Mounting Panel Attachment Holes



Mounting panel attachment illustration

Figure 3-13 Mounting Panel with UCNE Enclosure and Heater Assembly Attached

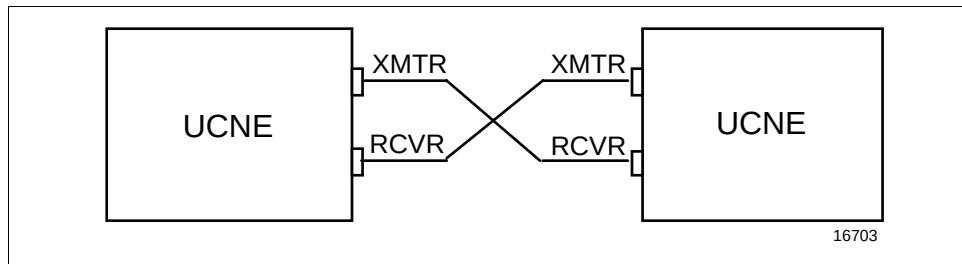


3.9 Universal Control Network Extender, Continued

Fiber optic cable connections

A pair of fiber optic cables connects to each UCNE's **RCVR** (Receive) and **XMTR** (Transmit) connectors at each end of the communication link as illustrated in Figure 3-14. The figure represents one of the redundant links, UCNE-A or UCNE-B. The same connection pattern is repeated for the other link in the redundant pair.

Figure 3-14 UCNE Fiber Optic Cable Interconnections



Cable drop taps

A UCN drop tap must be used between the UCNE's **UCN DROP (COAX)** connector and the HPMM's HPM UCN Interface Module's **UCN A** or **UCN B** connector to ensure proper attenuation of the signal. The UCNE device must never connect directly to the HPMM's HPM UCN Interface Module. RG6-type coaxial cable must be used for the connections.

The drop taps for both the A and B cables are mounted in the enclosure. A single HPMM requires 2-port taps, and redundant HPMMs require 4-port taps.

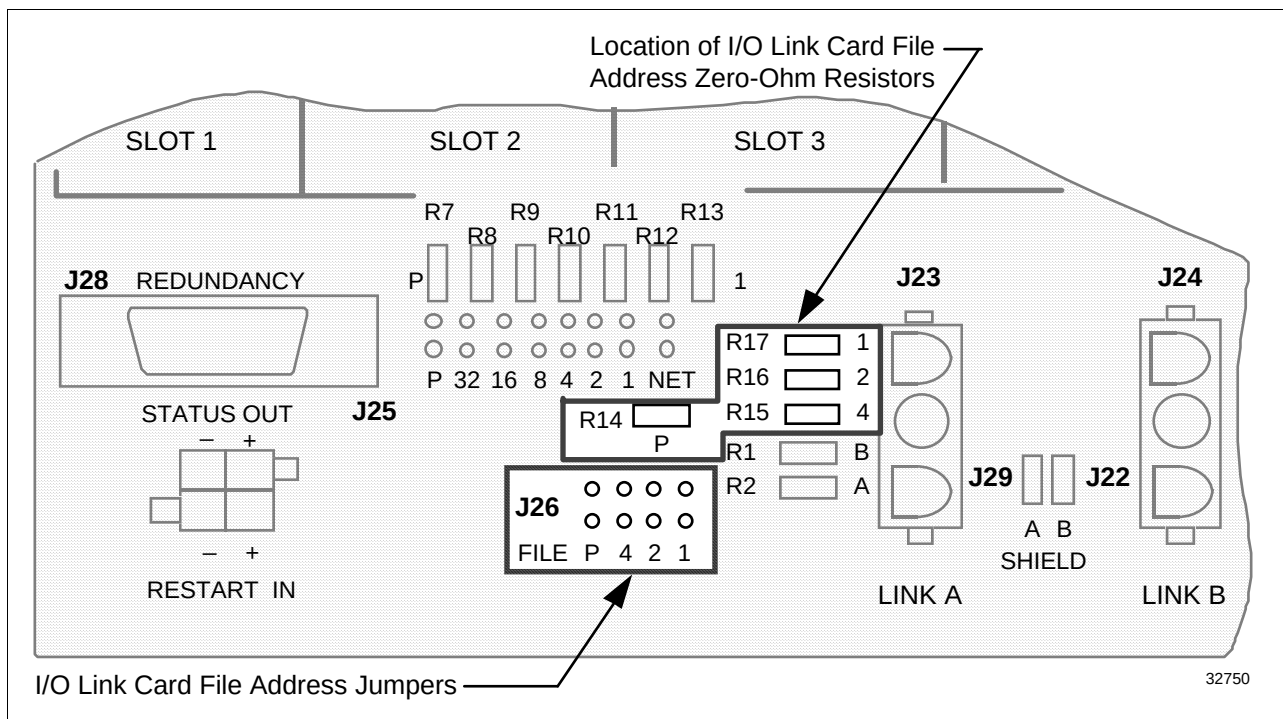
3.10 Card File I/O Link Interface Address Selection

Introduction

The assigned I/O Link Interface address for a card file is selected by pinning on the card file's backpanel. Generally, the card file(s) in the enclosure is pinned (configured) for the proper I/O Link Interface address in the factory before shipment. However, the assigned address can be changed on site.

Figure 3-15 is an illustration of a section of a Left 7-Slot card file's backpanel where the I/O Link Interface address is selected. The Right 7-Slot card file's backpanel is identical, except the slots are numbered 9 through 15. Pinning on a 15-Slot card file's backpanel is also identical, except the designated location is J41.

Figure 3-15 Left 7-Slot Card File I/O Link Interface Address Selection



3.10 Card File I/O Link Interface Address Selection, Continued

I/O Link Interface address selection

The selection is made at J26 on the Left and Right 7-Slot card files and J41 on the 15-Slot card file. There are four jumper positions that are labeled **1**, **2**, **4**, and **P**. The positions **1**, **2**, and **4** represent the binary equivalent of the I/O Link Interface address. The **P** position represents parity. An odd number of jumpers must be installed, so if the I/O Link Interface address is 3, jumpers must be installed in positions **1**, **2**, and **P**. If the I/O Link Interface address is 2, a single jumper is installed in position **2**.

3.11 Card File UCN Node Address Selection

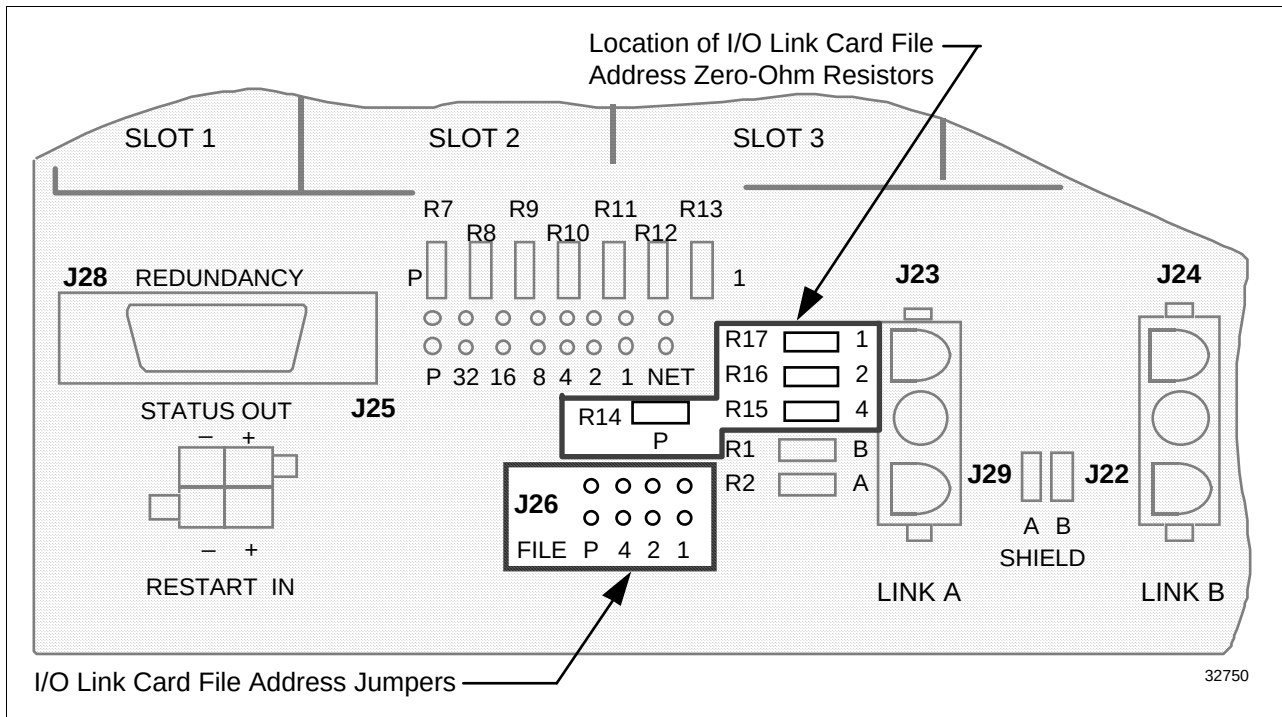
Introduction

The assigned UCN node address for an HPMM card file is selected by pinning on the card file's backpanel. Generally, the HPMM card file(s) in the enclosure is pinned (configured) for the proper UCN node address in the factory before shipment. However, the assigned address can be changed on site.

Figure 3-16 is an illustration of a section of a Left 7-Slot card file's backpanel where the I/O Link Interface address is selected. The Right 7-Slot card file's backpanel is identical, except the slots are numbered 9 through 15. Pinning on a 15-Slot card file's backpanel is also identical, except the designated location is J40.

3.11 Card File UCN Node Address Selection, Continued

Figure 3-16 Left 7-Slot Card File UCN Node Address Selection



UCN node address selection

The selection is made at J25 on the Left and Right 7-Slot card files and J40 on the 15-Slot card file. There are seven jumper positions that are labeled 1, 2, 4, 8, 16, 32, and P. The positions 1, 2, 4, 8, 16, 32 represent the binary equivalent of the UCN node address. The P position represents parity. An odd number of jumpers must be installed, so if the UCN node address is 3, jumpers must be installed in positions 1, 2, and P. If there are redundant HPMMs, both HPMMs will have the same UCN node address.

3.12 Optional Backup Battery Assembly

Introduction

The backup battery assembly that is used to power the 24 Vdc Power Supply Module(s) in a standard Power System consists of a 48 Vdc lead acid battery pack that provides backup power for approximately 20 minutes with a full charge in the normal ambient temperature range. The Power Supply Module charging circuit is active whenever ac power is applied to the Power Supply Module(s). At least one Power Supply Module must be present to allow the battery to supply power to the Power System.

48 volt battery switch

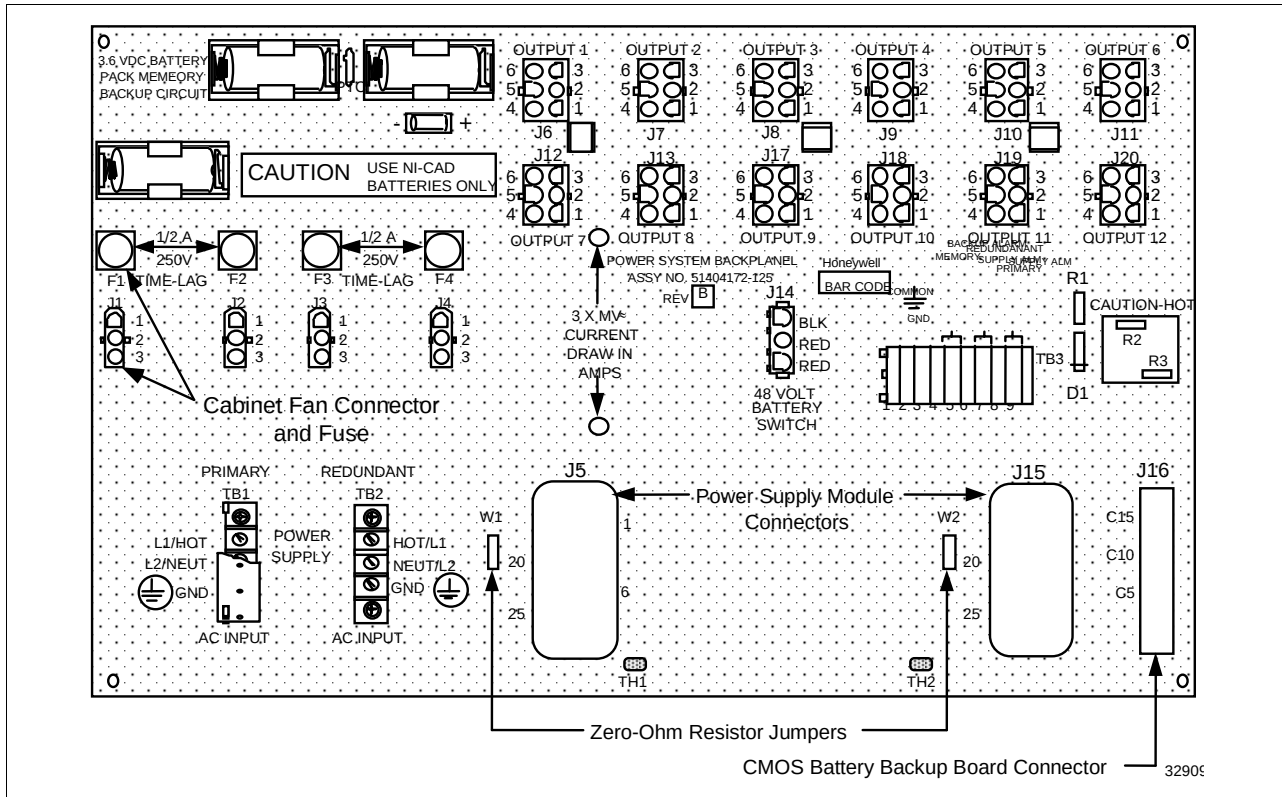
A 48 volt battery switch that is located at the left side of the standard Power System chassis provides a means of preventing the Power Supply Module(s) from discharging the backup battery when ac power is removed from one or both Power Supply Modules. Placing the switch in the off position disconnects the backup battery from the Power Supply Module(s).

No backup battery

When there is no backup battery, a failure is reported unless the zero-ohm resistor jumpers, W1 and W2, that are located next to the Power Supply Module connectors, are cut. This is done at the factory before shipping. If a backup battery assembly is installed on site, the connections must be restored. Figure 3-17 is a view of the Standard Power System backpanel.

3.12 Optional Backup Battery Assembly, Continued

Figure 3-17 Standard Power System Backpanel



Mounting the assembly

The backup battery assembly attaches to four mounting studs beneath the front of the enclosure between the RHC enclosure's mounting legs. Cabling from the Power System passes through two NEMA 4X sealed fittings in the enclosure that ensure atmospheric integrity.

3.12 Optional Backup Battery Assembly, Continued

Installation procedure Use the procedure in Table 3-4 to install the optional model MU-backup battery assembly kit. Refer to Figure 3-18, which is an exploded view of the backup battery assembly, during the installation.

Table 3-4 Backup Battery Assembly Installation Procedure

Step	Action
1	Place the Power System's BATTERY switch in the OFF position.
2	Except for the NEMA 4X fittings' retention nuts and the cable's connectors, the battery extension cable is preassembled. Route the 15-inch end of the cable that has male pins attached to the wire leads through the two openings in the side of the battery tray.
3	Slip a retention nut over each wire lead inside the battery tray and tighten the nuts.
4	Noting the 51190722-102 connector's orientation, which is shown in Figure 3-19, attach the connector to the wire leads.
5	Mount the battery tray assembly on the four mounting studs beneath the RHC enclosure and secure it in place with four lockwashers (51198912-007) and hex-nuts (51197575-007).
6	Route the 36 inch-end of the cable that has female pins attached to the wire leads through the two openings in the bottom of the RHC enclosure. Figure 3-20 shows the location of the two access holes.
7	Slip a retention nut over each wire lead inside the RHC enclosure and tighten the nuts.
8	Noting the 51190722-202 connector's orientation, which is shown in Figure 3-19, attach the connector to the wire leads.
9	Connect the battery extension cable connector to its mating connector at the side of the Power System.
10	Place the battery pack (51303948-100) partially on the battery tray with its connector at the back of the tray. Connect the battery pack's connector to the battery extension cable and slide the battery pack backward until it makes contact with the hold-down bracket at the rear of the tray.
11	Secure the battery pack in place with two lockwashers (51197521-006) and screws (51197520-608) at the front of the tray.

3.12 Optional Backup Battery Assembly, Continued

Installation procedure, continued

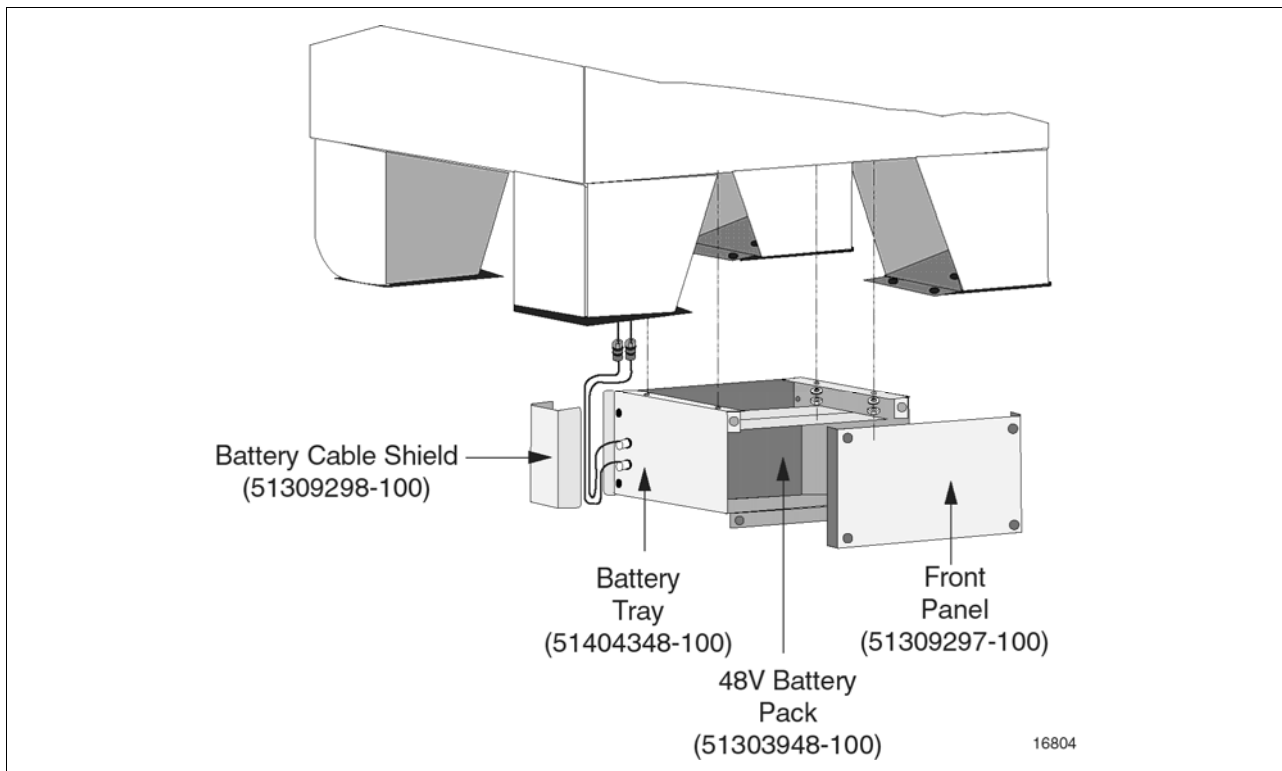
Table 3-4 Backup Battery Assembly Installation Procedure,
Continued

Step	Action
12	Place the cable shield (51309298-100) over the battery extension cable and secure it in place with two 5 mm x 10 mm cross-head screws that are included in the kit.
13	Attach the battery tray's front panel with four 5 mm x 10 mm cross-head screws that are included in the kit.

Backup battery mounting kit illustration

Figure 3-18 shows an exploded view of the model MU-RBMK01 backup battery mounting kit.

Figure 3-18 Exploded View of Model MU-RBMK01 Backup Battery Mounting Kit

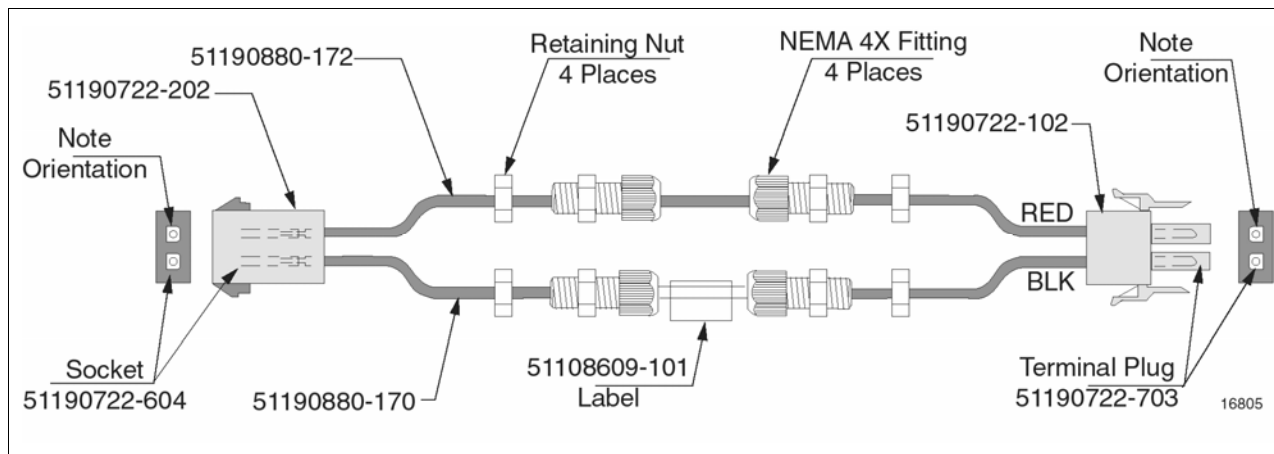


3.12 Optional Backup Battery Assembly, Continued

Battery extension cable assembly

Figure 3-19 illustrates the assembly of the battery extension cable.

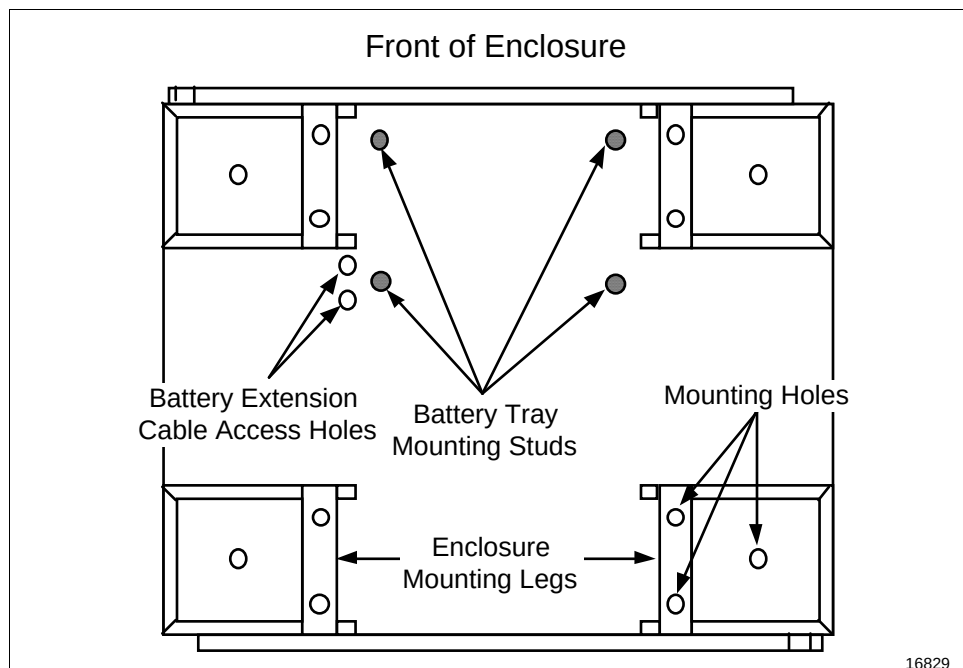
Figure 3-19 Battery Extension Cable Assembly



Bottom view of RHC enclosure

Figure 3-20 illustrates the battery extension cable openings in the bottom of the RHC enclosure.

Figure 3-20 Bottom View of RHC Enclosure



3.13 Optional Heater Assembly

Introduction

The optional thermostatically controlled electric heater has an operating temperature range between -18°C and 38°C (0°F and 100°F). The operating temperature is set by adjusting the rheostat at the front of the heater. The recommended setting for the rheostat is 30°C.

Heater kit

The heater is included in a kit. The model number of the kit is MU-RCHT01 for 120 Vac operation and model MU-RCHT02 for 240 Vac operation.

Table 3-5 is a list of the parts that are included in the kits.

Table 3-5 Model MU-RCHT01/MU-RCHT02 Heater Kits Parts List

Item Number	Part Number	Description	MU-RCHT01 Quantity	MU-RCHT02 Quantity
1	51204192-100	120 Vac Heater	1	0
2	51204192-200	240 Vac Heater	0	1
3	51192619-100	Thermal Switch	1	1
4	51192620-100	Female Terminal Block	1	1
5	51192621-100	Male Terminal Block	1	1
6	51108385-312	M3 x 12 mm Pan-Head Machine Screw	2	2
7	51108385-320	M3 x 20 mm Pan-Head Machine Screw	4	4
8	51108392-003	M3 Split Lockwasher	6	6
9	51190879-008	Cable Anchor	1	1
10	51190879-010	Cable Tie	1	1
11	51303378-115	WARNING Label	1	1
12	51198958-100	120/240 Vac Label	1	1
13	51198919-100	White Jumper Wire Assembly	1	1
14	51198919-101	Green Jumper Wire Assembly	1	1
15	51198919-102	Black Jumper Wire Assembly	1	1
16	51198919-103	Black Jumper Wire Assembly	1	1

3.13 Optional Heater Assembly, Continued

CAUTION—Power for the heater must be provided by a circuit breaker that is separate from the circuit breaker that provides power to the RHC Power System. Use a separate 15 amp circuit breaker for both 120 Vac and 240 Vac operation and 14-gauge wire for the connections.

Heater assembly installation procedure

Use the procedure in Table 3-6 to install the optional heater assembly. The item numbers that are mentioned in **bold** letters in the instructions refer to the parts that are listed in Table 3-5.

Table 3-6 Heater Kit Installation Procedure

Step	Action
1	Remove ac power from the enclosure. Provide a separate 120 Vac or 240 Vac power circuit breaker for the heater.
2	Use the four #10-32 pan-head screws that are included with the heater (51204192-100), item 1 or 2 , to secure the heater to the mounting panel which is located near the bottom front side of the enclosure. Note the orientation of the assembly in Figure 3-21. The locations of the mounting holes in the mounting panel are illustrated in Figure 3-11. Figure 3-12 illustrates the panel with the heater assembly attached.
3	Using two M3 x 12 mm pan-head machine screws (51108385-312), item 6 , and M3 split lockwashers (51108392-003), item 8 , mount the thermal switch (51192619-100), item 3 , to the right of the heater as illustrated in Figures 3-11 and 3-21.
4	Using two M3 x 20 mm pan-head machine screws (51108385-320), item 7 , and M3 split lockwashers (51108392-003), item 8 , mount the female terminal block (51192620-100), item 4 , to the left of the fan assembly as illustrated in Figures 3-11 and 3-21.
5	Slip the male terminal block (51192621-100), item 5 , into the female terminal block and attach it to the mounting plate with two M3 x 20 mm pan-head machine screws (51108385-320), item 7 , and M3 split lockwashers (51108392-003), item 8 .
6	Mount the self-adhesive cable anchor (51190879-008), item 9 , to the left of the fan assembly as illustrated in Figures 3-11 and 3-21. Center the anchor over the hole in the panel.
7	Mount the WARNING label (51303378-115), item 11 , above the thermal switch as illustrated in Figure 3-21.

3.13 Optional Heater Assembly, Continued

Heater assembly
installation
procedure, continued

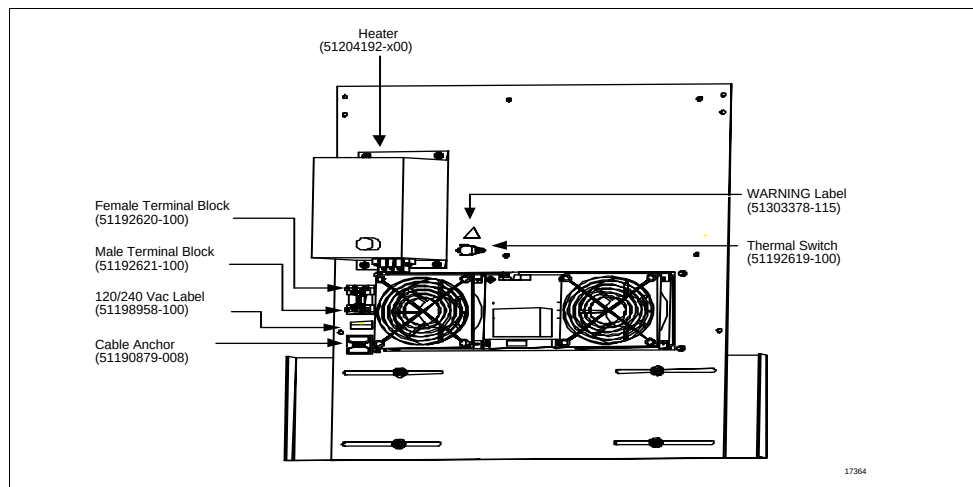
Table 3-6 Heater Kit Installation Procedure, Continued

Step	Action																				
8	Mount the 120/240 Vac label (51198958-100), item 12 , below the male terminal block as illustrated in Figure 3-21.																				
9	There are four wire jumpers that must be installed. The heater requires wire jumpers with a terminal lug and the thermal switch requires wire jumpers with a FASTON (spade) terminal. The terminal block has compression screws that accept wire jumpers that have a stripped end. Install the jumpers as follows. <table><tr><th><u>Jumper</u></th><th><u>Color</u></th><th><u>From</u></th><th><u>To</u></th></tr><tr><td>51198919-102</td><td>Black</td><td>Terminal Block-L1/HOT</td><td>Thermal Switch</td></tr><tr><td>51198919-103</td><td>Black</td><td>Thermal Switch</td><td>Heater-L1</td></tr><tr><td>51198919-100</td><td>White</td><td>Terminal Block-L2/NEUT</td><td>Heater-NEUT</td></tr><tr><td>51198919-101</td><td>Green</td><td>Terminal Block-GND</td><td>Heater-GND</td></tr></table>	<u>Jumper</u>	<u>Color</u>	<u>From</u>	<u>To</u>	51198919-102	Black	Terminal Block-L1/HOT	Thermal Switch	51198919-103	Black	Thermal Switch	Heater-L1	51198919-100	White	Terminal Block-L2/NEUT	Heater-NEUT	51198919-101	Green	Terminal Block-GND	Heater-GND
<u>Jumper</u>	<u>Color</u>	<u>From</u>	<u>To</u>																		
51198919-102	Black	Terminal Block-L1/HOT	Thermal Switch																		
51198919-103	Black	Thermal Switch	Heater-L1																		
51198919-100	White	Terminal Block-L2/NEUT	Heater-NEUT																		
51198919-101	Green	Terminal Block-GND	Heater-GND																		
10	The ac power source connections are made at the male terminal block. The terminal block compression terminals accept stripped wire ends.																				
11	Use a cable tie (51190879-010) to fasten the ac power cable to the cable anchor.																				

Heater kit parts
pictorial

Figure 3-21 is a pictorial of the heater kit parts pictorial.

Figure 3-21 Heater Kit Parts Pictorial



3.14 Enclosure Integrity

Enclosure security

At the completion of the installation, check all the openings made in the enclosures for conduit or cabling to ensure that they meet NEMA 4X specifications.

Section 4 – Service

4.1 Overview

Introduction

Use the following procedures if it becomes necessary to replace or add hardware to the enclosure.

Other than the fan assembly, heater assembly, and battery backup assembly, none of the electrical assemblies in the Remote Hardened Controller enclosure are unique. All other electrical assemblies are used extensively in HPM cabinets. Some mechanical assemblies may be different, but they normally do not require replacement.

HPMM cards, IOPs, FTAs, I/O Link Interface Extender cards, Fiber Optic Coupler modules, and UCNE modules are easily replaced. The servicing and replacement procedures for these assemblies are found in the *High-Performance Process Manager Service*, *High-Performance Process Manager Checkout*, and *Universal Control Network Installation* manuals. Service information for the unique assemblies, which were previously mentioned, is included in this section.

Be sure the inside of the enclosure is protected from adverse weather conditions while servicing or replacing any assemblies.

4.2 Fan Assembly Replacement

Introduction

Fan assemblies (51404376-150/120 Vac or 51404376-250/240 Vac) are located at both the top and bottom of the enclosure. They are not repairable in the field so the complete assembly must be replaced when a failure occurs.

The upper fan assembly circulates air from the front to the rear of the enclosure. The lower fan assembly circulates air from the rear to the front of the enclosure.

Power source

Fused 0.5 A, 120 Vac, or 240 Vac power to the fan assembly is provided by a 3-pin connector on the standard or AC Only Power System's backpanel.

Fan assembly description

The fan assembly has an LED indicator that lights when both fans are operating normally, and an alarm signal, represented by normally open contacts, that is available at the terminal connector TB1. The contacts are closed when the fan assembly is operating normally. The operation of the fan assembly can be monitored by an FTA, such as a Digital Input FTA, by providing a 24 Vdc signal source from the FTA and returning it to the FTA through the alarm contacts.

Replacement procedure

Use the procedure in Table 4-1 to replace a fan assembly.

Table 4-1 Fan Assembly Replacement Procedure

Step	Action
1	Disconnect the fan assembly's ac power cable (51109516-100) from J1 on the fan assembly.
2	Disconnect the fan assembly's signal cable's connector from TB1 on the fan assembly, if present.
3	Loosen and remove the four screws that secure the fan assembly. Remove the fan assembly from the enclosure. Note which side of the assembly faces the front or rear of the enclosure.
4	Install the replacement fan assembly by reversing the procedure in Steps 1 through 3, being sure that the assembly faces the correct direction (front or rear).

4.3 Backup Battery Replacement

CAUTION—Batteries may contain toxic materials that are not biodegradable. Batteries should be disposed of safely in accordance with local laws and regulations.

Introduction

The lead acid 48 volt backup battery provides approximately 20 minutes of standby power for the HPM Power System and has a normal standby service life of 4 to 5 years in a controlled environment. However, the amount of usage and high environmental temperatures will reduce its service life. For this reason, environmental temperature, battery storage time, and battery usage should be monitored and recorded. In the absence of these requirements, it is recommended that the battery performance be checked on a yearly basis, or be replaced every 3 years to ensure proper performance.

Battery performance

All batteries with the same capacity, rating, and specifications do not provide equivalent performance or reliability. For optimum system backup performance, use only Honeywell recommended and approved batteries.

4.3 Backup Battery Replacement, Continued

48 volt battery pack service life

The battery end-of-life is defined as the time when the battery capacity, meaning its ability to accept and hold a charge, has decayed to 60% of the original capacity. There are several factors that determine useful service life. The standby service life is directly reduced by events other than a continual 20°C standby condition and is defined as follows.

- Standby (float) service life means that the battery is connected to a charger and only occasionally discharged. The Honeywell 48 volt battery is designed to operate in standby (float) service for 4 to 5 years at 20°C. The service life of a battery can be shortened by an improper charging voltage. The charging voltage from the Power Supply Modules should be checked every 6 months for the proper voltage level.
- As shipped from Honeywell, the battery is fully charged. However, if a system is stored in a nonpowered state, the battery will self-discharge and can become unusable. The time required for this to happen is a function of storage temperature.

0° to 20°C	12 months
21° to 30°C	9 months
31° to 40°C	5 months
41° to 50°C	2.5 months
- The battery should not be left in a discharged condition for extended periods of time, as this strongly affects the ability of the battery to accept a charge thereafter. The battery should be charged periodically if it is stored or does not have power applied to it.
- The HPM Power System provides proper charging current to the battery while the subsystem is powered up and running. Whenever ac power is removed from the HPM, the Power System automatically draws power from the batteries to maintain the subsystem's 24 Vdc power. Anytime the system is intentionally shutdown and left without power, the **BATTERY** switch at the front of the Power System should be placed in the **OFF** position to prevent discharge of the batteries.
- The service life of the battery is reduced by high ambient temperatures. Heat will destroy a battery. While the operating temperature range of the battery is -15°C to 50°C, maximum service life will be realized when the batteries are operated with an ambient temperature of 20°C. "Brief usage," that is, a few days, at higher temperatures will have no adverse effect on the service life. However, if such usage is over extended periods of time or occurs repetitively, the service life will be reduced. This reduction may be as much as 20% of service life (1 year) for each 10°C above the reference ambient temperature of 20°C.

4.3 Backup Battery Replacement, Continued

48 volt battery pack service life, continued

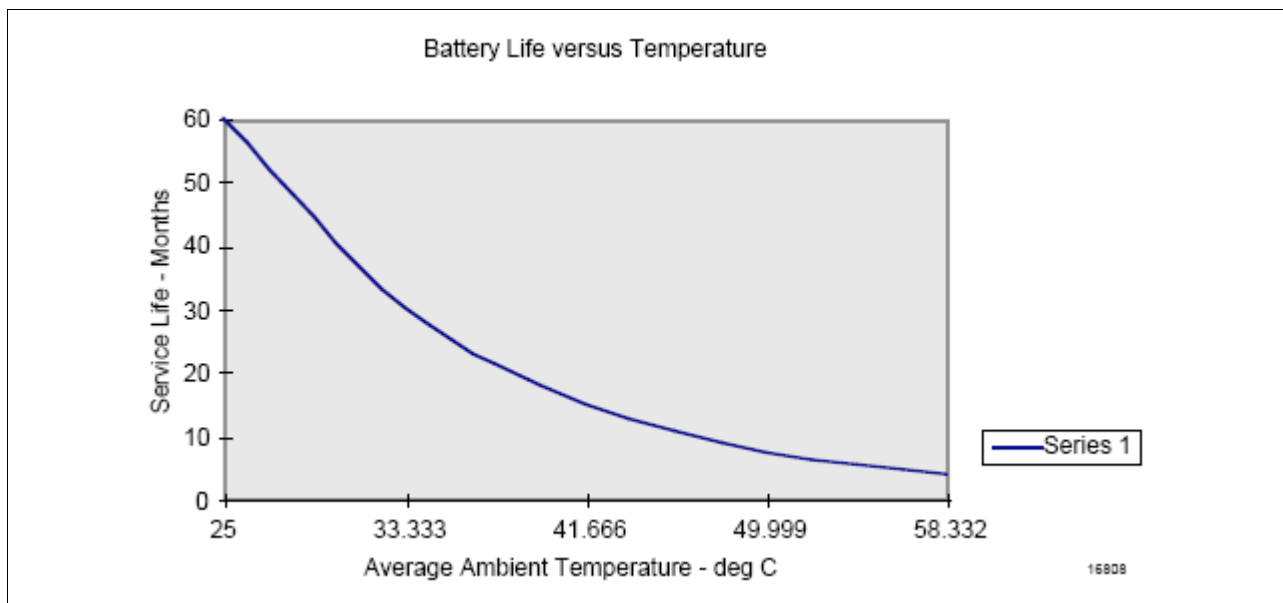
- Battery service life is reduced by the number of discharges and the depth of each discharge. The battery can provide 200 cycles at 100% depth of discharge, 550 cycles at 50% depth of discharge, 1200 cycles at 30% depth of discharge, or 2000 cycles at 10% depth of discharge. This effect is superimposed on the standby aging effect previously noted.

With an expected 5 year life, 200 cycles would represent one power outage approximately every 9 days, or approximately 2000 ampere-hours of 24 Vdc subsystem backup.

Battery life for average ambient temperatures

Figure 4-1 is a graph of the expected battery life for average ambient temperatures.

Figure 4-1 Expected Battery Life for Average Ambient Temperature



4.3 Backup Battery Replacement, Continued

Battery charge voltage check procedure

Use the procedure in Table 4-2 to verify that the output voltage from the Power System to the 48 volt battery is proper.

Table 4-2 Battery Charge Rate Verification Procedure

Step	Action
1	Locate the test points marked " COMMON " (-) and " BATTERY " (+) at the front of the primary Power Supply Module and connect a dc voltmeter with an accuracy of +/- 0.05 volts across the test points.
2	The voltage should read between 54.0 volts minimum to 55.2 volts maximum at 20°C. For each degree the battery enclosure temperature deviates from 20°C, the following adjustments in Power Supply Module's battery charge output voltage are applicable. Above 20°C, the battery charge voltage limits are decreased by 72 mv/deg C. Below 20°C, the battery charge voltage limits are increased by 72 mv/deg C. If the charge voltage is outside of these limits, replace the module.
3	This step and Step 5 should be performed during plant shutdowns since it requires disconnecting the battery and placing the redundant Power Supply Module's power switch in the OFF position. Disconnect the battery. With the voltmeter connected to the primary Power Supply Module's battery test points, place the redundant Power Supply Module's power switch in the OFF position. The voltage should not fall below the minimum voltage specified in Step 3. If it does, replace the redundant Power Supply Module.
4	Place the redundant Power Supply Module's power switch in the ON position. Move the test leads to the battery test points on the redundant Power Supply Module and place the primary Power Supply Module's power switch in the OFF position. The voltage should not fall below the minimum voltage specified in Step 3. If it does, replace the redundant Power Supply Module.
5	Reconnect the battery pack to the Power System.

4.3 Backup Battery Replacement, Continued

End of battery life check

Use the procedure in Table 4-3 to check the 48 volt backup battery pack for its end of life.

Table 4-3 Battery End of Life Check Procedure

Step	Action
1	Set the BATTERY switch on the Power System in the OFF position. Remove the Power System's lower cover. You can use either a screwdriver or a coin to remove the two screws that are holding it in place.
2	Disconnect the battery from the Power System.
3	Attach two pairs of leads to the battery terminals. One pair of leads are for connecting a load resistor across the battery's terminals and must be a minimum of 1.0 mm ² (18 AWG) wire. The second pair of leads are for connecting a voltmeter and can be a smaller gauge.
4	Select a voltmeter range of 60 Vdc for monitoring the battery voltage. Attach the voltmeter to the smaller gauge leads.
5	The battery load for the test must be 5 ohms and rated at 500 watts. Five 1-ohm, 100-watt resistors placed in series is recommended.
6	Attach the resistors to the larger test leads. Record the time that they are attached.
7	The battery voltage must remain above 40 Vdc for a minimum of 20 minutes. If the minimum battery voltage is not maintained, the battery is considered at the end of its life and must be replaced.
8	All batteries of the same capacity, rating, and specifications, do not provide equivalent performance or reliability. For optimum battery backup performance, use only Honeywell recommended and approved batteries.

4.3 Backup Battery Replacement, Continued

48 V battery pack replacement procedure

Use the procedure in Table 4-4 to replace the 48 volt battery pack.
Table 4-4 48 Volt Battery Pack Replacement Procedure

Step	Action
1	Place the BATTERY switch on the Power System in the OFF position.
2	Remove the backup battery enclosure's front cover.
3	Loosen and remove the two screws that secure the battery pack in the backup battery enclosure.
4	Move the battery pack toward you until you have access to its power connector. Disconnect the battery pack from the battery extension cable.
5	Move the battery extension cable out of the way so that it will not be damaged while the battery pack is being replaced.
6	The battery is heavy. Carefully slide the battery toward you and remove it from its assembly.
7	Slide the replacement battery pack partially into the backup battery enclosure while being careful not to damage the battery extension cable. Connect the battery pack to the battery extension cable.
8	Secure the battery in place by replacing the two screws that were previously removed.
9	Reinstall the backup battery enclosure's front cover.
10	Apply power to the battery pack by placing the BATTERY switch on the Power System in the ON position. Unless it is already fully charged, the battery may need to be initially charged for 16 hours before it will be capable of providing 20 minutes of standby power for the High-Performance Process Manager.

4.4 Electrical Heater Assembly Replacement

Introduction

The optional thermostatically controlled electric heater has an operating temperature range between -18°C and 38°C (0°F and 100°F). The desired operating temperature is set by adjusting the rheostat on the front of the heater.

Heater assembly replacement procedure

Use the procedure in Table 4-5 to replace the optional electrical heater assembly.

Table 4-5 Heater Assembly Replacement Procedure

Step	Action
1	Remove ac power from the enclosure.
2	Disconnect the power cable from the terminal strip on the heater assembly. Be sure that the cable leads are properly labeled so that they can be properly connected to the replacement heater assembly.
3	Four #10-32 screws are used to secure the heater to a mounting plate which is located near bottom front side of the enclosure. Loosen and remove the four screws. Remove the heater assembly from the enclosure.
4	To install the replacement heater assembly, reverse the procedure in Steps 1 through 3.

4.5 Preventative Maintenance

CAUTION—Batteries may contain toxic materials that are not biodegradable. Batteries should be disposed of safely in accordance with local laws and regulations.

NiCad batteries

The three C-size NiCad batteries that are installed in the standard Power System that provide standby power for CMOS memory have a tendency to age over a period of time. Aging occurs even if the batteries have been regularly recharged throughout their lifetime. For this reason, the NiCad batteries should be replaced after every two years of operation.

Be sure that you observe the correct battery polarity when replacing the batteries. The battery holders indicate the correct polarity.

Alkaline batteries

The three AA-size alkaline batteries that are installed in the AC Only Power System provide standby power for CMOS memory. They are not rechargeable and must be replaced periodically. The period should not exceed 6 months, provided there is no sustained loss of power to the CMOS Memory. If there is, replace the batteries after dc power is restored to the enclosure.

Be sure that you observe the correct battery polarity when replacing the batteries. The battery holder indicates the correct polarity.

Backup battery pack

The standard Power System's 48 volt backup battery pack should be tested periodically and replaced as necessary. Extremely high ambient operating temperatures will drastically shorten the battery pack's life.

Refer to subsection 4.3 for service life information and testing procedures.

Section 5 – Spare Parts

5.1 Overview

Introduction

This section contains an abbreviated list of the Optimum Replaceable Units (ORU) and periodic maintenance replacement parts for the Remote Hardened Controller enclosure. The parts are listed by their Honeywell part number and where applicable, by their model number. Part numbers and model numbers for Field Termination Assemblies (FTAs), IOPs, I/O Link Interface Extenders, FTA to IOP cables, DC Power cables, and I/O Link Interface cables are listed because they can be added to the enclosure after the initial installation.

For additional or optional equipment part numbers that are not listed in the table, contact your Honeywell Sales Representative or reference the *High-Performance Process Manager Service* manual.

5.2 Periodic Maintenance Parts

Periodic maintenance parts list Table 5-1 lists components that may be required during the life of the Remote Hardened Controller enclosure complex.

Table 5-1 Periodic Maintenance Parts List

Name	Description		Part Number
BATTERIES			
Alkaline Battery	One AA-size Alkaline battery		51190467-100
NiCad Rechargeable Batteries	One C-size NiCad battery		51192060-100
FUSES			
Assembly	Quantity	Description	
Left 7-Slot Card File (51402594-100 or -200)	8	2.0 A Medium Time-delay fuse (F1-8)	30754661-021
Right 7-Slot Card File (51401629)	8	2.0 A Subminiature fuse (F1-8)	30754661-021
15-Slot Card File (51401629)	16	2.0 A Subminiature fuse (F1-16)	30754661-021
3-30 Vdc SS DO FTA (51304423)	16	5.0 A Fast-action fuse 5 x 20 mm	51190582-250
3-30 Vdc SS DO FTA (51304650)	16	2.5 A Time-delay fuse 5 x 20 mm	51190584-225
3-30 Vdc SS DO FTA (51309153)	16	2.5 A Time-delay fuse 5 x 20 mm	51190584-225
31-200 Vdc SS DO FTA (51304428)	16	1.0 A Time-delay fuse 5 x 20 mm	51190584-210
31-200 Vdc SS DO FTA (51309154)	16	1.0 A Time-delay fuse 5 x 20 mm	51190584-210
24-240 Vac SS DO FTA (51304408)	16	3.0 A Time-delay fuse 5 x 20 mm	51190584-230
120/240 Vac SS DO FTA (51304648)	16	2.5 A Time-delay fuse 5 x 20 mm	51190584-225
120 Vac Relay DO FTA (51304443)	16	6.0 A Time-delay fuse 5 x 20 mm	51190584-260
120 Vac Relay DO FTA (51309148)	16	6.0 A Time-delay fuse 5 x 20 mm	51190584-260

5.2 Periodic Maintenance Parts, Continued

Periodic maintenance parts list, continued

Table 5-1 Periodic Maintenance Parts List, Continued

Name	Description		Part Number
FUSES, continued			
<u>Assembly</u>	<u>Quantity</u>	<u>Description</u>	
240 Vac Relay DO FTA (51304427)	16	3.0 A Time-delay fuse 5 x 20 mm	51190584-230
240 Vac Relay DO FTA (51309150)	16	3.0 A Time-delay fuse 5 x 20 mm	51190584-230
24 Vdc Nonisolated DO FTA (51304446)	16	0.2 A Fast-action fuse 5 x 20 mm (F1-16)	51190582-120
	1	2.0 A Fast-action fuse 5 x 20 mm (F17)	51190582-220
24 Vdc Isolated DO FTA (80364013)	4	1.6 A Fast-action fuse MP16 (F1-4)	83957061-004
24 Vdc Isolated DO FTA (80366183)	16	1.6 A Fast-action fuse MP16 (F1-16)	83957061-004
DI FTA (80364010, 80366180)	2	0.5 A Fast-action fuse MP05 (F1, F3)	83957061-002
DI FTA (80364010, 80366180)	1	1.0 A Fast-action fuse MP10 (F2)	83957061-003
LLAI FTA (51304437)	4	0.5 A Subminiature fuse (F1-4)	30754661-014
DI Power Distribution Assy (51304425)	12	1.0 A Fast-action fuse 5 x 20 mm (F1-12)	51190582-210
SI/SDI FTA (51303932)	2	0.25 A Subminiature fuse (F1-2)	30754661-011
Nonredundant AO FTA (51304476)	2	1.0 A Fast-action fuse 5 x 20 mm (F1-2)	51190582-210
Redundant AO FTA (51304335)	3	1.0 A Fast-action fuse 5 x 20 mm (F1-3)	51190582-210
Redundant AO FTA (80364007, 80366177, 80366481, 80366484)	1	1.0 A Fast-action fuse MP10 (F1)	83957061-003

5.2 Periodic Maintenance Parts, Continued

Periodic maintenance parts list, continued

Table 5-1 Periodic Maintenance Parts List, Continued

Name	Description		Part Number
FUSES, continued			
<u>Assembly</u>	<u>Quantity</u>	<u>Description</u>	
Pulse Input FTA (51304084)	8	0.25 A Fast-action fuse 5 x 20 mm (F1-8)	51190582-125
Galvanic AO FTA (51304638)	3	1.0 A Subminiature fuse (F1-3)	30754661-019
High-Perform I/O Link Card	1	3.0 A Fast-action fuse (F1)	30754661-022
Fuseholder Puller		Fuseholder extractor	51190586-100

5.3 Optimum Replaceable Units

ORU parts Lists

This subsection identifies the assemblies that may be required for maintenance during the life of the Remote Hardened I/O or FTA enclosure. Maintenance procedures are found in this manual. Additional information can be found in the *Process Manager/Advanced Process Manager Installation* and *Process Manager I/O Installation* manuals.

When applicable, all replacement assemblies must be corrosion protected.

5.3 Optimum Replaceable Units, Continued

Cables and corrosion protected ORU parts list

Table 5-2 is a parts list of cables and corrosion protected ORU assemblies.

Table 5-2 Cables and Corrosion Protected ORU Parts List

Name	Description	Part Number
CABLES		
Power Cable – Nonshielded (For internal cabinet use)	Power System to card file cable – Set of two (With surge protection network for I/O Link Interface cable)	
	1 meter	51204126-001
	1.5 meters	51204126-915
	2 meters	51204126-002
	3 meters	51204126-003
	4 meters	51204126-004
	5 meters	51204126-005
	6 meters	51204126-006
Fan Assembly AC Power Cable	Power System to Fan Assembly	51109516-100
Power Adapter Cable	Power Adapter to LLMux/SDI/SI FTA cable (External to the cabinet)	
	Power Adapter to RHMUX FTA (Indoor cable)	
	MU-KLX076 76 meters	51190696-076
	MU-KLX152 152 meters	51190696-152
	MU-KLX305 305 meters	51190696-305
Power Adapter Cable	Power Adapter to RHMUX FTA (Outdoor cable)	
	MU-KLO305	51197597-305

5.3 Optimum Replaceable Units, Continued

Cables and corrosion protected ORU parts list

Table 5-2 Cables and Corrosion Protected ORU Parts List

Name	Description	Part Number
CABLES, continued		
Power Adapter Cable	Power Adapter to LLMux/SDI/SI FTA cable (Internal to the cabinet)	
	30 centimeters	51304465-100
	66 centimeters	51304465-200
	100 centimeters	51304465-300
	200 centimeters	51304465-400
	300 centimeters	51304465-500
Galvanic Isolation FTA Power Cable (For internal cabinet use)	GI Power Distribution assembly to FTA cable	
	0.50 meter	51109620-020
	0.75 meter	51109620-030
	1 meter	51109620-040
	1.25 meters	51109620-050
	2 meters	51109620-080
	3 meters	51109620-120
Bridge Cable	Bridge cable for model MC-TDOY23 and MC-TDOY63 FTAs	
	1 meter	80366198-100
	2 meters	80366198-200
I/O Link Interface Cable (Set of two)	HPMM to IOP card file communication cable	
	Two drop in cabinet	51204042-100
	Three drop in cabinet	51204042-200
	Four drop in cabinet	51204042-300
	Five drop in cabinet	51204042-400
	Six drop in cabinet	51204042-500
HPMM Redundancy Cable	HPMM redundancy interconnect cable (2 m)	51201667-100
IOP to FTA Cable	IOP to FTA cable	
	1 meter	51201420-001
	1.5 meters	51201420-915
	MU-KFTA02 2 meters	51201420-002
	MU-KFTA03 3 meters	51201420-003
	MU-KFTA04 4 meters	51201420-004
	MU-KFTA05 5 meters	51201420-005

5.3 Optimum Replaceable Units, Continued

Cables and corrosion protected ORU parts list, continued

Table 5-2 Cables and Corrosion Protected ORU Parts List, Continued

Name	Description	Part Number
POWER SYSTEMS		
Power Supply Module	Standard 20 A Power Supply Module	51109456-200
Power Supply Module	AC Only 8 A Power Supply Module	51190465-150
Power Supply Module	AC Only 16 A Power Supply Module	51190465-250
48 V Battery Backup	48 V battery assembly	51303948-100
Power System Backpanel Assy	HPM Power System backpanel	51404172-150
AC/DC Distribution Assembly	AC Only power control and distribution board 115 Vac dual PSMs	51401135-150
AC/DC Distribution Assembly	AC Only power control and distribution board 115 Vac single PSM	51401135-250
AC/DC Distribution Assembly	AC Only power control and distribution board 230 Vac dual PSMs	51401135-350
AC/DC Distribution Assembly	AC Only power control and distribution board 230 Vac single PSM	51401135-450
CMOS Battery Backup Assembly	HPM PS Charger/monitor for CMOS memory	51309206-150

5.3 Optimum Replaceable Units, Continued

Cables and corrosion protected ORU parts list, continued

Table 5-2 Cables and Corrosion Protected ORU Parts List, Continued

Name	Description	Part Number
CARD FILE ASSEMBLIES		
Left 7- Slot Card File (MC-HPFH01)	Left 7-Slot HPMM/IOP card file assy (slots 1-7)	51404125-150
Left 7- Slot Card File (MC-HPFH03)	Left 7-Slot HPMM card file assy (slots 1-7)	51404191-175
Left 7- Slot Card File (MC-HPFI03)	Left 7-Slot IOP card file assy (slots 1-7)	51404191-275
Left 7- Slot Card File Backpanel	Left 7-Slot card file backpanel assy (slots 1-7)	51401626-150
Right 7- Slot Card File (MC-HPFH11)	Right 7- Slot HPMM/IOP card file assy (9-15)	51404126-150
Right 7- Slot Card File (MC-HPFH13)	Right 7- Slot HPMM card file assy (slots 9-15)	51404192-175
Right 7- Slot Card File (MC-HPFI13)	Right 7- Slot IOP card file assy (slots 9-15)	51404192-275
Right 7- Slot Card File Backpanel	Right 7- Slot card file backpanel assy (slots 9-15)	51401629-150
15-Slot Card File (MC-HPFX02)	15-Slot HPMM/IOP card file assy (slots 1-15)	51404127-150
15-Slot Card File (MC-HPFX03)	15-Slot HPMM card file assy (slots 1-15)	51404193-175
15-Slot Card File (MC-HPFI23)	15-Slot IOP card file assy (slots 1-15)	51404193-275
15-Slot Card File Backpanel	15-Slot card file backpanel assy (slots 1-15)	51401632-150
IOP CARDS		51304489-150
HLAI IOP (MC-PAIH02)	High Level Analog Input card	51304754-150
HLAI IOP (MC-PAIH03)	High Level Analog Input card	51304481-150
LLAI IOP (MC-PAIL02)	Low Level Analog Input card	51304362-150
LLMux IOP (MC-PLAM02)	Low Level Multiplexer card	51404109-175
RHMUX IOP (MC-PLAM03)	Remote Hardened Low Level Multiplexer card	51304485-150
DI IOP (MC-PDIX02)	Digital Input card	80363972-150
DI IOP (MC-PDIY22)	Digital Input card	51304483-150
AO IOP (MC-PAOX02)	Analog Output card	51309152-175
AO IOP (MC-PAOX03)	Analog Output card	80363969-150
AO IOP (MC-PAOY22)	Analog Output card	51304487-150
DO IOP (MC-PDOX02)	Digital Output card	80363975-150
DO IOP (MC-PDOY22)	Digital Output card	51304516-250
STI IOP (MC-PSTX02)	Smart Transmitter Interface	51304281-250
STIM IOP (MC-PSTX03)	Smart Transmitter Interface Multivariable	51304386-150
PI IOP (MC-PPIX02)	Pulse Input card	51304362-250
SDI IOP (MC-PSDX02)	Serial Device Interface card	51304362-350
SI IOP (MC-PSIM11)	Serial Interface card	51402625-175
DISOE IOP (MC-PDIS12)	Digital Input Sequence Of Events card	51304260-250
Slot Filler Card	Slot filler card	

5.3 Optimum Replaceable Units, Continued

Cables and corrosion protected ORU parts list, continued

Table 5-2 Cables and Corrosion Protected ORU Parts List, Continued

Name	Description	Part Number
I/O LINK EXTENDER COMPONENTS		
Standard I/O Link Extender (MC-IOLM02)	Standard I/O Link Extender card Standard fiber optic Tx/Rx coupler (1) Standard fiber optic Tx/Rx coupler (3)	51304419-150 51201557-150 51201557-350
Long Distance I/O Link Extender (MC-ILDX03)	Long Distance I/O Link Extender card Long Distance fiber optic Tx/Rx coupler (1)	51304532-150 51309208-150
FTA COMPONENTS		
HLAI/STI FTA (MC-TAIH02)	HLAI/STI with compression terminals	51304453-150
HLAI/STI FTA (MC-TAIH12)	HLAI/STI with compression terminals	51304337-150
HLAI/STI FTA (MC-TAIH22)	HLAI/STI with compression terminals AI Adapter Module for the above FTA	80366195-150 80367163-001
HLAI/STI FTA (MC-TAIH52)	HLAI/STI with screw terminals	51304337-250
HLAI/STI FTA (MC-TAIH62)	HLAI/STI with screw terminals AI Adapter Module for the above FTA	80366192-150 80360713-001
HLAI FTA (MC-TAIH03)	HLAI with compression terminals	51309136-175
HLAI FTA (MC-TAIH13)	HLAI with compression terminals	51309138-175
HLAI FTA (MC-TAIH23)	HLAI with compression terminals AI Adapter Module for the above FTA	80369165-175 80367163-001
HLAI FTA (MC-TAIH53)	HLAI with screw terminals	51309138-275
STI FTA (MC-TSTX03)	STI with compression terminals	51309140-175
STI FTA (MC-TSTX13)	STI with compression terminals	51309142-175
STI FTA (MC-TSTX53)	STI with screw terminals	51309142-275
LLAI FTA (MC-TAIL03)	LLAI FTA with compression terminals	51309202-175
LLMux FTA (MC-TAMT03)	LLMux local TC with compression terminals	51309223-175
LLMux FTA (MC-TAMT12)	LLMux remote TC with compression terminals (Replaced by MC-TAMT13) LLMux Analog board for above FTA	51401573-150 51401564-100
LLMux FTA (MC-TAMT13)	LLMux remote sensor board for above FTA LLMux remote TC with compression terminals	51304589-150 51309213-175
LLMux FTA (MC-TAMR03)	LLMux RTD with compression terminals	51309218-125

5.3 Optimum Replaceable Units, Continued

Cables and corrosion protected ORU parts list, continued

Table 5-2 Cables and Corrosion Protected ORU Parts List, Continued

Name	Description	Part Number
FTA COMPONENTS, continued		
AO FTA (MC-TAOX02)	Analog Output with compression terminals	51304476-175
AO FTA (MC-TAOX12)	Analog Output with compression terminals	51304335-175
AO FTA (MC-TAOX52)	Analog Output with screw terminals	51304335-275
	Plug-in module for above FTAs	51304338-100
AO FTA (MC-TAOY22)	Analog Output with comp term, Stdby Manual	80366481-175
AO FTA (MC-TAOY23)	Analog Output with comp term, no Stdby Manual	80366177-250
AO FTA (MC-TAOY52)	Analog Output with screw term, Stdby Manual	80366484-175
AO FTA (MC-TAOY53)	Analog Output with screw term, no Stdby Manual	80364007-250
DI 120 Vac FTA (MC-TDIA12)	Digital Input 120 Vac with compression terminals	51304439-175
DI 120 Vac FTA (MC-TDIA52)	Digital Input 120 Vac with fixed-screw terminals	51304439-275
	Plug-in module for above FTAs	51190523-200
DI 120 Vac FTA (MC-TDIA72)	Digital Input 120 Vac with removable-screw term	51303930-150
DI 240 Vac FTA (MC-TDIA22)	Digital Input 240 Vac with compression terminals	51304431-175
DI 240 Vac FTA (MC-TDIA62)	Digital Input 240 Vac with fixed-screw terminals	51304431-275
	Plug-in module for above FTAs	51190523-300
DI 24 Vdc FTA (MC-TDID12)	Digital Input 24 Vdc with compression terminals	51304441-175
DI 24 Vdc FTA (MC-TDID52)	Digital Input 24 Vdc with fixed-screw terminals	51304441-275
	Plug-in module for above FTAs	51190523-100
DI 24 Vdc FTA (MC-TDID72)	Digital Input 24 Vdc with removable-screw term	51303928-150
DI 24 Vdc FTA (MC-TDIY22)	Digital Input 24 Vdc with compression terminals	80366180-175
DI 24 Vdc FTA (MC-TDIY62)	Digital Input 24 Vdc with screw terminals	80364010-175
DO 120/240 Vac FTA (MC-TDOA13)	DO 24-240 Vac SS with removable-comp term	51304648-175
DO 120/240 Vac FTA (MC-TDOA53)	DO 24-240 Vac SS with removable-screw term	51304648-275
	Solid-state relay plug-in module for above FTAs	51190516-332
DO 3-30 Vdc FTA (MC-TDOD13)	DO 3-30 Vdc SS with removable-comp terminals	51304650-150
DO 3-30 Vdc FTA (MC-TDOD53)	DO 3-30 Vdc SS with removable-screw terminals	51304650-250
	Solid-state relay plug-in module for above FTAs	51190516-132
DO 3-30 Vdc FTA (MC-TDOD14)	DO 3-30 Vdc SS with fixed-compression terminals	51309153-175
DO 3-30 Vdc FTA (MC-TDOD54)	DO 3-30 Vdc SS with fixed-screw terminals	51309153-275
	Solid-state relay plug-in module for above FTAs	51190516-134

5.3 Optimum Replaceable Units, Continued

**Cables and corrosion
protected ORU parts
list, continued**

Table 5-2 Cables and Corrosion Protected ORU Parts List, Continued

Name	Description	Part Number
FTA COMPONENTS, continued		
DO 31-200 Vac FTA (MC-TDOD22)	DO 3-30 Vdc SS with fixed-compression terminals	51304428-150
DO 31-200 Vac FTA (MC-TDOD62)	DO 3-30 Vdc SS with fixed-screw terminals	51304428-250
	Solid-state relay plug-in module for above FTAs	51190516-402
	DO 31-200 Vdc SS with compression terminals	51309154-175
	DO 31-200 Vdc SS with screw terminals	51309154-275
DO 31-200 Vac FTA (MC-TDOD23)	Solid-state relay plug-in module for above FTAs	51190516-404
DO 31-200 Vac FTA (MC-TDOD63)		51304446-150
	DO 24 Vdc 100 ma nonisolated with comp. term	51304446-250
	DO 24 Vdc 100 ma nonisolated with screw term	51191945-100
	Relay plug-in module for above assemblies	
DO 24 Vdc FTA (MC-TDON12)		51309148-175
DO 24 Vdc FTA (MC-TDON52)	DO 120 Vac/125 Vdc with compression terminals	51309148-275
	DO 120 Vac/125 Vdc with screw terminals	51191945-100
	Relay plug-in module for above assemblies	
DO 120 Vac Relay (MC-TDOR12)		51309150-175
DO 120 Vac Relay (MC-TDOR52)	DO 240 Vac/125 Vdc with compression terminals	51309150-275
	DO 240 Vac/125 Vdc with screw terminals	51191945-100
	Relay plug-in module for above assemblies	80366189-175
DO 240 Vac Relay (MC-TDOR22)	DO 240 Vac/125 Vdc with compression terminals	80366185-175
DO 240 Vac Relay (MC-TDOR62)	DO 240 Vac/125 Vdc with screw terminals	83970001-110
	Relay (NT1) for above assemblies	51190610-101
DO 240 Vac Relay (MC-TDOY23)	Jumper for above assemblies	
DO 240 Vac Relay (MC-TDOY63)		51304084-175
		51304084-275
		51201602-100
Pulse Input FTA (MC-TPIX12)		
Pulse Input FTA (MC-TPIX52)	Pulse Input with compression terminals	
	Pulse Input with screw terminals	
	Plug-in module for above FTAs	

5.3 Optimum Replaceable Units, Continued

Cables and corrosion protected ORU parts list, continued

Table 5-2 Cables and Corrosion Protected ORU Parts List, Continued

Name	Description	Part Number
FTA COMPONENTS, continued		
Serial Device Interface FTA (MC-TSDT02)	Serial Device Interface (Toledo 8142) FTA Plug-in module for above FTA	51303932-251 51303934-201
Serial Device Interface FTA (MC-TSDM02)	Serial Device Interface (Manual/Auto Station) Plug-in module for above FTA	51303932-252 51303934-202
Serial Device Interface FTA (MC-TSDU02)	Serial Device Interface (UDC 6000) Plug-in module for above FTA	51303932-253 51303934-203
Serial Interface (MC-TSIM12)	Serial Interface (Modbus) Plug-in module for above FTA	51303932-476 51303934-401
Serial Interface (MC-TSIA12)	Serial Interface (Allen-Bradley) Plug-in module for above FTA	51303932-478 51303934-403
Power Adapter (MC-TLPA02)	Power Adapter (used with LLMux, SDI, and SI)	51309204-175
DI Power Dist Assy (MC-TDPR02)	Digital Input Power Distribution Assembly	51304425-175
DO Standby Manual (MC-SMDC02)	Digital Output Standby Manual Device with case	51304526-150
DO Standby Manual (MC-SMDX02)	Digital Output Standby Manual Device with case and cable	51304527-150
AO Standby Manual (MC-SMAC02)	Analog Output Standby Manual Device Assy with cable	51401277-150
GALVANICALLY ISOLATED FTA COMPONENTS		
RHMUX FTA (MC-GAMT01)	RHMUX TC with screw terminals	51404106-175
RHMUX IS Power Adapter (MC-GLP01)	RHMUX IS Power Adapter with compression term	51304724-175
RHMUX NI Power Adapter (MC-TLPA01)	RHMUX NI Power Adapter with compression term	51304722-175
HLAI FTA (MC-GAIH13)	HLAI FTA with GI Modules (Compress terminals)	51304718-175
HLAI FTA (MC-GAIH83)	HLAI FTA with GI Modules (Crimp terminals)	51304718-375
	HLAI Galvanic Isolation Module (MTL4041B)	51191697-200
	HLAI FTA with GI Modules (Compress terminals)	51304730-175
	HLAI FTA with GI Modules (Crimp terminals)	51304730-375
	HLAI Galvanic Isolation Module (MTL4041P)	51191928-100
HLAI FTA (MC-GAIH14)	HLAI FTA with GI Modules (Compress terminals)	51304748-175
HLAI FTA (MC-GAIH84)	HLAI FTA with GI Modules (Crimp terminals)	51304748-375
	HLAI Galvanic Isolation Module (MTL4041B)	51191697-200
HLAI FTA (MC-GAIH22)	HLAI calibration tool	51201450-100
HLAI FTA (MC-GAIH92)		
HLAI Calibration Tool		

5.3 Optimum Replaceable Units, Continued

**Cables and corrosion
protected ORU parts
list, continued**

Table 5-2 Cables and Corrosion Protected ORU Parts List, Continued

Name	Description	Part Number
GALVANICALLY ISOLATED FTA COMPONENTS, continued		
24 Vdc DI FTA (MC-GDID12)	24 Vdc DI FTA with GI Modules (Compress Term)	51304640-175
24 Vdc DI FTA (MC-GDID82)	24 Vdc DI FTA with GI Modules (Crimp Terminals)	51304640-375
24 Vdc DI FTA (MC-GDID13)	24 Vdc DI Galvanic Isolation Module (MTL4016)	51191699-100
24 Vdc DI FTA (MC-GDID83)	24 Vdc DI FTA with GI Modules (Compress Term)	51304728-175
AO FTA (MC-GAOX02)	24 Vdc DI Galvanic Isolation Module (MTL4013)	51304728-375
AO FTA (MC-GAOX72)	24 Vdc DI FTA with GI Modules (Crimp Terminals)	51191926-100
AO FTA (MC-GAOX12)	24 Vdc DI Galvanic Isolation Module (MTL4013)	51304638-175
AO FTA (MC-GAOX82)	AO FTA with GI Modules (Comp Term)	51304638-375
	AO FTA with GI Modules (Crimp Term)	51304638-575
	AO FTA with GI Modules (Comp Term)	51304638-775
	AO FTA with GI Modules (Crimp Term)	51192097-200
DO FTA (MC-GDOD12)	AO Galvanic Isolation Module (MTL4045C)	51304338-100
DO FTA (MC-GDOD82)	Redundancy Module for FTA	
DO FTA (MC-GDOL12)	DO FTA with GI Modules (Compression Terminals)	51304642-175
DO FTA (MC-GDOL82)	DO FTA with GI Modules (Crimp Terminals)	51304642-375
LFD Combiner Panel (MC-GLFD02)	DO Galvanic Isolation Module (MTL4021)	51191698-100
Power Distribution Assembly (MC-GPRD02)	DO FTA with GI Modules (Compression Terminals)	51304736-175
Marshalling Panel (MC-GMAR52)	DO FTA with GI Modules (Crimp Terminals)	51304736-375
	DO Galvanic Isolation Module (MTL4023)	51191927-100
MISCELLANEOUS	Line Fault Detection Combiner Panel	51304732-175
Fan Assembly	Galvanic Isolation Power Distribution Assy	
Fan Assembly	Marshalling Panel	51304644-175
Heater Assembly (MU-RCHT01)		51309156-175
Heater Assembly (MU-RCHT02)		
	Fan assembly – 120 Vac	51404376-150
	Fan assembly – 240 Vac	51404376-250
	800-Watt heater assembly – 120 Vac	51204192-100
	800-Watt heater assembly – 240 Vac	51204192-200

5.4 UCN Parts List

UCN Parts list

The UCN spare parts are listed in Table 5-3 for easy reference. This same list of spare parts can also be found in the *High-Performance Process Manager Service* manual. Information about these parts can be found in the *High-Performance Process Manager Installation* and the *Universal Control Network (UCN) Planning* manuals.

Table 5-3 UCN Parts List

Name	Description	Part Number
UCN Drop Cable	RG6-type UCN drop cable (pair)	
	0.5 meter	51195153-900
	1.5 meters	51195153-901
	MU-NKD002	2.0 meters
		2.5 meters
		3.5 meters
		4.5 meters
	MU-NKD005	5.0 meters
	MU-NKD010	10.0 meters
	MU-NKD020	20.0 meters
	MU-NKD030	30.0 meters
	MU-NKD040	40.0 meters
	MU-NKD050	50.0 meters
		51195153-050
UCN Trunk Cable	RG11-type UCN trunk cable (pair) – internal application	
		51195199-002
	MU-NKT002	2.6 meters
	MU-NKT005	5.2 meters
	MU-NKT010	10.4 meters
	MU-NKT020	20.8 meters
	MU-NKT030	31.2 meters
	MU-NKT040	41.6 meters
	MU-NKT050	52.0 meters
	MU-NKT100	101.4 meters
	MU-NKT200	200.2 meters
	MU-NKT400	400.4 meters
	MU-NKT600	600.6 meters
		51195199-600
UCN Trunk Cable	UCN trunk cable RG11-type – external application	
		51191607-152
		51191607-305
		51191607-762
UCN Trunk Cable		152.4 meters
		304.9 meters
		762.4 meters
		51190899-xxx
	UCN trunk cable RG11-type – internal application	
		51190899-152
UCN Trunk Cable		152.4 meters
		304.9 meters
		51190899-762

5.4 UCN Parts List, Continued

UCN Parts list, continued

Table 5-3 UCN Parts List, Continued

Name	Description	Part Number
UCN Trunk Tap Kit	2-Drop tap assembly kit (pair) (MU-NTAP02)	51201394-100
	4-Drop tap assembly kit (pair) (MU-NTAP04)	51201395-100
	8-Drop tap assembly kit (pair) (MU-NTAP08)	51109485-100
UCN Tap Terminator	UCN tap terminator	51190782-100
Trunk Splice Kit	Splice kit for RG11-type cable (MU-NCSK01)	51195160-100
Trunk Cable Connector Kit	RG6 drop cable connect kit W/O tool (MU-NKDK01)	51195161-100
	RG11 trunk cable connect kit W/O tool (MU-NKTK01)	51195162-100
Connector Tool	RG6 drop cable connector tool (MU-NKDT01)	51190797-200
	RG11 trunk cable connector tool (MU-NKTT01)	51190795-200
Connector Torque Wrench	25 inch/pound torque wrench kit with three wrench heads for UCN connectors (MU-NKTQ01)	51109612-100

Index

- 15-Slot card file, 13
- 24 Vdc power cables, 43
 - model numbers, 43
- 48 V battery pack
 - charge voltage check procedure, 88
 - end of life check procedure, 89
 - replacement, 85
 - replacement procedure, 90
 - service life, 86
- 48 volt battery switch, 75
- 7-slot card files, 13
- AC Only Power System, 38, 58, 60
 - ac power connections, 58
 - alarm connections, 60
- AC Only Power System, 13
- AC Only Power System safety ground connections, 58
- AC Power Distribution
 - APM, 25, 26
 - Conduit, 24
 - PM, 25, 26
- alarm annunciator contacts, 59, 60
- alkaline battery, 13
- alkaline battery replacement, 92
- approval body certification, 1
- backup battery, 12
- backup battery assembly
 - installation, 75
 - installation procedure, 76
- backup battery assembly, 75
- basic enclosure, 2
- cable
 - fiber optic, 37
 - I/O Link Extender, 63
- cable entry, 8, 51
- cabling considerations, 50
- card file
 - 15-Slot, 13
 - 7-Slot, 13
 - combinations, 39
 - I/O Link address, 73
 - I/O Link Interface address selection, 73
 - pinning, 73, 74
 - UCN node address, 74
 - UCN node address selection, 74
- card file
 - selection, 39
- CE Compliance, 7
- certification, 1
- Circuit Breakers, 20
- conduit
 - recommended vendor, 46
 - size selection, 47
 - type recommended, 46
- conduit fittings, 46
- conduit installation, 55
- conduit selection, 46
- conformal coating, 1, 5
- Distribution Box Sizing, 24
- Distribution Wiring Guidelines, 21
- electrical heater, 47
- enclosure
 - basic, 2
 - clearance space, 15
 - complexing, 5
 - door grounding, 8
 - door integrity, 8
 - door security, 8
 - hardware options, 12
 - indoor/outdoor use, 8
 - interior hardware, 9
 - mounting procedure, 54
 - security, 82
 - size, 5
 - transport, 15
 - weather protection, 15
 - weight, 6
- enclosure complex, 50
 - two, 49
- enclosure complex thermal analysis, 49
- enclosure complexing, 48
- enclosure internal temperature formula, 48
- enclosure model numbers, 3
- enclosure mounting, 53
- enclosure mounting legs, 53
- entry fittings, 51
- entry holes, 51
- fan assembly, 61
 - alarm connections, 61
 - replacement procedure, 84
- fiber optic cable
 - installation, 45
- fiber optic cables, 37, 45, 52
- field servicing, 15
- FTA model numbers, 41
- FTA Mounting Channel model numbers, 42
- FTA Mounting Channel selection, 42
- FTA Mounting Channels, 12
- FTA selection, 41
- FTA sizes, 41
- FTA types, 41
- FTAs, 14
- Grounding Systems (with MRG)
 - PM/APM Cabinet, 34
 - PM/APM FTA Wiring Shield, 36
 - Star Plate Rules, 33
 - TDC 3000^X, 30
- Guidelines
 - Power System, 20
- hardware selection, 37
- heater, 47
 - model numbers, 47

Index

- mounting, 47
- specifications, 47
- heater assembly
 - installation, 79
 - installation procedure, 80
 - replacement procedure, 91
- high temperature operation, 7
- HPMM card set selection, 40
- I/O Link address
 - pinning, 52
 - selection, 52, 73
- I/O Link Extender, 5, 14
 - cable A and B, 63
 - introduction, 42
 - Long Distance, 65
 - link selection, 65
 - termination, 65
 - model numbers, 42
 - Standard, 64
 - termination, 64
- I/O Link Extender cable
 - connection, 63
 - connection rules, 63
- I/O Link Extender cables, 44
 - installation, 44
- I/O Link Interface
 - installation, 63
- I/O Link Interface cables, 43
 - installation, 62
 - model numbers, 43
- installation, 51
- installation considerations, 52
- IOP model numbers, 40
- IOP selection, 40
- IOP to FTA cables, 43
 - model numbers, 43
- IOP types, 40
- IOPs, 13
- Lightning Ground, 31
- low temperature operation, 6
- Master Reference Ground, 27
- multi-ground system, 27
- NiCad battery replacement, 92
- parts, 93
 - ORU, 96
 - periodic maintenance, 94
 - UCN, 106
- Planned Expansions, 17
- planning
 - guidelines, 15
- Planning the Power System, 17
- Power Distribution
 - Conduit, 24
- Power Distribution System Losses, 19
- Power Entry Box, 24
- power source, 52
- Power Supply Module
 - circuit breaker, 52
 - protection, 52
- Power System, 57
 - AC Only, 13, 38
 - standard, 12, 38
- Power System Design Guidelines, 20
- Power System selection, 5, 38
- preventative maintenance, 92
- procedure
 - UCNE enclosure installation, 69
- Required Grounds, 28
- Rules for star plates, 33
- Safety Ground
 - Definition, 28
- selection
 - card file, 39
 - FTA, 41
 - FTA Mounting Channels, 42
 - hardware, 37
 - HPMM card set, 40
 - IOP, 40
 - Power System, 38
- service, 83
- Shield grounding methods, 36
- spare parts, 93
- standard Power System, 12, 38, 57, 59
 - ac power connections, 57
 - alarm connections, 59
 - safety ground connections, 57
- System Grounds, 28
- Technical Assistance, 17
- temperature
 - high, 7
 - low, 6
- temperatures
 - external ambient, 2
- TotalPlant** Configurator, 4, 48
- UCN coaxial cables, 44
- UCN drop cables, 44, 45
- UCN drop tap kits, 44, 45
- UCN Extender, 5, 14
- UCN node address
 - selection, 74
- UCNE, 14
 - fiber optic cable installation, 72
 - installation, 67
 - introduction, 42
 - UCN drop cable installation, 72
- UCNE cabling, 45
- UCNE enclosure
 - installation procedure, 69
- UCNE kits, 45
- Universal Control Network Extender
 - installation, 67
 - introduction, 42
 - model numbers, 42
- Universal Control Network Extender cabling, 45
- Wire Size Table, 22, 23

Honeywell

Honeywell

Process Solutions

2500 West Union Hills Drive

Phoenix, AZ 85027